

Spacetime and Geometry – Sean M. Carroll

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1 Special Relativity and Flat Spacetime (pp. 1 – 47)

1.1 Introductory Concepts

General relativity: basically gravity and geodesics.

- Most forces are embedded in spacetime (e.g. Electromagnetic field).
- “Gravity” isn’t a force, but the *curvature* of spacetime. It’s an *inherent* feature.

Note: We adopt the convention $c = 1$ for the speed of light.

We begin from Newton’s law of gravitation: $F = GMm/r^2$. This can be rewritten as Poisson’s Equation, describing gravity in terms of a potential.

$$\underbrace{\nabla^2 \Phi}_{\text{curvature}} = \underbrace{4\pi G \rho}_{\text{source}}$$

- Φ is the gravitational potential, a scalar field of potential energy per mass.
- ρ is the mass density, i.e. the distribution of mass in space. This is the source that “generates” the gravitational potential.
- Acceleration is simple: $a = \nabla \Phi$, the gradient.

Roughly, mass affects the shape of the gravitational potential field. Similarly, in general relativity, spacetime curves in the presence of matter and energy.

However, Poisson’s equation (and Newton’s law) describes an instantaneous transfer of information: a change in mass immediately affects gravity. This can’t be true, if special relativity is accepted. So the equations must be modified.

Later we’ll derive

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0$$

The first equation is Einstein’s equation. The second describes a geodesic, where $x^\mu(\lambda)$ is the parametrized path.

1.2 Special Relativity

1.2.1 Basics

(Largely a review) Special relativity is a theory of 4D spacetime, the “Minkowski Space” (a special case of 4D manifolds). The distinction between “space” and “time” is arbitrary — you have to think of “spacetime” as a whole entity, not the sum of its parts.

In Newtonian dynamics, we have a well-defined notion of “simultaneity.” In SR, we don’t. Particles stay within their light cones in a spacetime diagram.

In Cartesian coordinates, distance is invariant under coordinate changes:

$$(\Delta s)^2 = (\Delta x)^2 + (\Delta y)^2$$

In SR, the “spacetime interval” is invariant under coordinate changes:

$$(\Delta s)^2 = -(c\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

We sum over Greek indices for spacetime coordinates and Latin indices for space coordinates. We also define the spacetime metric $\eta_{\mu\nu}$.

$$x^\mu : \begin{cases} x^0 = ct \\ x^1 = x \\ x^2 = y \\ x^3 = z \end{cases}, \quad x^i : \begin{cases} x^1 = x \\ x^2 = y \\ x^3 = z \end{cases}$$

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Then we can rewrite the spacetime interval more compactly. The summation convention is that indices in both superscript & subscript are summed over.

$$(\Delta s)^2 = \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu$$

$$(\Delta s)^2 < 0 : \text{timelike separated}$$

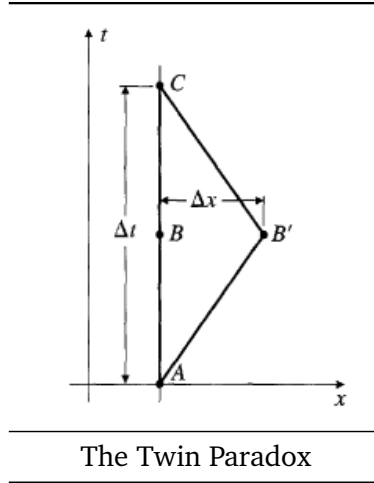
$$(\Delta s)^2 = 0 : \text{null}$$

$$(\Delta s)^2 > 0 : \text{spacelike separated}$$

Define “proper time” τ to measure the time elapsed as seen by an observer moving in a straight path between two events. I.e. This is the time recorded by the *observer*. We add a minus sign because otherwise, timelike separated events would be negative and that’s weird.

$$(\Delta \tau)^2 = -(\Delta s)^2 = -\eta_{\mu\nu} \Delta x^\mu \Delta x^\nu$$

1.2.2 The Twin Paradox



Let $B = \left(\frac{1}{2}\Delta t, 0\right)$ and $B' = \left(\frac{1}{2}\Delta t, \Delta x\right)$ with $\Delta x = \frac{1}{2}v\Delta t$.

Alice travels straight from A to B . Bob travels from A to B' . The proper times (with $c = 1$ and $v \leq c$) are

$$\begin{aligned} \text{Alice : } \Delta\tau_{AB} &= \frac{1}{2}\Delta t \\ \text{Bob : } \Delta\tau_{AB'} &= \sqrt{\left(\frac{1}{2}\Delta t\right)^2 - (\Delta x)^2} = \frac{1}{2}\Delta t\sqrt{1-v^2} \end{aligned}$$

Bob has aged less than Alice! Because a nonstraight path in *spacetime* has a different interval than a straight path. Additionally, the elapsed time for the observer depends on the path you take.

1.2.3 Lorentz Transformations

Transformations in spacetime. The basic example is a translation with δ as the potential. We “shift” one variable, either spatial or temporal. Note that the distance Δx^μ between two points remains the same, thus the interval $(\Delta s)^2$ is invariant.

$$x^\mu \longrightarrow x^{\mu'} = \delta_\mu^{\mu'} (x^\mu + a^\mu), \quad \delta_\mu^{\mu'} = \begin{cases} 1 & \mu' = \mu \\ 0 & \mu' \neq \mu \end{cases}$$

Other examples are rotations and boosts (offsets by a constant velocity vector). We model these with a matrix Λ .

$$x^{\mu'} = \Lambda_\nu^{\mu'} x^\nu, \quad \text{i.e. } x' = \Lambda x$$

The interval $(\Delta s)^2$ must remain invariant, so Δx^μ must not change. Thus we have

$$\begin{aligned} (\Delta s)^2 &= (\Delta x)^T \eta (\Delta x) = (\Delta x')^T \eta (\Delta x') \\ (\Delta x)^T \eta (\Delta x) &= (\Delta x)^T \Lambda^T \eta \Lambda (\Delta x) \\ \eta &= \Lambda^T \eta \Lambda \end{aligned}$$

The Λ that satisfies the above is a Lorentz transformation. Under matrix multiplication, the Lorentz transformations form a Lorentz group. The set of translations and Lorentz transformations form a 10-parameter, non-Abelian group known as the *Poincaré group*.

As an example, consider a boost in the x -direction (which can be thought of as rotations between space and time directions).

$$\Lambda_{\nu}^{\mu'} = \begin{pmatrix} \cosh \phi & -\sinh \phi & 0 & 0 \\ -\sinh \phi & \cosh \phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \phi \in (-\infty, \infty)$$

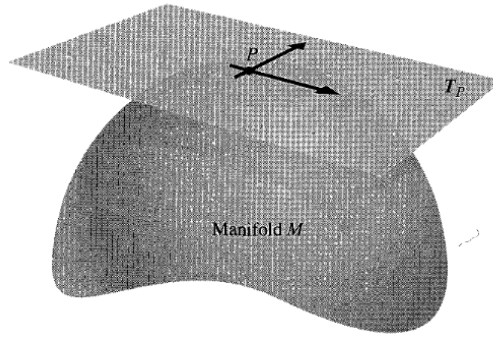
By applying Λ , we get $t' = t \cosh \phi - x \sinh \phi$ and $x' = -t \sinh \phi + x \cosh \phi$. Focusing on the point at x' , we see that it moves according to $t \sinh \phi = x \cosh \phi$. Thus $v = x/t = \tanh \phi$.

Using $\phi = \tanh^{-1} v$, we can reconstruct the usual Lorentz transformations. Set $\gamma = \frac{1}{\sqrt{1-v^2}}$ and $c = 1$ to get

$$\begin{aligned} t' &= \gamma(t - vx) \\ x' &= \gamma(x - vt) \end{aligned}$$

1.2.4 4-Vectors

4-vectors are located at a single point in spacetime, not stretched from A to B. At each point p , the set of all possible vectors located at p is T_p , the tangent space of p . The set of all tangent spaces of an n -dimensional manifold M can be assembled into a $2n$ -dimensional manifold $T(M)$, the “tangent bundle.”



Tangent Space and Manifold Diagram (it doesn't exactly look like this though)

A vector A can be represented as a linear combination of basis vectors $\hat{e}_{(\mu)}$ and coefficients A^μ :

$$A = A^\mu \hat{e}_{(\mu)}$$

The basis vectors transform as expected with Lorentz transformations:

$$\hat{e}_{(\mu)} = \Lambda_{\mu}^{\nu'} \hat{e}_{(\nu')}$$

A tangent vector $V(\lambda)$ to a curve in spacetime parametrized by $x^\mu(\lambda)$ can be represented

$$\begin{aligned} V^\mu &= \frac{dx^\mu}{d\lambda} \\ V &= V^\mu \hat{e}_{(\mu)} \end{aligned} \quad (\text{components})$$

1.2.5 Dual Vectors

Kind of a complement to a vector space. The dual vector of the tangent space T_p is denoted T_p^* , the “cotangent space.” A vector ω is in T_p^* iff:

$$\begin{aligned} \omega(aV + bW) &= a\omega(V) + b\omega(W) \in \mathbb{R} \\ V, W &\in T_p \end{aligned}$$

Essentially, the dual space is the space of all linear maps from T_p to $\mathbb{R}$. A quick and dirty way to think about it: upper indices denote vectors, lower indices denote dual vectors.

Examples of dual vectors include the gradient, the row vector (corresponding to column vectors), etc.

We can define a “basis dual vector” $\hat{\theta}^{(\nu)}$ in order to represent the dual vector ω :

$$\begin{aligned} \hat{\theta}^{(\nu)}(\hat{e}_{(\mu)}) &= \delta_\mu^\nu \\ \omega &= \omega_\mu \hat{\theta}^{(\mu)} \end{aligned}$$

Then we can find the effect of a dual vector on a vector:

$$\begin{aligned} \omega(V) &= \omega_\mu \hat{\theta}^{(\mu)}(V^\nu \hat{e}_{(\nu)}) \\ &= \omega_\mu V^\nu \hat{\theta}^{(\mu)}(\hat{e}_{(\nu)}) \\ &= \omega_\mu V^\nu \delta_\nu^\mu \\ &= \omega_\mu V^\mu \in \mathbb{R} \end{aligned}$$

A similar calculation shows that vectors are also linear maps on dual vectors, which is a nice symmetry.

$$V(\omega) = \omega(V) = \omega_\mu V^\mu \in \mathbb{R}$$

A simple example of a dual vector in spacetime is the gradient of a scalar function. We denote $d\phi$ as the set of partial derivatives w.r.t. spacetime coordinates.

$$d\phi = \frac{\partial \phi}{\partial x^\mu} \hat{\theta}^{(\mu)}$$

The chain rule is intuitive.

$$\frac{\partial \phi}{\partial x^{\mu'}} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial \phi}{\partial x^\mu} = \Lambda_{\mu'}^\mu \frac{\partial \phi}{\partial x^\mu}$$

We define the shorthand $\partial_\mu \phi = \frac{\partial \phi}{\partial x^\mu}$. The gradient of a tangent vector along the parametrized curve $x^\mu(\lambda)$ is

$$\frac{d\phi}{d\lambda} = \partial_\mu \phi \frac{\partial x^\mu}{\partial \lambda}$$

1.3 Tensors

Tensors are a generalization of vectors and dual vectors. Recall that a dual vector is a map from vectors to $\mathbb{R}$. Tensors are a map from a collection of vectors & dual vectors to $\mathbb{R}$. For a tensor of rank (k, l) , we map the vectors in T_p^* and dual vectors in T_p to $\mathbb{R}$.

$$T : \underbrace{T_p^* \times \cdots \times T_p^*}_{k \text{ terms}} \times \underbrace{T_p \times \cdots \times T_p}_{l \text{ terms}} \longrightarrow \mathbb{R}$$

$\times$ is the Cartesian product. $T_p \times T_p$ is the space of the ordered pair of vectors.

A tensor is *multilinear*, meaning it acts linearly in each argument. So for example, a tensor of rank $(1, 1)$ would act like

$$T(a\omega + b\eta, cV + dW) = acT(\omega, V) + adT(\omega, W) + bcT(\eta, V) + bdT(\eta, W)$$

We can further say that a scalar is a tensor of rank $(0, 0)$; a vector is a tensor of rank $(1, 0)$; and a dual vector is a tensor of rank $(0, 1)$. The space of all tensors of rank (k, l) form a vector space, where we can define the *tensor product* $\otimes$ as

$$\begin{aligned} T &: \text{tensor of rank } (k, l) \\ S &: \text{tensor of rank } (m, n) \\ T \otimes S &\left(\omega^{(1)}, \dots, \omega^{(k)}, \dots, \omega^{(k+m)}, V^{(1)}, \dots, V^{(l)}, \dots, V^{(l+n)} \right) \\ &= T \left(\omega^{(1)}, \dots, \omega^{(k)}, \dots, \omega^{(k+m)} \right) \times \\ &\quad S \left(V^{(1)}, \dots, V^{(l)}, \dots, V^{(l+n)} \right) \end{aligned}$$

$T \otimes S$ is a rank $(k + m, l + n)$ tensor.

In spacetime, for a rank (k, l) tensor, there's 4^{k+l} basis tensors. We can represent this simply by summing over each bases.

$$T = T^{\mu_1 \cdots \mu_k}_{\nu_1 \cdots \nu_l} \hat{e}_{(\mu_1)} \otimes \cdots \otimes \hat{e}_{(\mu_k)} \otimes \hat{\theta}^{(\nu_1)} \otimes \cdots \otimes \hat{\theta}^{(\nu_k)}$$

Similarly, the action of tensors on a st of vectors and dual vectors is given below. Note that each a (k, l) tensor has k upper and l lower indices. Upper indices transform like vectors, while lower indices transform like dual vectors.

$$T \left(\omega^{(1)}, \dots, \omega^{(k)}, V^{(1)}, \dots, V^{(l)} \right) = T^{\mu_1 \cdots \mu_k}_{\nu_1 \cdots \nu_l} \omega_{\mu_1}^{(1)} \cdots \omega_{\mu_k}^{(k)} V^{(1)\nu_1} \cdots V^{(l)\nu_l}$$

Tensors transform as expected under Lorentz:

$$T^{\mu'_1 \cdots \mu'_k}_{\nu'_1 \cdots \nu'_l} = \Lambda^{\mu'_1}_{\mu_1} \cdots \Lambda^{\mu'_k}_{\mu_k} \Lambda^{\nu_1}_{\nu'_1} \cdots \Lambda^{\nu_l}_{\nu'_l} T^{\mu_1 \cdots \mu_k}_{\nu_1 \cdots \nu_l}$$

We're also under no obligation to transform every component. For example, we could just transform a single component of a $(1, 1)$ tensor to make it a vector $\rightarrow$ vector map.

$$T^\mu{}_\nu : V^\nu \longrightarrow T^\mu{}_\nu V^\nu$$

It gets complicated, but just try to keep track of the indices. Also note that tensors $\neq$ vectors. Tensors are geometric entities independent of the coordinate system.

1.3.1 Examples of Tensors

Metric The $\eta_{\mu\nu}$ is a $(0, 2)$ tensor. Its action on two vectors is known as the “inner product”:

$$\eta(V, W) = \eta_{\mu\nu} V^\mu W^\nu = V \cdot W$$

If the inner product vanishes, the two vectors are orthogonal. Note that the inner product is a scalar, thus invariant under Lorentz transformations. So a set of orthogonal vectors in one inertial Cartesian frame is also orthogonal after some Lorentz transforms, even though they may not look like it.

The norm of a vector is its inner product with itself:

$$\eta(V, V) = \eta_{\mu\nu} V^\mu V^\nu \begin{cases} < 0 & V^\mu \text{ is timelike} \\ = 0 & V^\mu \text{ is lightlike/null} \\ > 0 & V^\mu \text{ is spacelike} \end{cases}$$

Kronecker Delta The Kronecker delta δ_ρ^μ is another example of a tensor. It’s an identity map from a vector to the same vector. Note that we put the indices in the same “column.” That’s because order doesn’t matter, $\delta_\rho^\mu = \delta^\mu_\rho = \delta_\rho^\mu$.

The Kronecker delta is defined as the metric times its inverse:

$$\delta_\rho^\mu = \eta^{\mu\nu} \eta_{\nu\rho} = \eta_{\rho\nu} \eta^{\nu\mu}$$

Levi-Civita The Levi-Civita tensor has rank $(0, 4)$ in spacetime. We define it as

$$\tilde{\epsilon}_{\mu\nu\rho\sigma} = \begin{cases} +1 & \text{if } \mu\nu\rho\sigma \text{ is an even permutation of } 0123 \\ -1 & \text{if } \mu\nu\rho\sigma \text{ is an odd permutation of } 0123 \\ 0 & \text{otherwise} \end{cases}$$

Electromagnetic Field Strength This is a rank $(0, 2)$ tensor composed of the E_i and B_i vectors. Note the Latin indices indicating spacelike 1, 2, 3 components. By Lorentz transforming this tensor, we can get the E and B fields in a different reference frame.

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix} = -F_{\nu\mu}$$

So for example, the field tensor transforms under a Lorentz boost as

$$F^{\mu'\nu'} = \Lambda^{\mu'}_\mu \Lambda^{\nu'}_\nu F^{\mu\nu}$$

1.3.2 Manipulating Tensors

Contraction A contraction of a tensor reduces its rank from $(k, l) \rightarrow (k - 1, l - 1)$. To do this, we sum over the upper and lower indices.

$$S^{\mu\rho}{}_\sigma = T^{\mu\nu\rho}{}_{\sigma\nu}$$

Order matters. In general $T^{\mu\nu\rho}{}_{\sigma\nu} \neq T^{\mu\rho\nu}{}_{\sigma\nu}$.

Raising & Lowering Indices We can also raise and lower indices in tensors using the metric and its inverse. For example,

$$\begin{aligned} T^{\alpha\beta\mu}{}_{\delta} &= \eta^{\mu\gamma} T^{\alpha\beta}{}_{\gamma\delta} \\ T_{\mu}{}^{\beta}{}_{\gamma\delta} &= \eta_{\mu\alpha} T^{\alpha\beta}{}_{\gamma\delta} \end{aligned}$$

Since the metric and its inverse are, well, inverse of each other, we also have the following:

$$A^{\lambda}B_{\lambda} = \eta^{\lambda\rho}A_{\rho}\eta_{\lambda\sigma}B^{\sigma} = \delta^{\rho}_{\sigma}A_{\rho}B^{\sigma} = A_{\sigma}B^{\sigma}$$

Symmetric & Antisymmetric Tensors A tensor is called symmetric if it remains unchanged under exchange of indices. In the following, S is symmetric in μ and ν .

$$S_{\mu\nu\rho} = S_{\nu\mu\rho}$$

We can symmetrize a tensor by summing over its permutations. The symmetric part is denoted by parentheses.

$$T_{(\mu_1\mu_2\cdots\mu_n)\rho}{}^{\sigma} = \frac{1}{n!} (T_{\mu_1\mu_2\cdots\mu_n\rho}{}^{\sigma} + \text{sum over permutations of } \mu_1 \cdots \mu_n)$$

A tensor is called antisymmetric if it changes sign under exchange of indices. In the following, A is symmetric in μ and ρ .

$$A_{\mu\nu\rho} = -A_{\rho\nu\mu}$$

We can antisymmetrize a tensor by taking the alternating sum over its permutations. The antisymmetric part is denoted by a bracket. “Alternating sum” means that the permutations that result from an odd number of exchanges are given a negative sign.

$$T_{[\mu_1\mu_2\cdots\mu_n]\rho}{}^{\sigma} = \frac{1}{n!} (T_{\mu_1\mu_2\cdots\mu_n\rho}{}^{\sigma} + \text{alternating sum over permutations of } \mu_1 \cdots \mu_n)$$

When contracting over a symmetric tensor, only the symmetric parts matter. Similar for antisymmetric tensors.

$$\begin{aligned} X^{(\mu\nu)}Y_{\mu\nu} &= X^{(\mu\nu)}Y_{(\mu\nu)} \\ X^{[\mu\nu]}Y_{\mu\nu} &= X^{[\mu\nu]}Y_{[\mu\nu]} \end{aligned}$$

We can also decompose a tensor into a sum for *two indices*. For 3 and above, we generally can’t.

$$T_{\mu\nu\rho\sigma} = T_{(\mu\nu)\rho\sigma} + T_{[\mu\nu]\rho\sigma}$$

Trace The trace of a tensor, denoted X is given by

$$X = X^{\lambda}{}_{\lambda}$$

That’s the simple case of a (1, 1) tensor. For a (0, 2) tensor $Y_{\mu\nu}$, we raise indices first.

$$Y = Y^{\lambda}{}_{\lambda} = \eta^{\mu\nu}Y_{\mu\nu}$$

1.4 Tensor Form of Maxwell's Equations

The classical differential form of Maxwell's Equations:

$$\begin{cases} \nabla \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J} & (1) \\ \nabla \cdot \mathbf{E} = \rho & (2) \\ \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0 & (3) \\ \nabla \cdot \mathbf{B} = 0 & (4) \end{cases}$$

We convert to component notation. Spatial indices (Latin i, j, k) are raised and lowered at will since the metric δ_{ij} is an identity on flat 3D space. We also replace J with the current 4-vector $J^\mu = (\rho, J^x, J^y, J^z)$.

$$\begin{cases} \epsilon^{ijk} \partial_j B_k - \partial_0 E^i = J^i & (5) \\ \partial_i E^i = J^0 & (6) \\ \epsilon^{ijk} \partial_j E_k + \partial_0 B^i = 0 & (7) \\ \partial_i B^i = 0 & (8) \end{cases}$$

We use the electromagnetic field strength tensor from before.

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix} = -F_{\nu\mu}$$

Now note that $F^{0i} = \eta^{00}\eta^{ii}F_{0i} = E^i$ (we raise and lower indices at will). Also, $F^{ij} = \epsilon^{ijk}B_k$. Thus the first two of Maxwell's equations can be rewritten as

$$\begin{aligned} \partial_j F^{ij} - \partial_0 F^{0i} &= J^i \\ \partial_i F^{0i} &= J^0 \end{aligned}$$

We can merge the two equations into one:

$$\partial_\mu F^{\nu\mu} = J^\nu$$

Similarly, the 3rd and 4th of Maxwell's equations can be merged into one, giving us the two equations that describe E and B :

$$\begin{aligned} \partial_\mu F^{\nu\mu} &= J^\nu \\ \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} + \partial_\lambda F_{\mu\nu} &= 0 \end{aligned}$$

1.5 Energy & Momentum

Consider the worldline of a single particle. The path is a parametrized curve $x^\mu(\lambda)$ for some variable λ . The tangent vector is $\frac{dx^\mu}{d\lambda}$. If this tangent vector is timelike/spacelike/null at some value of λ , the path is also timelike/spacelike/null at that point.

Consider paths of massive particles. They'll be timelike, slower than c . We'll attempt to parametrize them in terms of their proper time τ (the time seen on the clock of an observer on that path). Recall that

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$$

$$d\tau^2 = -\eta_{\mu\nu} dx^\mu dx^\nu$$

We integrate over these to get the total path:

$$\Delta s = \int \sqrt{\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda$$

$$\Delta \tau = \int \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda$$

We can use the equation above to parametrize $\tau(\lambda)$ in terms of λ . Afterwards, we can invert to parametrize $\lambda(\tau)$ in terms of τ . Great, now we can parametrize the path in terms of τ : $x^\mu(\tau)$.

1.5.1 4-Velocity

We define the tangent vector of the path (parametrized w.r.t τ) as the 4-velocity.

$$U^\mu = \frac{dx^\mu}{d\tau}$$

Note that because $d\tau^2 = -\eta_{\mu\nu} dx^\mu dx^\nu$, the 4-velocity is automatically normalized.

$$\eta_{\mu\nu} U^\mu U^\nu = -1 \quad (\text{with } c = 1)$$

Note that the 4-velocity isn't the velocity through space, which can have different magnitudes. It's the velocity through spacetime, with the same magnitude for all observers.

Here we were working with timelike paths. Spacelike paths are weirder because it would mean the observer is travelling "faster than c ." For null paths, τ vanishes, so more care is needed.

In the rest frame of a particle, the 4-velocity has components $U^\mu = (1, 0, 0, 0)$ assuming $c = 1$.

1.5.2 4-Momentum

Let the rest mass of a particle be m . Its 4-momentum is defined as

$$p^\mu = mU^\mu$$

The energy of a particle is given by

$$E = p^0$$

For particles at rest, $p^0 = m$. But we were setting $c = 1$, so technically it's $E = p^0 = mc^2$. We see Einstein's famous equation for a particle at rest.

For a moving frame, Lorentz transform the 4-momentum. For example, consider a particle moving in the $+x$ direction with velocity $v = \frac{dx}{dt}$. Keep in mind that $c = 1$, so $\beta = v/c = v$. Get used to it.

$$p^\mu = \begin{pmatrix} \gamma & \gamma v & 0 & 0 \\ \gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} E \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma E \\ \gamma v E \\ 0 \\ 0 \end{pmatrix}$$

For small v , we know that $\gamma = (1 - v^2)^{-1/2} \approx 1 + \frac{1}{2}v^2$. Thus $p^0 \approx m + \frac{1}{2}mv^2$ and $p^1 \approx mv$, which is the Newtonian energy and momentum, respectively.

We also see that $p_\mu p^\mu = -m^2$. Tweaking this expression gives the full relativistic energy equation:

$$E^2 = m^2 + \mathbf{p}^2, \quad \mathbf{p}^2 = \delta_{ij} p^i p^j$$

Classically, Newton's 2nd Law can be written as $\vec{F} = m\vec{a} = \frac{d\vec{p}}{dt}$. Relativistically, we use the 4-vector F^μ :

$$F^\mu = m \frac{d^2 x^\mu}{d\tau^2} = \frac{dp^\mu}{d\tau}$$

For example, the Lorentz force $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ becomes

$$F^\mu = qU^\mu F_\lambda{}^\mu$$

1.5.3 Stress-Energy Tensor

The 4-momentum works great for the case of 1 particle. For multi-particle systems, we describe the state as a “fluid” – a continuum characterized by density, pressure, etc. Instead of a 4-momentum vector field, we define an energy-momentum tensor (also stress-energy tensor) $T^{\mu\nu}$. This is a $(2, 0)$ symmetric tensor that tells us the system's energy density, pressure, etc.

We can interpret $T^{\mu\nu}$ as “The flux of 4-momentum p^μ across a surface of constant x^ν .” From this we can see that

- i) T^{00} : “Flux of p^0 (energy) across surface x^0 (time),” i.e. energy density
- ii) $T^{0i} = T^{i0}$: Momentum density
- iii) $T^{ij}, i \neq j$: “Shearing” force between neighboring elements of fluid.
- iv) T^{ii} : Force exerted (per unit area) by fluid in x^i -direction

To start off, we consider dust (or “matter” to cosmologists). It's a collection of particles at rest w.r.t each other in flat spacetime. Define the number-flux 4-momentum N^μ :

$$N^\mu = nU^\mu$$

Here n is the number density of particles in the rest frame. Note that N^0 is the number density of particles in any frame, while N^i is the number density of particles in the x^i -direction. Let m be the mass of 1 particle. In the rest frame, the energy density is

$$\rho = mn$$

Note that this is the combination of $N^\mu = (n, 0, 0, 0)$ and $p^\mu = (m, 0, 0, 0)$ – the number-density and 4-momentum. Thus ρ is the $\mu = \nu = 0$ component of the tensor $p \otimes N$. We therefore define the energy-momentum tensor for dust as

$$T_{\text{dust}}^{\mu\nu} = p^\mu N^\nu = mnU^\mu U^\nu = \rho U^\mu U^\nu$$

Now we generalize for *perfect fluids*: the “fluids” completely specified by the rest-frame energy density ρ and the isotropic rest-frame pressure p .

Due to isotropy, $T^{\mu\nu}$ is diagonal in the rest frame. There’s no net flux of any component of momentum in an orthogonal direction. Further, $T^{11} = T^{22} = T^{33}$. Thus the only independent numbers are $\rho = T^{00}$ and $p = T^{ii}$.

It turns out that the general energy-momentum tensor of a perfect fluid in any frame is

$$T^{\mu\nu} = (\rho + p)U^\mu U^\nu + p\eta^{\mu\nu}$$

To determine the evolution of a system, we relate the pressure to density: $p = p(\rho)$. Dust is the special case of $p = 0$. An isotropic photon gas has $p = \frac{1}{3}\rho$.

Note: we’re in flat spacetime. Things change with curvature.

Also note that the stress-energy tensor is conserved: the divergence vanishes.

$$\partial_\mu T^{\mu\nu} = 0$$

The $\nu = 0$ case expresses the conservation of energy. Otherwise, $\nu = k$ expresses the conservation of the k -th component of momentum.

1.6 Classical Field Theory

1.6.1 Background

This will be a crude overview, because most of it went over my head.

To go from special relativity (flat spacetime) to general relativity (curvature), we promote the metric tensor $\eta_{\mu\nu}$ to a dynamical tensor field $g_{\mu\nu}(x)$.

Consider a particle in 1D with coordinate $q(t)$ in classical mechanics. Let the Lagrangian be $L(q, \dot{q})$. Then we define the action S as

$$S = \int L(q, \dot{q}) dt$$

And the Euler-Lagrange equations

$$\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) = 0$$

In field theory, we replace the single coordinate $q(t)$ with a set of spacetime-dependent fields $\phi^i(x^\mu)$. Each ϕ^i is a function on spacetime, and i is an index for each individual field. The action S becomes a functional of these fields (a function of ∞ variables).

The Lagrangian is an integral over the space of Lagrange density $\mathcal{L}$. Here, $\mathcal{L}$ is the function of fields ϕ^i and spacetime derivatives $\partial_\mu \phi^i$:

$$L = \int \mathcal{L}(\phi^i, \partial_\mu \phi^i) d^3x$$

The action is the integral of the Lagrangian:

$$S = \int L dt = \int \mathcal{L}(\phi^i, \partial_\mu \phi^i) d^4x$$

Here we use natural units: $c = 1$, $\hbar = 1$, $k_B = 1$. Thus

$$[\text{energy}] = [\text{mass}] = [\text{length}^{-1}] = [\text{time}^{-1}]$$

Considering infinitesimal variations, and doing lots of calculations I couldn't follow, we arrive at the Euler-Lagrange equations for field theory in flat spacetime:

$$\frac{\delta S}{\delta \phi^i} = \frac{\partial \mathcal{L}}{\partial \phi^i} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^i)} \right) = 0$$

The simplest example of a field is a real scalar field.

$$\phi(x^\mu) : (\text{spacetime}) \longrightarrow \mathbb{R}$$

Upon quantization, “excitations” of a field are observable as particles. Different fields correspond to different particle types.

- i) Scalar fields: spinless particles
- ii) Vector fields & other tensors: higher-spin particles
- iii) Complex-valued fields (2 degrees of freedom): particle + antiparticle

1.6.2 Motion & Properties

For the field ϕ , we can apply classical mechanics:

$$\begin{aligned} E_{\text{kinetic}} &= \frac{1}{2} \dot{\phi}^2 \\ E_{\text{gradient}} &= \frac{1}{2} (\nabla \phi)^2 \\ E_{\text{potential}} &= V(\phi) \end{aligned}$$

Potential energy is Lorentz invariant. Kinetic and gradient energy aren't Lorentz invariant by themselves, but together they are:

$$-\frac{1}{2}\eta^{\mu\nu}(\partial_\mu\phi)(\partial_\nu\phi) = \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2$$

Thus our Lagrangian (Lagrange density) is

$$\mathcal{L} = -\frac{1}{2}\eta^{\mu\nu}(\partial_\mu\phi)(\partial_\nu\phi) - V(\phi)$$

Using $\square \equiv \eta^{\mu\nu}\partial_\mu\partial_\nu$, the d'Alembertian, our equation of motion is

$$\square\phi - \frac{dV}{d\phi} = 0$$

In flat spacetime, we have $\ddot{\phi} - \nabla^2\phi + \frac{dV}{d\phi} = 0$.

For a simple harmonic oscillator with $V(\phi) = \frac{1}{2}m^2\phi^2$, our equation of motion is $\square\phi - m^2\phi = 0$. This is the famous Klein-Gordon equation.

Note that the signs in our equation are dependent on the sign convention. In this book, it's $(-, +, +, +)$.

For electromagnetic fields, the relevant field is the vector potential A_μ . A_0 is the electrostatic potential ϕ , while the spacelike components are traditional vector potentials A (e.g. magnetic field is $B = \nabla \times A$). We can express the field strength tensor $F_{\mu\nu}$ as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

This exhibits *gauge invariance*: for a small change $A_\mu \rightarrow A_\mu + \partial_\mu\lambda(x)$,

$$F_{\mu\nu} \rightarrow F_{\mu\nu} + \partial_\mu\partial_\nu\lambda - \partial_\nu\partial_\mu\lambda = F_{\mu\nu}$$

Gauge invariance, and the symmetries that come with it, will become important in quantum field theory. But that's way too advanced.

The energy-momentum tensor for scalar field theory is

$$T_{\text{scalar}}^{\mu\nu} = \eta^{\mu\lambda}\eta^{\lambda\sigma}\partial_\lambda\phi\partial_\sigma\phi - \eta^{\mu\nu}\left[\frac{1}{2}\eta^{\lambda\sigma}\partial_\lambda\phi\partial_\sigma\phi + V(\phi)\right]$$

The energy-momentum tensor for an electromagnetic field is

$$T_{\text{EM}}^{\mu\nu} = F^{\mu\nu}F^\nu{}_\lambda - \frac{1}{4}\eta^{\mu\nu}F^{\lambda\sigma}F_{\lambda\sigma}$$

2 Manifolds (pp. 48 – 92)

2.1 Gravity as Geometry

The field giving rise to gravity is the metric tensor itself (which describes the curvature of spacetime). It's not an additional field propagating through spacetime, which is interesting.

The evidence for this is the *universality* of gravitational interaction: The “Principle of Equivalence.” Every object is influenced by gravity in the same way. Compare that to an electromagnetic field: particles with different charges behave differently.

2.1.1 Weak Equivalence Principle

(Form 1) Inertial mass and gravitational mass are equal.

Inertial mass m_i is given by Newton's 2nd law: $\vec{F} = m_i \vec{a}$. Gravitational mass is given by Newton's law of gravitation, which can be restated with the gravitational-potential scalar field Φ : $\vec{F}_g = -m_g \nabla \Phi$. The WEP thus states that

$$\begin{aligned} m_i &= m_g \\ \vec{a} &= -\nabla \Phi \end{aligned}$$

In other words, there's a preferred class of trajectories through spacetime on which unaccelerated particles travel, which we call *inertial trajectories*. This can be seen above, where the acceleration depends only on the gradient of the gravitational potential. By “unaccelerated,” we mean the particle is subject only to gravity. The WEP has been verified to high precision by the Eötvös Experiment.

Compare this to electromagnetism, where particles with different charges have different trajectories even in the same field.

2.1.2 Einstein Equivalence Principle

The WEP in another form is that

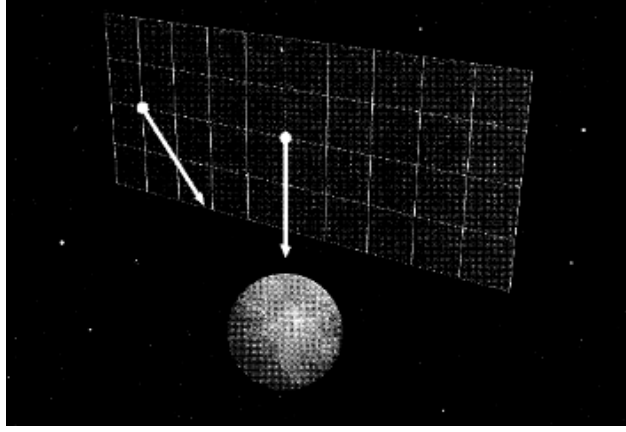
The motion of freely-falling particles are same in a gravitational field and in a uniformly accelerated frame, in small enough regions of spacetime.

Special relativity showed that mass is just a manifestation of energy. Thus Einstein strengthened the WEP to formulate the Einstein Equivalence Principle (EEP):

In small enough regions of spacetime, the laws of physics reduce to those of special relativity. It's impossible to detect the existence of gravitational fields by a local experiment.

The EEP accounts for *all* laws of physics, not just free fall. A consequence of the EEP is that “gravitational acceleration” is hard to define: we can't determine the gravitational field by a local experiment. Instead, we let “unaccelerated” mean “freely falling.” Thus gravity is not viewed as a force, since no “acceleration” is caused in this view.

We must also give up the notion of inertial frames extending throughout spacetime. We can only have *locally inertial frames*. At some distance away, “acceleration” appears to be observed, i.e. the particle doesn’t seem to be stationary in the reference frame:



(Carroll 51) At some distance away, it appears that the left particle is not stationary in the reference frame defined by the right particle.

As an example of the EEP, consider 2 rockets separated by a distance z . Both rockets accelerate at the same rate a . The trailing rocket emits a photon with wavelength λ_0 at time t_0 . We consider the speed of the rockets to be much less than c , $v \ll c$. Then the photon reaches the front rocket after $\Delta t = \frac{z}{c}$. The rockets also pick up $\Delta v = a\Delta t = \frac{az}{c}$.

The light wave should be redshifted by the Doppler effect. Assuming $v \ll c$,

$$\frac{\Delta\lambda}{\lambda_0} \approx \frac{\Delta v}{c} = \frac{az}{c^2}$$

Now let’s consider a *gravitational redshift*. Consider a tower at height z on Earth. A photon is emitted from the ground at time t_0 . Let $\vec{a}_g$ be the strength of the gravitational field, what we call gravitational acceleration in Newtonian terms. Then by the equivalence principle, there’s no difference between this and the rockets. Though the tower seems “stationary” on Earth, it’s actually being accelerated against the gravitational field (which pulls it towards the center). The same redshift occurs:

$$\frac{\Delta\lambda}{\lambda_0} \approx \frac{az}{c^2}$$

The redshift is usually stated as $\vec{a}_s = \nabla\Phi$, where Φ is the Newtonian potential and $\vec{a}_s$ is the acceleration of the reference frame (not the particle — hence why the signs are reversed). We combine this with $\frac{\Delta\lambda}{\lambda_0} \approx \frac{\Delta v}{c}$ from before to get

$$\begin{aligned} \frac{\Delta\lambda}{\lambda_0} &\approx \int \nabla\Phi \, dt \\ &= \frac{1}{c^2} \int \partial_z \Phi \, dz && \text{(reparametrize with } z) \\ &= \Delta\Phi && \text{(with } c = 1) \end{aligned}$$

Thus the gravitational redshift depends on the change in gravitational potential. This is because the path in spacetime through which the photons travelled was curved.

2.2 Manifolds

A manifold is a space that may be curved with complicated topology, but locally looks like $\mathbb{R}^n$. For example:

- i) $\mathbb{R}^n$ itself (line $\mathbb{R}^1$, plane $\mathbb{R}^2$, etc.)
- ii) n -sphere S^n (circle S^1 , sphere S^2 , etc.)
- iii) n -torus T^n (from folding an n -dimensional cube)
- iv) Riemann surface of genus g (2-torus with g holes)

Manifolds aren't limited to shapes. For example, $SO(2)$, the set of rotations in 2D, is a manifold. It turns out to be the same manifold as S^1 .

The product of 2 manifolds, e.g. $M \times M'$ is also a manifold. If M has dimension n with $p \in M$ and M' has dimension n' with $p' \in M'$, then $M \times M'$ has dimension $n + n'$ with ordered pairs (p, p') .

2.2.1 A More Rigorous Definition

Rigor is mathematicians propaganda. But it's helpful to get an idea. First, some basic definitions for maps:

- A map between two sets is a generalized function, denoted $\phi : M \rightarrow N$.
- A map is injective (one-to-one) if each element in N has *at most* one element in M mapped to it.
- A map is surjective (onto) if each element in N has *at least* one element in M mapped to it.
- If a map is both injective and surjective, it's bijective (invertible) with the inverse map $\phi^{-1} : N \rightarrow M$.

For the map $\phi : \mathbb{R} \rightarrow \mathbb{R}$, $\phi(x) = e^x$ is injective but not surjective. $\phi(x) = x^3 - x$ is surjective but not injective. $\phi(x) = x^3$ is both.

For the map $\phi : M \rightarrow N$,

- M is the domain of the map.
- The image of ϕ is the set of points in N that M gets mapped to.
- The preimage of U under ϕ , denoted $\phi^{-1}(U)$, is the set of elements in M that get mapped to $U \subset N$.

For a general map $\phi : \mathbb{R}^m \rightarrow \mathbb{R}^n$, we're mapping the coordinates $(x^1, x^2, \dots, x^m)$ to $(y^1, y^2, \dots, y^n)$:

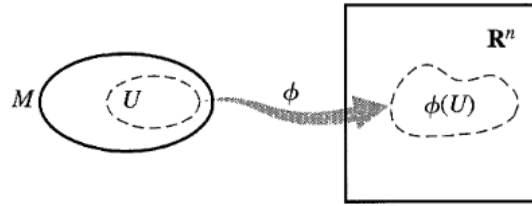
$$\begin{aligned} y^1 &= \phi^1(x^1, x^2, \dots, x^m) \\ y^2 &= \phi^2(x^1, x^2, \dots, x^m) \\ &\vdots \\ y^n &= \phi^n(x^1, x^2, \dots, x^m) \end{aligned}$$

Refer to ϕ as C^p if the p -th derivative exists for all $\phi^1 \dots \phi^n$. Then a C^0 map is continuous but not necessarily differentiable, for example. A C^∞ map is both continuous and infinitely differentiable, and we call this “smooth.”

Additionally, M, N are *diffeomorphic* if there exists a C^∞ map $\phi : M \rightarrow N$ with a C^∞ inverse $\phi^{-1} : N \rightarrow M$. Here ϕ is the *diffeomorphism*. This is the mathematical statement of our notion that two spaces are the “same” manifolds. For example, $SO(2)$ is diffeomorphic to S^1 .

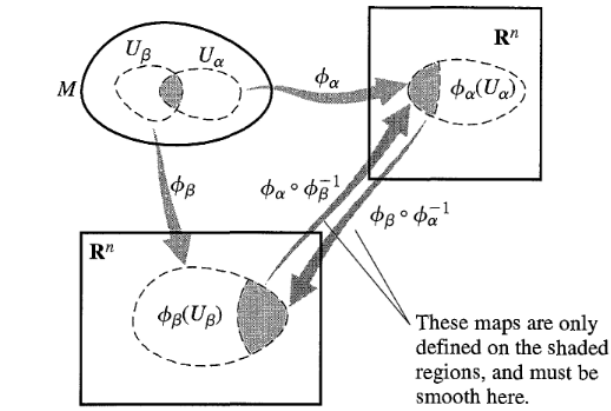
Now let's start to define open sets:

- The *open ball*: a set of points $\{x\} \in \mathbb{R}^n$ such that $|x - y| < r$ for some fixed $y \in \mathbb{R}^n$ and $r \in \mathbb{R}$. Here $|x - y|$ is the distance, $|x - y| = \sqrt{\sum_i (x_i - y_i)^2}$
- The *open set*: a set in $\mathbb{R}^n$ constructed from the arbitrary union of open balls
 - I.e. $V \subset \mathbb{R}^n$ is open if $\forall v \in V$, there exists an open ball centered at y that's completely in V .
- Charts/coordinate systems: a subset U of M along with a one-to-one map $\phi : U \rightarrow \mathbb{R}^n$ such that the image $\phi(U)$ is open in $\mathbb{R}^n$
 - Note that any map is onto its image, so ϕ is invertible.
 - Then U is an *open set* in M .



Charts/coordinate systems diagram (Carroll p. 59)

- C^∞ atlas: indexed collection of charts $\{(U_\alpha, \phi_\alpha)\}$ such that a) The union of U_α is M , b) The charts are smoothly sewn together.
 - By “smoothly sewn together,” we mean that if $U_\alpha \cap U_\beta \neq \emptyset$, the map $(\phi_\alpha \circ \phi_\beta^{-1})$ takes points in $\phi_\beta(U_\alpha \cap U_\beta) \subset \mathbb{R}^n$ onto the open set $\phi_\alpha(U_\alpha \cap U_\beta)$, and all of these maps are C^∞ whenever they're defined.



“Smoothly sewn together” diagram (Carroll p. 60)

Now we can finally say: a C^∞ n -dimensional manifold is a set M along with a maximal atlas: one that contains every possible compatible chart.

This is our notion of a set that locally looks like $\mathbb{R}^n$.

Why do we need charts & atlases? Many manifolds can't be covered by a single chart. Consider S^1 . The conventional description might be $\theta : S^1 \rightarrow \mathbb{R}$ with $\theta \in [0, 2\pi)$. But we require $\theta(S^1)$ to be open in $\mathbb{R}$, so we can't include both $\theta = 0$ and $\theta = 2\pi$. Thus we need at least 2 charts to fully cover S^1 .

2.2.2 The Chain Rule

Consider the maps $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$ and $g : \mathbb{R}^n \rightarrow \mathbb{R}^l$. The composition is $(f \circ g) : \mathbb{R}^m \rightarrow \mathbb{R}^l$. Label points x^a on $\mathbb{R}^m$, y^b on $\mathbb{R}^n$, and z^c on $\mathbb{R}^l$.

The chain rule is

$$\frac{\partial}{\partial x^a} (g \circ f)^c = \sum_b \frac{\partial f^b}{\partial x^a} \frac{\partial g^c}{\partial y^b}$$

Its simplified form is

$$\frac{\partial}{\partial x^a} = \sum_b \frac{\partial y^b}{\partial x^a} \frac{\partial}{\partial y^b}$$

2.3 Vectors Revisited

Recall that the tangent space was the set of all vectors at a point in spacetime. Here we revisit that notion a bit more rigorously.

The main problem is, how do we construct a tangent space at a point p in the manifold M *independently of coordinates*?

Let $\mathcal{F}$ be the set of all smooth functions f on M , i.e. the set of all all C^∞ maps $f : M \rightarrow \mathbb{R}$. Each curve through p defines the “directional derivative” operator, mapping $f \rightarrow \frac{df}{d\lambda}$ at p . Here λ is the

curve parametrization.

The tangent space T_p can be identified as the space of directional derivative operators along curves through p .

Proof: First, we have to show that the space of directional derivatives is a vector space. This is straightforward: consider operators $\frac{d}{d\lambda}$ and $\frac{d}{d\eta}$, representing derivatives along $x^\mu(\lambda)$ and $x^\mu(\eta)$ through p . These are obviously linear. They also obey the Leibniz (product) rule:

$$\begin{aligned} \left(a \frac{d}{d\lambda} + b \frac{d}{d\eta} \right) (fg) &= af \frac{dg}{d\lambda} + ag \frac{df}{d\lambda} + bf \frac{dg}{d\eta} + bg \frac{df}{d\eta} \\ &= \left(a \frac{df}{d\lambda} + b \frac{df}{d\eta} \right) g + \left(a \frac{dg}{d\lambda} + b \frac{dg}{d\eta} \right) f \end{aligned}$$

So we're justified in saying that the space of directional derivatives is a vector space.

Now we have to show that this is the vector space we want: it should have the same dimension as M , and yield the natural idea of a vector pointing in a certain direction. The easiest way to do this is by finding a basis for the space. Consider the coordinate chart to be x^μ . This has a set of n directional derivatives at p , the *partial derivatives*.

Partial derivatives: directional derivatives parametrized by x^μ itself along the curve defined by $x^\nu = \text{const.}$, for all $x^\nu \neq x^\mu$.

Our claim is that the partial derivative operators $\{\partial_\mu\}$ at p form a basis for the tangent space T_p .

Proof: It suffice to show that any directional derivative can be decomposed into partial derivatives. Consider the n -manifold M , coordinate chart $\phi : M \rightarrow \mathbb{R}^n$, the curve $\gamma : \mathbb{R} \rightarrow M$, and a function $f : M \rightarrow \mathbb{R}$.

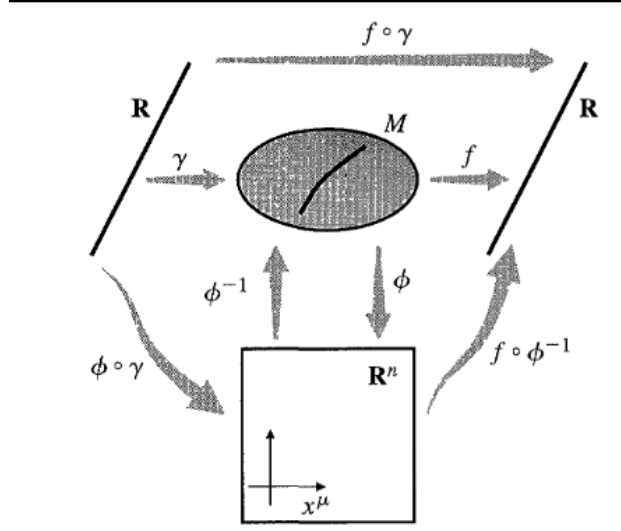


Diagram for above conditions (Carroll 65)

Let λ be the parameter for the curve γ . Then express $\frac{d}{d\lambda}$ in terms of partials:

$$\begin{aligned}
 \frac{d}{d\lambda} f &= \frac{d}{d\lambda} (f \circ \gamma) \\
 &= \frac{d}{d\lambda} [(f \circ \phi^{-1}) \circ (\phi \circ \gamma)] \\
 &= \frac{d(\phi \circ \gamma)^\mu}{d\lambda} \frac{\partial (f \circ \phi^{-1})^\mu}{\partial x^\mu} && \text{(chain rule)} \\
 &= \frac{dx^\mu}{d\lambda} \partial_\mu f
 \end{aligned}$$

We've shown that the partials are a good basis for the vector space of directional derivatives. In fact, the particular basis $\hat{e}_{(\mu)} \equiv \partial_\mu$ is the *coordinate basis* for T_p . However, note that for curved manifolds, the coordinate basis vectors usually aren't orthonormal.

The transformation law under a change of coordinates is intuitive. Assume basis vectors $\hat{e}_{(\mu)} \equiv \partial_\mu$. Now the basis vectors become

$$\partial_{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu$$

Thus for a general $V = V^\mu \partial_\mu$,

$$V^{\mu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} V^\mu$$

2.3.1 Commutators

A vector is a directional derivative operator at a curve through a point, Thus a *vector field* is a map from smooth functions to smooth functions all over the manifold.

For vector fields X and Y , define the commutator by its action on a function $f(x^\mu)$:

$$[X, Y](f) \equiv X(Y(f)) - Y(X(f))$$

This is itself a vector field: it's linear and obeys the Leibniz rule.

$$\begin{aligned} [X, Y](af + bg) &= a[X, Y](f) + b[X, Y](g) && \text{(linear)} \\ [X, Y](fg) &= f[X, Y](g) + g[X, Y](f) && \text{(Leibniz)} \end{aligned}$$

The specific coordinates are given by

$$[X, Y]^\mu = X^\lambda \partial_\lambda Y^\mu - Y^\lambda \partial_\lambda X^\mu$$

The commutator is sometimes called a *Lie bracket* and it's a special case of the Lie derivative. As partials commute, the commutator of the vector field given by the partial derivatives $\{\partial_\mu\}$ of a coordinate function always vanishes.

2.4 Tensors Revisited

Consider dual vectors (one-forms). The space of dual vectors is the cotangent space T_p^* , or the set of linear maps $\omega : T_p \rightarrow \mathbb{R}$. The classic example is the gradient df . Its action on $\frac{d}{d\lambda}$ is

$$df\left(\frac{d}{d\lambda}\right) = \frac{df}{d\lambda}$$

The gradient encodes the necessary information to find the directional derivatives along any curve through $p \in M$, fulfilling the role of a dual vector.

For the tangent space T_p , the partials along the coordinate axes were a natural basis. For the cotangent space T_p^* , the gradients of the coordinate function x^μ are a natural basis. In flat spacetime, we made a basis for T_p^* by requiring that $\hat{\theta}^{(\mu)}(\hat{e}_{(\nu)}) = \delta_\nu^\mu$. Now, we have the condition

$$dx^\mu(\partial_\nu) = \frac{\partial x^\mu}{\partial x^\nu} = \delta_\nu^\mu$$

And an arbitrary dual vector (one-form) can be expanded into its components as $\omega = \omega_\mu dx^\mu$. The transformation laws for tensors are intuitive. For a (k, l) tensor,

$$T^{\mu'_1 \dots \mu'_k}_{\nu'_1 \dots \nu'_l} = \frac{\partial x^{\mu'_1}}{\partial x^{\mu_1}} \dots \frac{\partial x^{\mu'_k}}{\partial x^{\mu_k}} \frac{\partial x^{\nu_1}}{\partial x^{\nu'_1}} \dots \frac{\partial x^{\nu_l}}{\partial x^{\nu'_l}} T^{\mu_1 \dots \mu_k}_{\nu_1 \dots \nu_l}$$

Note: the partial derivatives of a higher-rank tensor aren't tensorial, generally. Later, we'll define new tensorial operators — exterior derivatives, covariant derivatives, Lie derivatives.

2.5 The Metric & Canonical Form

The metric is obviously pretty important in relativity. Some uses are (Sachs & Wu 1977):

1. Supplies notion of “past” & “future”
2. Allows computation of path length & proper time
3. Determines “shortest distance” between 2 points
4. Replaces Newtonian gravitational field ϕ
5. Provides notion of locally inertial frames

6. Determines causality by defining speed of light
7. Replaces traditional Euclidean 3D dot product

Recall that line elements in special relativity are $ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu$ with the Minkowski metric $\eta_{\mu\nu}$. In general relativity, we use the metric tensor $g_{\mu\nu}$ to write

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

Note 1: dx isn't the "infinitesimal displacement" from calculus, but a rigorous notion of a *basis one-form* given by the gradient function. Generally, we can treat it like an infinitesimal and operate as such. Still good know that it's something else.

Note 2: ds^2 isn't a square of any quantity, nor a displacement. It's just a conventional shorthand for the metric tensor, a map of 2 vectors to one real:

$$g_{\mu\nu} V^\mu W^\nu = g(V, W) = ds^2(V, W)$$

Meanwhile, $(dx)^2$ refers to $dx \otimes dx$, a $(0, 2)$ tensor.

The **canonical form** of $g_{\mu\nu}$ is a diagonal matrix:

$$g_{\mu\nu} = \text{diag}(-1, \dots, -1, +1, \dots, +1, 0, \dots, 0)$$

The **signature** is the number of both positive and negative eigenvalues. For example, Minkowski spacetime has a "- + + +" signature.

- If all eigenvalues are > 0 , the metric is Euclidean/Riemannian.
- If there's a single -1 , the metric is Lorentzian/pseudo-Riemannian.
- If there's a mix of $+1$ and -1 , the metric is indefinite.
- If any eigenvalue is 0, the metric is degenerate.

At any $p \in M$, there exists a coordinate system $x^{\hat{\mu}}$ in which $g_{\hat{\mu}\hat{\nu}}$ has a canonical form, and its first derivatives $\partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}}$ vanish. This is useful because it allows us to treat small enough regions of spacetime as flat.

$$g_{\hat{\mu}\hat{\nu}}(\rho) = \eta_{\hat{\mu}\hat{\nu}} \quad , \quad \partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}}(\rho) = 0$$

We call this coordinate system $x^{\hat{\mu}}$ **locally inert coordinates**. As an example, consider an observer with 4-velocity $U^{\hat{\mu}}$ and a rocket flying past with 4-velocity $V^{\hat{\mu}}$. In special relativity, we can work in the observer's reference frame, in which

$$U^{\hat{\mu}} = (1, 0, 0, 0) \quad , \quad V^{\hat{\mu}} = (\gamma, \gamma\beta, 0, 0)$$

We see that $\gamma = -\eta_{\hat{\mu}\hat{\nu}} U^{\hat{\mu}} V^{\hat{\nu}} = -U_{\hat{\mu}} V^{\hat{\mu}}$ using the - + + + signature for the Minkowski metric. Then, we know that $\gamma = 1/\sqrt{1 - \beta^2}$, so $\beta = \sqrt{1 - \gamma^{-2}}$. Thus we have

$$\beta = \sqrt{1 - (U_{\hat{\mu}} V^{\hat{\mu}})^{-2}}$$

In curved spacetime, the metric isn't flat. But at the point of measurement, we can use locally inertial coordinates where $g_{\hat{\mu}\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}}$. Thus the same equation holds in general relativity, which shouldn't be a surprise since the equation is purely tensorial (doesn't depend on coordinates).

2.6 Expanding Universe

Consider the metric below (special case of the Robertson–Walker metric):

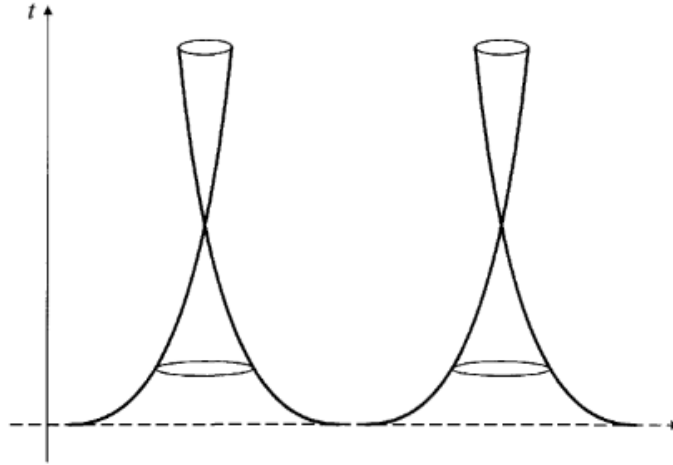
$$ds^2 = -dt^2 + a(t)^2 [dx^2 + dy^2 + dz^2]$$

This models an expanding universe with scale factor $a(t)$. The solutions for the scale factor can be of the form $a(t) = t^q$, with $0 < q < 1$ and $0 < t < \infty$.

Now consider null paths, where $ds^2 = 0$. Hold y, z as constant. Then

$$\begin{aligned} 0 &= -dt^2 + t^{2q} dx^2 \\ \frac{dx}{dt} &= \pm t^q \\ t &= (1 - q)^{1/(1-q)} (\pm x - x_0)^{1/(1-q)} \end{aligned}$$

Note again, to do this rigorously we should've treated dx as a one-form. But handwaving it with differentials works, and we're physicists not mathematicians.



Spacetime diagram of expanding universe (Carroll 78). Note the singularity at $t = 0$ and two light cones, for two events. Unlike Minkowski spacetime, light cones here need not intersect in the past.

2.7 Causality

2.7.1 Definitions Bash

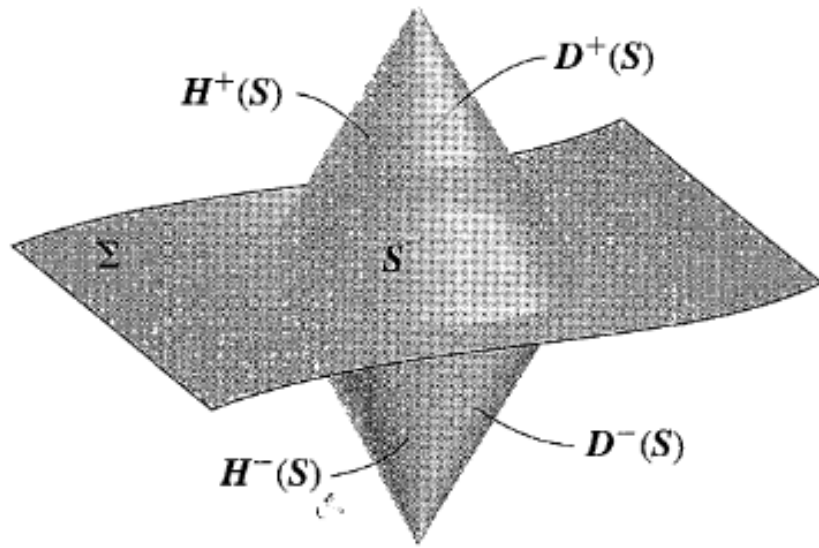
A common question in physics is “given the state at time t_1 , can we predict the state at time t_2 ?” In general relativity, causality must be handled more carefully.

Define a **causal curve** to be one that's timelike or null everywhere. Then for some subset S of the manifold M , define the **causal future** of S , $J^+(S)$, as the set of points that can be reached from S by following a future-directed causal curve. Similarly, $J^-(S)$ is the set of points that can be reached from S by following a past-directed causal curve.

Define the **chronological future** of S , $I^+(S)$ as the set of points that can be reached from S by following a future-directed *timelike curve* only.

We call a subset $S \subset M$ **achronal** if no two points in S are connected by a timelike curve. Given a closed achronal set S , define the **future domain of dependence** of S , $D^+(S)$, as the set of all points p such that every past-moving inextendible causal curve through p intersects S . The boundary of $D^+(S)$ is the **future Cauchy horizon**, $H^+(S)$. This is a null surface. We define the past domain of dependence $D^-(S)$ and the past Cauchy horizon $H^-(S)$ similarly.

That's difficult to understand, so here's a diagram:



Future domain of dependence and Cauchy horizon on S

The **domain of dependence** is the set of all points for which we can predict what happens by knowing what happens on S :

$$D(S) = D^+(S) \cup D^-(S)$$

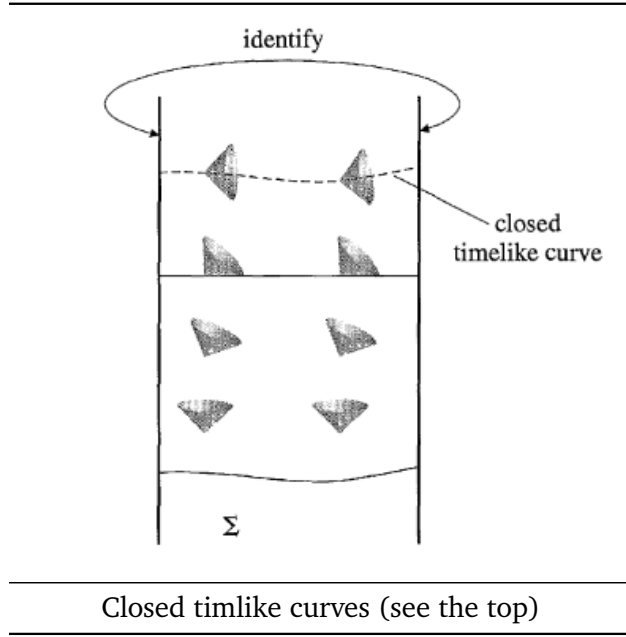
A closed achronal surface Σ is a **Cauchy surface** if $D(\Sigma)$ is the entire manifold. From information on the Cauchy surface alone, we can predict anything on the manifold. If some region of spacetime has a Cauchy surface, it's **globally hyperbolic** on that region.

A set Σ that's closed, achronal, and has no edge is a **partial Cauchy surface**. This could fail to be a Cauchy surface by its own fault or by some quirk of the spacetime region. For example, singularities will affect how the domain of dependence looks.

2.7.2 Closed Timelike Curves

Now this is an interesting result. In special relativity, an observer is restricted to their own light cone. In general relativity, that becomes a local notion. The curvature of spacetime might “tilt” light cones.

If sufficiently distorted, it's possible for an observer to move on a timelike path but intersect himself at a point in the “past.”



2.8 Tensor Densities

Recall the Levi-Civita symbol:

$$\tilde{\epsilon}_{\mu_1 \mu_2 \dots \mu_n} = \begin{cases} +1 & \text{If } \mu_1 \mu_2 \dots \mu_n \text{ is an even permutation of } 0, 1, \dots, n-1 \\ -1 & \text{If } \mu_1 \mu_2 \dots \mu_n \text{ is an odd permutation of } 0, 1, \dots, n-1 \\ 0 & \text{otherwise} \end{cases}$$

The Levi-Civita symbol isn't a tensor because it's defined not to change under coordinate transformations. But its behavior can be made tensor-like. Note that given an $n \times n$ matrix $M^\mu_{\mu'}$, the determinant obeys

$$\tilde{\epsilon}_{\mu'_1 \mu'_2 \dots \mu'_n} |M| = \tilde{\epsilon}_{\mu_1 \mu_2 \dots \mu_n} M^{\mu_1}_{\mu'_1} M^{\mu_2}_{\mu'_2} \dots M^{\mu_n}_{\mu'_n}$$

By setting $M^\mu_{\mu'} = \partial x^\mu / \partial x^{\mu'}$, we get

$$\tilde{\epsilon}_{\mu'_1 \mu'_2 \dots \mu'_n} = \left| \frac{\partial x^{\mu'}}{\partial x^\mu} \right| \tilde{\epsilon}_{\mu_1 \mu_2 \dots \mu_n} \frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}} \frac{\partial x^{\mu_2}}{\partial x^{\mu'_2}} \dots \frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}}$$

The Levi-Civita symbol transforms like a tensor, except for the determinant in front. Objects transforming like these are called *tensor densities*. The power to which the Jacobian is raised is called the *weight* of the tensor density.

To convert tensor densities to tensors, multiply by $|g|^{w/2}$, where w is the weight of the density. Thus, the *Levi-Civita tensor* can be defined as

$$\epsilon_{\mu_1 \mu_2 \dots \mu_n} = \sqrt{|g|} \tilde{\epsilon}_{\mu_1 \mu_2 \dots \mu_n}$$

We can raise indices on this and so on.

2.9 Differential Forms

These are a special kind of tensors. The **differential p -form** is a $(0, p)$ tensor, completely antisymmetric. For example, scalars are 0-forms and dual vectors are 1-forms (hence why we called them “one-forms” earlier).

The space of all p -forms is Λ^p . The space of all p -form fields over a manifold M is $\Lambda^p(M)$.

The basic idea is that differential forms can be differentiated/integrated without the help of additional geometric structures. That’ll be clarified later, hopefully.

2.9.1 Wedge Product

Given a p -form A and q -form B , we can form the $(p + q)$ -form $A \wedge B$, the **wedge product** of A and B .

$$(A \wedge B)_{\mu_1 \dots \mu_{p+q}} = \frac{(p+q)!}{p!q!} A_{[\mu_1 \dots \mu_p} B_{\mu_{p+1} \dots \mu_{p+q}]}$$

For example, for 1-forms,

$$(A \wedge B)_{\mu\nu} = 2A_{[\mu} B_{\nu]} = A_\mu B_\nu - A_\nu B_\mu$$

2.9.2 Exterior Derivative

This allows us to differentiate p -forms to get $(p + 1)$ -form fields. Define the exterior derivative as

$$(dA)_{\mu_1 \dots \mu_{p+1}} = (p+1) \partial_{[\mu_1} A_{\mu_2 \dots \mu_{p+1}]}$$

For example, the gradient is the exterior derivative of a 1-form:

$$(d\phi)_\mu = \partial_\mu \phi$$

The exterior derivative satisfies a modified Leibniz rule. For p -form ω and q -form η , we have

$$d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p \omega \wedge (d\eta)$$

Above, $d(\omega \wedge \eta)$ is a tensor even in curved spacetime. This is in contrast to partial derivatives, which generally aren’t. But exterior derivatives don’t cover the full range of partial derivatives. They’re only defined for differential p -forms.

We call a p -form **closed** if $dA = 0$ and **exact** if $A = dB$ for some $(p - 1)$ -form B .

Exterior derivatives have the interesting property that for any differential form A , $d(dA) = 0$. This is sometimes written as $d^2 = 0$.

2.9.3 Hodge Duality

The Hodge star operator is an n -dimensional manifold defined on differential forms. It maps p -forms to $(n - p)$ -forms:

$$(*A)_{\mu_1 \dots \mu_{n-p}} = \frac{1}{p!} \epsilon^{\nu_1 \dots \nu_p}{}_{\mu_1 \dots \mu_{n-p}} A_{\nu_2 \dots \nu_p}$$

$$*(*A) = (-1)^{s+p(n-p)} A, \quad s = \# \text{ of negative eigenvalues in metric}$$

In 3D Euclidean space, the Hodge dual of the wedge product of 2 1-forms gives another 1-form. This is the conventional cross product, and only works in 3D.

$$*(U \wedge V)_i = \epsilon_i^{jk} U_j V_k$$

Recall the electrodynamic tensor $F : F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix}$

And the tensor form of Maxwell's equations:

$$\begin{cases} \partial_\mu F^{\nu\mu} = J^\nu \\ \partial_{[\mu} F_{\nu\lambda]} = 0 \end{cases}$$

From the definition of the exterior derivative, we see that $dF = 0$. Thus there's a 1-form A_μ such that $F = dA$. Here, A is the electromagnetic vector potential.

The other of Maxwell's equations is $d(*F) = *J$. The one-form J is the current 4-vector with lowered indices. If we set $J_\mu = 0$, the equations are invariant under “duality transformations”:

$$\begin{aligned} F &\rightarrow *F \\ *F &\rightarrow -F \end{aligned}$$

Thus the vacuum Maxwell's equations are duality-invariant. But in the presence of charges, the invariance is spoiled.

2.10 Integration

Tensor densities and differential forms allow us to integrate on manifolds. Recall that in ordinary calculus (in $\mathbb{R}$), the “volume element” $d^n x$ picks up a Jacobian under coordinate transformations:

$$d^n x' = \left| \frac{\partial x^{\mu'}}{\partial x^\mu} \right| d^n x$$

Mathematically, this is because on an n -dimensional manifold, the integrand is understood as an n -form. That doesn't make a lot of sense yet. But for an n -dimensional region $\Sigma \subset M$ on the manifold M , the integral can be regarded as a map of n -form fields to real numbers.

$$\int_\Sigma : \omega \rightarrow \mathbb{R}$$

For example, in 1 dimension, the one-form is $\omega = \omega(x)dx$. The line integral of this does return a number in $\mathbb{R}$. Here, dx is a differential form, though it helps to handwave it as “infinitesimal distance” in calculations.

Why is the integrand an n -form? To be an n -form, it needs to be an antisymmetric $(0, n)$ tensor. First, write the integral as

$$\int f(x) d\mu \quad \begin{cases} f(x) : \text{scalar function on manifold} \\ d\mu : \text{volume element (“measure”)} \end{cases}$$

The volume element assigns a real number to every infinitesimal region, given n vectors: $d\mu(U_1, U_2, \dots, U_n) \in \mathbb{R}$. Addition and scalar multiplication work out as expected for vectors and thus tensors. Furthermore, $d\mu$ is antisymmetric. If two vectors are switched, the sign flips.

So we can consider the integrand to be an n -form.

To be more precise, the volume element $d\mu \equiv d^n x$ is an antisymmetric **tensor density** constructed from wedge products.

$$d^n x = dx^0 \wedge \dots \wedge dx^{n-1}$$

The right-hand side is a coordinate-dependent quantity, because it depends on the coordinate functions themselves. This is a pretty subtle result, but it's the reason $d^n x$ is considered a tensor density, not a tensor.

To see why $d^n x$ transforms like a tensor density, note the following:

$$\begin{aligned} 1) \quad dx^0 \wedge \dots \wedge dx^{n-1} &= \frac{1}{n!} \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} \\ 2) \quad \tilde{\epsilon}_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} &= \tilde{\epsilon}_{\mu'_1 \dots \mu'_n} \frac{\partial x^{\mu_1}}{\partial x^{\mu'_1}} \dots \frac{\partial x^{\mu_n}}{\partial x^{\mu'_n}} dx^{\mu'_1} \wedge \dots \wedge dx^{\mu'_n} \\ &= \left| \frac{\partial x^{\mu}}{\partial x^{\mu'}} \right| \tilde{\epsilon}_{\mu'_1 \dots \mu'_n} dx^{\mu'_1} \wedge \dots \wedge dx^{\mu'_n} \end{aligned}$$

Combining the two and cleaning up any coefficients, we see that

$$d^n x' = \left| \frac{\partial x^{\mu'}}{\partial x^{\mu}} \right| d^n x$$

Thus the volume element $d^n x$ transforms like a tensor density.

Recall that we can form tensors from tensor densities by multiplying with $|g|^{w/2}$, where w is the weight of the tensor. Here, that's just 1. So we can create an invariant volume element by multiplying with $\sqrt{|g|}$:

$$\begin{aligned} \sqrt{|g'|} dx^{0'} \wedge \dots \wedge dx^{(n-1)'} &= \sqrt{|g|} dx^0 \wedge \dots \wedge dx^{(n-1)} \\ &= \frac{1}{n!} \epsilon_{\mu_1 \dots \mu_n} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_n} \end{aligned}$$

Thus the integral of a scalar function ϕ over an n -dimensional manifold is

$$\int \phi(x) \sqrt{|g|} d^n x$$

3 Curvature (pp. 93 – 150)

3.1 Overview

In this chapter we want to explore **curvature**. Intuitively, this should depend on the metric of the manifold. There are 4 main equations to be studied:

1. Curvature manifests through a **connection** (which is a way to relate vectors in tangent spaces of nearby points). From the metric, a unique connection can be constructed, the **Christoffel symbol**.

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2}g^{\lambda\sigma}(\partial_{\mu}g_{\nu\sigma} + \partial_{\nu}g_{\sigma\mu} - \partial_{\sigma}g_{\mu\nu})$$

2. Connections are used to take **covariant derivatives**, a generalization of the partial derivative. For a vector field V^{ν} , the covariant derivative is

$$\nabla_{\mu}V^{\nu} = \partial_{\mu}V^{\nu} + \Gamma_{\mu\sigma}^{\nu}V^{\sigma}$$

3. Connections are also used to define geodesics. A parametrized curve $x^{\mu}(\lambda)$ is a geodesic if

$$\frac{d^2x^{\mu}}{d\lambda^2} + \Gamma_{\rho\sigma}^{\mu}\frac{dx^{\rho}}{d\lambda}\frac{dx^{\sigma}}{d\lambda} = 0$$

4. Connections are used to define the Riemann tensor, a $(1, 3)$ tensor that encodes the curvature. It vanishes on a flat metric.

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma_{\nu\sigma}^{\rho} - \partial_{\nu}\Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho}\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho}\Gamma_{\mu\sigma}^{\lambda}$$

3.2 Covariant Derivatives

Recall the problem with partial derivatives: in general, they're not tensorial. We seek a generalization of the partial derivative on manifolds, satisfying:

1. Like partial derivatives on flat space with inertial coordinates.
2. Tensorial, $\nabla_{\mu} : (k, l) \rightarrow (k, l + 1)$ tensor for all arbitrary manifolds.
3. Linear, $\nabla(T + S) = \nabla T + \nabla S$
4. Leibniz, $\nabla(T \otimes S) = (\nabla T) \otimes S + T \otimes (\nabla S)$
5. Commutes with contractions, $\nabla_{\mu}(T^{\lambda}{}_{\lambda\rho}) = (\nabla T)_{\mu}{}^{\lambda}{}_{\lambda\rho}$
6. Reduces to partial derivative on scalars, $\nabla_{\mu}\phi = \partial_{\mu}\phi$

We can try to build on the partial derivative (which is linear and Leibniz) by adding a correction term. For an n -dimensional vector V^{ν} , we use the partial derivatives ∂_{μ} as well as the “correction” given by the matrices Γ_{μ} . The form we seek is

$$\nabla_{\mu}V^{\nu} = \partial_{\mu}V^{\nu} + \Gamma_{\mu\lambda}^{\nu}V^{\lambda}$$

Note that the connection coefficients $\Gamma_{\mu\lambda}^{\nu}$ have been collapsed. The coefficient is nontensorial.

By applying requirements (5) and (6) on the scalar $\omega_{\lambda}V^{\lambda}$, we get the covariant derivative forms for both a vector and a 1-form:

$$\begin{aligned}\nabla_{\mu}V^{\nu} &= \partial_{\mu}V^{\nu} + \Gamma_{\mu\lambda}^{\nu}V^{\lambda} \\ \nabla_{\mu}\omega_{\nu} &= \partial_{\mu}\omega_{\nu} - \Gamma_{\mu\nu}^{\lambda}\omega_{\lambda}\end{aligned}$$

Note the subtle differences in signs and indices. For tensors of arbitrary rank, use $+\Gamma$ on all upper indices and $-\Gamma$ on all lower indices. E.g.

$$\begin{aligned}\nabla_\sigma T^{\mu_1\mu_2\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l} &= \partial_\sigma T^{\mu_1\mu_2\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l} \\ &+ \Gamma_{\sigma\lambda}^{\mu_1} T^{\lambda\mu_2\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l} + \Gamma_{\sigma\lambda}^{\mu_2} T^{\mu_1\lambda\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l} + \cdots \\ &- \Gamma_{\sigma\nu_1}^\lambda T^{\mu_1\mu_2\cdots\mu_k}_{\lambda\nu_2\cdots\nu_l} - \Gamma_{\sigma\nu_2}^\lambda T^{\mu_1\mu_2\cdots\mu_k}_{\nu_1\lambda\cdots\nu_l} - \cdots\end{aligned}$$

3.2.1 Specific Form

Define the torsion tensor

$$T^\lambda_{\mu\nu} = \Gamma^\lambda_{\mu\nu} - \Gamma^\lambda_{\nu\mu} = 2\Gamma^\lambda_{[\mu\nu]}$$

We define a connection on a manifold with metric $g_{\mu\nu}$ that has the following properties:

1. Torsion-free, $\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\nu\mu}$
2. Metric compatibility, $\nabla_\rho g_{\mu\nu} = 0$

It turns out, there's only one connection that satisfies these conditions. If the metric is $g_{\mu\nu}$, the form of that connection is

$$\Gamma^\sigma_{\mu\nu} = \frac{1}{2}g^{\sigma\rho}(\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})$$

This is known as the **Christoffel connection**. The coefficients are called **Christoffel symbols**.

3.2.2 Transformations Laws

The variables should transform tensorially, thus

$$\nabla_{\mu'} V^{\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \nabla_\mu V^\nu$$

Expanding the right and left sides, respectfully, gives

$$\begin{aligned}\text{(Right)} \quad \nabla_{\mu'} V^{\nu'} &= \partial_{\mu'} V^{\nu'} + \Gamma_{\mu'\lambda'}^{\nu'} V^{\lambda'} \\ &= \frac{\partial x^\mu}{\partial x^{\mu'}} \left(\frac{\partial x^{\nu'}}{\partial x^\nu} \partial_\mu V^\nu + V^\nu \frac{\partial}{\partial x^\mu} \frac{\partial x^{\nu'}}{\partial x^\nu} \right) + \Gamma_{\mu'\lambda'}^{\nu'} \frac{\partial x^{\lambda'}}{\partial x^\lambda} V^\lambda \\ \text{(Left)} \quad \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \nabla_\mu V^\nu &= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \partial_\mu V^\nu + \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma_{\mu\lambda}^\nu V^\lambda\end{aligned}$$

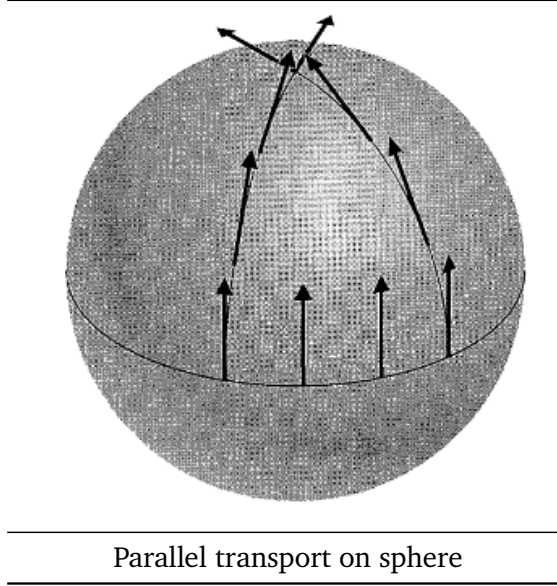
Some manipulation gives

$$\Gamma_{\mu'\lambda'}^{\nu'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\lambda'}}{\partial x^{\lambda'}} \frac{\partial x^{\nu'}}{\partial x^\nu} \Gamma_{\mu\lambda}^\nu - \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^{\lambda'}}{\partial x^{\lambda'}} \frac{\partial^2 x^{\nu'}}{\partial x^\mu \partial x^{\lambda'}}$$

Note that this is non-tensorial. It's okay since the connection coefficient $\Gamma_{\mu\lambda}^\nu$ isn't a tensor anyway.

3.3 Parallel Transport

Intuitively, we want to move a vector along a path and keep it constant. In flat spacetime, we shift vectors around freely. In curved manifolds, the result depends on the path taken.



I.e. two vectors can only be compared if they're **members of the same tangent space**. Two particles passing each other, for example, have a well-defined relative velocity. Two particles far away don't. This is also why naively applying the Doppler formula to gravitational redshift doesn't work: we can't have a well-defined velocity between faraway galaxies. The metric of spacetime between us and the observed galaxy has changed, because the universe has expanded.

Given a curve $x^\mu(\lambda)$ and a tensor $T^{\mu_1\mu_2\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l}$ on it, we seek to keep the tensor constant. In flat spacetime, this just means that the components are constant.

$$\frac{d}{d\lambda} T^{\mu_1\mu_2\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l} = \frac{dx^\mu}{d\lambda} \frac{\partial}{\partial x^\mu} T^{\mu_1\mu_2\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l} = 0$$

The partial derivative makes this expression non-tensorial. We replace this with the covariant derivative. Define the directional covariant derivative

$$\frac{D}{d\lambda} \equiv \frac{dx^\mu}{d\lambda} \nabla_\mu$$

Then the **parallel transport** of T along $x^\mu(\lambda)$ must follow the path on which the covariant derivative of T vanishes:

$$\left(\frac{D}{d\lambda} T \right)^{\mu_1\mu_2\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l} = \frac{dx^\sigma}{d\lambda} \nabla_\sigma T^{\mu_1\mu_2\cdots\mu_k}_{\nu_1\nu_2\cdots\nu_l} = 0$$

For vectors, we have

$$\frac{d}{d\lambda} V^\mu + \Gamma^\mu_{\sigma\rho} \frac{dx^\sigma}{d\lambda} V^\rho = 0$$

The parallel transport obviously depends on the connection, which in turn depends on the metric. If the connection is metric-compatible (no torsion), the metric is always parallel transported with respect to that connection. Additionally, the inner product of two parallel-transported vectors is preserved.

$$\begin{aligned} 1) \quad & \frac{D}{d\lambda} g_{\mu\nu} = \frac{dx^\sigma}{d\lambda} \nabla_\sigma g_{\mu\nu} = 0 \\ 2) \quad & \frac{D}{d\lambda} (g_{\mu\nu} V^\mu W^\nu) = 0 \end{aligned}$$

3.4 Geodesics

In flat space, a “straight line” is the path of shortest distance between two points. In curved spacetime, a “straight line” is a path that **parallel-transport its own tangent vector**. The tangent vector to the path $x^\mu(\lambda)$ is $dx^\mu/d\lambda$. We require the directional covariant derivative of this to be zero. Thus the **geodesic equation** is

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0$$

Geodesics are the paths followed by **unaccelerated test particles**. A test particle is an ideal one that doesn’t influence the geometry through which it moves. This usually isn’t true, but it’s a good approximation.

The geodesic equation is like a generalization of $\mathbf{F} = m\mathbf{a}$.

The path we parametrize for the geodesic equation was $x^\mu(\lambda)$. In the geodesic equation above, τ (proper time) appears, not λ . The specific condition is that

$$\tau \rightarrow \lambda = a\tau + b$$

As long as the relationship is linear like this, any parametrization is fine.

For **timelike paths**, use the 4-velocity $U^\mu = dx^\mu/d\tau$. The geodesic equation becomes

$$U^\lambda \nabla_\lambda U^\mu = 0$$

Additionally, for 4-momentum $p^\mu = mU^\mu$, we have

$$p^\lambda \nabla_\lambda p^\mu = 0$$

I.e. freely falling particles keep moving in the direction pointed to by their 4-velocity/4-momentum. For null paths, proper time is 0, not a good parameter. More work is needed there.

Timelike geodesics are maxima of proper time. For any timelike path, approximate it with jagged null paths. The timelike path is close to the total null path (of 0 proper time), but not quite. It turns out that the timelike path **maximizes** proper time. Thus in the twin paradox, the twin that stays on Earth is on a geodesic, maximizing his proper time.

3.4.1 The Exponential Map

Geodesics are useful for mapping the tangent space T_p to the region of the manifold M that contains p . This map is called the **exponential map**. It defines a coordinate set for the region that is automatically locally inertial.

Recall: locally inertial coordinates were the ones that looked like flat spacetime close up. I.e. coordinates $x^{\hat{\mu}}$ such that

$$\begin{aligned} g_{\hat{\mu}\hat{\nu}}(p) &= \eta_{\hat{\mu}\hat{\nu}} \\ \partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}}(p) &= 0 \end{aligned}$$

At a point $p \in M$, a tangent vector $k \in T_p$ defines a unique geodesic through it. Parameterizing as λ with $\lambda(p) = 0$,

$$\left. \frac{dx^\mu}{d\lambda} \right|_{\lambda=0} = k^\mu$$

Now we can adjust the parameter such that $\lambda = 1$ somewhere on the manifold. The the exponential map $\exp_p : T_p \rightarrow M$ is defined as

$$\exp_p(k) = x^\nu|_{\lambda=1}$$

I.e. the exponential map takes a tangent vector to the point of its corresponding geodesic where $\lambda = 1$. Sometimes this might not be possible because the manifold has a singularity.

3.4.2 Locally Inertial Coordinates

From the exponential map, we want to construct locally inertial coordinates. We need basis vectors such that the local space looks flat.

$$g_{\hat{\mu}\hat{\nu}} = g(\hat{e}_{(\hat{\mu})}, \hat{e}_{(\hat{\nu})}) = \eta_{\hat{\mu}\hat{\nu}}$$

Additionally, the coordinate system $x^{\hat{\mu}}$ should satisfy

$$\begin{aligned} \hat{e}_{(\hat{\mu})} &= \partial_{\hat{\mu}} \\ \partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}} &= 0 \end{aligned}$$

Remember, at p , the tangent vector was $k = k^{\hat{\mu}} \hat{e}_{(\hat{\mu})}$. From here, define the coordinates

$$x^{\hat{\mu}}(q) = k^{\hat{\mu}}$$

for q sufficiently close to p , which we can achieve with an appropriate parameterization of λ . I.e. define the components of $x^{\hat{\mu}}(q)$ to be the components of the tangent vector k that gets mapped to q by $\exp_p$. Coordinates constructed like this are called **Riemann normal coordinates** at p . These can be seen to satisfy

$$\begin{aligned} g_{\hat{\mu}\hat{\nu}}(p) &= \eta_{\hat{\mu}\hat{\nu}} \\ \partial_{\hat{\sigma}} g_{\hat{\mu}\hat{\nu}}(p) &= 0 \end{aligned}$$

3.5 The Expanding Universe Revisited

3.5.1 Calculating the Christoffel connection

Recall the geodesic equation

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0$$

Finding the Christoffel symbols from this is complicated. An easier way is to vary the proper time functional. An extremal of this is a path that maximizes proper time. That's a property of the geodesic.

$$\tau = \int \frac{d\tau}{d\lambda} d\lambda = \int \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda$$

1. We let $f = g_{\mu\nu}(dx^\mu/d\lambda)(dx^\nu/d\lambda)$, then vary the integral to find the extremal of proper time.

$$\begin{aligned} \delta\tau &= \int \delta\sqrt{-f} d\lambda \\ &= \int -\frac{1}{2}(-f)^{-1/2} \delta f d\lambda \end{aligned}$$

2. Assume that the particle is **timelike**, and set the parameter to $\lambda = \tau$ (proper time). Then $f = g_{\mu\nu}U^\mu U^\nu = -1$.

$$\delta\tau = -\frac{1}{2} \int \delta f d\tau$$

3. Thus, the paths for which $\delta\tau = 0$ (extremal of proper time) is the same timelike path for which $\delta I = 0$, where

$$I = \frac{1}{2} \int f d\tau = \frac{1}{2} \int g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} d\tau$$

4. Vary I and keep the first order terms in δx^μ

$$\begin{aligned} x^\mu &\rightarrow x^\mu + \delta x^\mu \\ g_{\mu\nu} &\rightarrow g_{\mu\nu} + (\partial_\sigma g_{\mu\nu})\delta x^\sigma \end{aligned}$$

$$\begin{aligned} \delta I &= \frac{1}{2} \int \delta \left(g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right) d\tau \\ &= \frac{1}{2} \int \left[(\partial_\sigma g_{\mu\nu})\delta x^\sigma \cdot \left(\frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right) + g_{\mu\nu} \cdot \delta \left(\frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right) \right] d\tau \\ &= \frac{1}{2} \int \left[\partial_\sigma g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \delta x^\sigma + g_{\mu\nu} \frac{d(\delta x^\mu)}{d\tau} \frac{dx^\nu}{d\tau} + g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{d(\delta x^\nu)}{d\tau} \right] d\tau \end{aligned}$$

5. Integrate the last two by parts and note that the variation δx^μ vanishes at the boundaries. After rearranging and renaming indices,

$$\delta I = - \int \left[g_{\mu\sigma} \frac{d^2 x^\mu}{d\tau^2} + \frac{1}{2} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right] \delta x^\sigma d\tau$$

6. We want this to vanish for all variations δx^σ . Thus

$$\begin{aligned} g_{\mu\sigma} \frac{d^2 x^\mu}{d\tau^2} + \frac{1}{2} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} &= 0 \\ \frac{d^2 x^\rho}{d\tau^2} + \frac{1}{2} g^{\rho\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}) \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} &= 0 \end{aligned}$$

This is the geodesic equation with the specific form of the Christoffel connection. By varying the functional I , we can get the Christoffel connection for any metric, and this is usually easier than using the definition.

*There's a lot of variations (δ) going on, and it almost seems to act like a normal derivative. here's a way to think about this: First, variations are taken over a functional, derivatives over functions. The variation is a function that takes in a family of functions on the time interval that transition into each other, i.e. the functions $F : [t_0, t_1] \times [-1, 1]$ that are restricted to $[t_0, t_1] \times 0$ at which the variation is being evaluated. Define L to be

$$L : (t, u) \rightarrow F(t, u)$$

then the partial derivative $\partial L / \partial u$ is the variation. Assuming the functions are smooth, it's safe to swap variations with normal derivatives ("equality of mixed partials"). It is also safe to swap variations and integrals.

The explanation is courtesy of my friend Radu, who is much smarter than me. If anything is off, it means I didn't convey his ideas properly.

3.5.2 The Expanding Metric

Use the universe-expanding metric

$$\begin{aligned} ds^2 &= -dt^2 + a^2(t) [dx^2 + dy^2 + dz^2] \\ &= -dt^2 + a^2(t) \delta_{ij} dx^i dx^j \end{aligned}$$

We want to find the Christoffel symbols and eventually the geodesic equation. Using the procedure described, vary the integral I .

$$I = \frac{1}{2} \int \left[- \left(\frac{dt}{d\tau} \right)^2 + a^2(t) \delta_{ij} \frac{dx^i}{d\tau} \frac{dx^j}{d\tau} \right] d\tau$$

We need $x^\mu \rightarrow x^\mu + \delta x^\mu$, and $\delta I = 0$. The geodesic equation found by varying each x^μ will reveal the coefficient $\Gamma_{\rho\sigma}^\mu$.

For the time coordinate ($\mu = 0$), note that

$$\begin{aligned} t &\rightarrow t + \delta t \\ a(t + \delta t) &= a(t) + \dot{a} \delta t \end{aligned}$$

Then vary the integral, integrate by parts, etc (δt vanishes at boundary).

$$\begin{aligned}\delta I &= \frac{1}{2} \int \left[-2 \frac{dt}{d\tau} \frac{d(\delta t)}{d\tau} + 2a\dot{a}\delta_{ij} \frac{dx^i}{d\tau} \frac{dx^j}{d\tau} \delta t \right] d\tau \\ &= \int \left[\frac{d^2 t}{d\tau^2} + a\dot{a}\delta_{ij} \frac{dx^i}{d\tau} \frac{dx^j}{d\tau} \right] \delta t d\tau\end{aligned}$$

This implies that

$$\begin{aligned}\frac{d^2 x^0}{d\tau^2} + a\dot{a}\delta_{ij} \frac{dx^i}{d\tau} \frac{dx^j}{d\tau} &= 0 \\ \Gamma_{00}^0 &= 0, \quad \Gamma_{i0}^0 = \Gamma_{0i}^0 = 0, \quad \Gamma_{ij}^0 = a\dot{a}\delta_{ij}\end{aligned}\tag{1}$$

For spatial coordinates, repeat with $x^i \rightarrow x^i + \delta x^i$. The variation (after integrating by parts) is

$$\begin{aligned}\delta I &= \frac{1}{2} \int a^2 \left(2\delta_{ij} \frac{dx^i}{d\tau} \frac{d(\delta x^j)}{d\tau} \right) d\tau \\ &= - \int \left(a^2 \frac{d^2 x^i}{d\tau^2} + 2a \frac{da}{d\tau} \frac{dx^i}{d\tau} \right) \delta_{ij} \delta x^j d\tau\end{aligned}$$

Set $da/d\tau = \dot{a}(dt/d\tau)$. This implies that

$$\begin{aligned}\frac{d^2 x^i}{d\tau^2} + 2 \frac{\dot{a}}{a} \frac{dt}{d\tau} \frac{dx^i}{d\tau} &= 0 \\ \Gamma_{00}^i &= 0, \quad \Gamma_{j0}^i = \Gamma_{0j}^i = \frac{\dot{a}}{a} \delta_j^i, \quad \Gamma_{jk}^i = 0\end{aligned}\tag{2}$$

So yay, we've found all the connection coefficients. Pretty remarkable how we got this just from varying the integral.

3.5.3 Cosmological Redshift

We've found the Christoffel connection coefficients. Now consider the case of null geodesics. WLOG, model this as $x^\mu(\lambda) = \{t(\lambda), x(\lambda), 0, 0\}$. Note the parameter is λ , not τ (proper time). From the metric, we have

$$\begin{aligned}0 &= -dt^2 + a^2(t)dx^2 \\ \frac{dx}{d\lambda} &= \frac{1}{a} \frac{dt}{d\lambda}\end{aligned}$$

We plug into our geodesic equation for x^μ (timelike component) found above. But use λ instead of τ .

$$\begin{aligned}\frac{d^2 x^0}{d\tau^2} + a\dot{a}\delta_{ij} \frac{dx^i}{d\tau} \frac{dx^j}{d\tau} &= 0 \\ \frac{d^2 t}{d\lambda^2} + \frac{\dot{a}}{a} \left(\frac{dt}{d\lambda} \right)^2 &= 0\end{aligned}$$

Solve the differential equation to get $dt/d\lambda = \omega_0/a$. Consider the energy of the photon measured by a co-moving observer, i.e. one at fixed x^i . This observer has 4-velocity $U^\mu = (1, 0, 0, 0)$, where $c = 1$ for natural units.

$$U^\mu = (1, 0, 0, 0)$$

$$p^m{}_u = \frac{dx^\mu}{d\lambda}$$

$$E = -p_\mu U^\mu = -g_{00} \frac{dx^0}{d\lambda} U^0 = \frac{\omega_0}{a}$$

Usually, $E = \hbar\omega$, but we use natural units where $\hbar = 1$. So a is just the scaling factor.

This is called a **cosmological redshift**. A photon emitted at E_1 with scale factor a_1 and observed at E_2 with scale factor a_2 will have

$$\frac{E_2}{E_1} = \frac{a_1}{a_2}$$

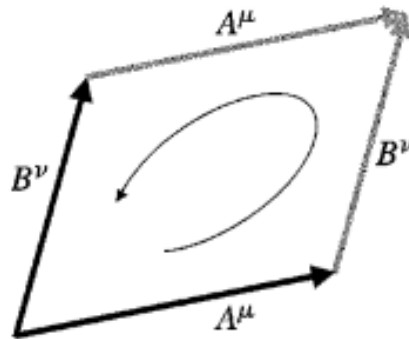
I.e. in an expanding universe, the wavelength grows with time. A greater redshift indicates a greater distance between two points.

Note that this isn't like a conventional Doppler shift. A galaxy that starts moving while the photon is moving, then stops right before the photon reaches it, will still have a cosmological redshift. But not a Doppler shift.

3.6 The Riemann Curvature Tensor

The Riemann curvature tensor quantifies a manifold's curvature (as its name suggests). The idea is that the Christoffel connection has properties of "flatness":

1. Parallel transport around closed loops leaves a vector unchanged.
2. Covariant derivatives of tensors commute.
3. Initially parallel geodesics remain parallel.



Infinitesimal loop defined by A^μ and B^ν (Carroll 121)

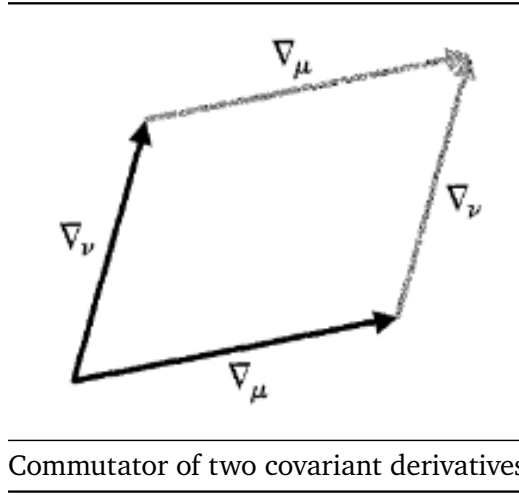
Consider an infinitesimal loop constructed above. For a vector V^σ , we seek to transport it along A^μ , then B^ν , etc. Since the overall transformation acts on a vector, it must have 1 lower and 1 upper

index. Additionally, the transformation is influenced by A^μ and B^ν , so it must have 2 lower indices to contract with those. Then the infinitesimal loop δV^ρ is given by

$$\delta V^\rho = R^\rho_{\sigma\mu\nu} V^\sigma A^\mu B^\nu$$

$R^\rho_{\sigma\mu\nu}$ is called the **Riemann metric tensor**. Note that since changing the order of A^μ, B^ν only changes the sign, the Riemann metric tensor is antisymmetric in the last two lower indices.

$$R^\rho_{\sigma\mu\nu} = -R^\rho_{\sigma\nu\mu}$$



Now consider the infinitesimal loop generated by two covariant derivative operators. By symmetry, transporting V^ρ along this is like the commutator.

$$\begin{aligned} [\nabla_\mu, \nabla_\nu] V^\rho &= \nabla_\mu \nabla_\nu V^\rho - \nabla_\nu \nabla_\mu V^\rho \\ &= \partial_\mu (\nabla_\nu V^\rho) - \Gamma^\lambda_{\mu\nu} \nabla_\lambda V^\rho + \Gamma^\rho_{\mu\sigma} \nabla_\nu V^\sigma - (\mu \leftrightarrow \nu) \\ &= \partial_\mu \partial_\nu V^\rho + (\partial_\mu \Gamma^\rho_{\nu\sigma}) V^\sigma + \Gamma^\rho_{\nu\sigma} \partial_\mu V^\sigma - \Gamma^\lambda_{\mu\nu} \partial_\lambda V^\rho - \Gamma^\lambda_{\mu\nu} \Gamma^\rho_{\lambda\sigma} V^\sigma \\ &\quad + \Gamma^\rho_{\mu\sigma} \partial_\nu V^\sigma + \Gamma^\rho_{\mu\sigma} \Gamma^\sigma_{\nu\lambda} V^\lambda - (\mu \leftrightarrow \nu) \\ &= (\partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}) V^\sigma - 2\Gamma^\lambda_{[\mu\nu]} \nabla_\lambda V^\rho \end{aligned}$$

Let the Riemann tensor be

$$R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}$$

Note that if the connection is torsion-free,

$$[\nabla_\mu, \nabla_\nu] V^\rho = R^\rho_{\sigma\mu\nu} V^\sigma$$

The Riemann metric tensor helps us recognize flat or curved manifolds:

1. If there exists a coordinate system in which the components of the metric are constant, $R^\rho_{\sigma\mu\nu}$ vanishes.
2. If $R^\rho_{\sigma\mu\nu}$ vanishes, we can construct a coordinate system in which the metric components are constant.

3.7 Properties of the Riemann Tensor

Examine the Riemann tensor with lower indices

$$R_{\rho\sigma\mu\nu} = g_{\rho\lambda} R^{\lambda}_{\sigma\mu\nu}$$

Establish locally inertial coordinates $X^{\hat{\mu}}$ at point p . The Christoffel symbols vanish, though their derivatives remain. The Riemann tensor at this point is

$$\begin{aligned} R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}}(p) &= g_{\hat{\rho}\hat{\lambda}} \left(\partial_{\hat{\mu}} \Gamma^{\hat{\lambda}}_{\hat{\nu}\hat{\sigma}} - \partial_{\hat{\nu}} \Gamma^{\hat{\lambda}}_{\hat{\mu}\hat{\sigma}} \right) \\ &= \frac{1}{2} g_{\hat{\rho}\hat{\lambda}} g^{\hat{\lambda}\hat{\tau}} (\partial_{\hat{\mu}} \partial_{\hat{\nu}} g_{\hat{\sigma}\hat{\tau}} + \partial_{\hat{\mu}} \partial_{\hat{\sigma}} g_{\hat{\tau}\hat{\nu}} - \partial_{\hat{\mu}} \partial_{\hat{\tau}} g_{\hat{\nu}\hat{\sigma}} - \\ &\quad \partial_{\hat{\nu}} \partial_{\hat{\mu}} g_{\hat{\sigma}\hat{\tau}} - \partial_{\hat{\nu}} \partial_{\hat{\sigma}} g_{\hat{\tau}\hat{\mu}} + \partial_{\hat{\nu}} \partial_{\hat{\tau}} g_{\hat{\mu}\hat{\sigma}}) \\ &= \frac{1}{2} (\partial_{\hat{\mu}} \partial_{\hat{\sigma}} g_{\hat{\rho}\hat{\nu}} - \partial_{\hat{\mu}} \partial_{\hat{\rho}} g_{\hat{\nu}\hat{\sigma}} - \partial_{\hat{\nu}} \partial_{\hat{\sigma}} g_{\hat{\rho}\hat{\mu}} + \partial_{\hat{\nu}} \partial_{\hat{\rho}} g_{\hat{\mu}\hat{\sigma}}) \end{aligned}$$

Properties that can be deduced:

1. Antisymmetric in first two indices

$$R_{\rho\sigma\mu\nu} = -R_{\sigma\rho\mu\nu}$$

2. Antisymmetric in last two indices

$$R_{\rho\sigma\mu\nu} = -R_{\rho\sigma\nu\mu}$$

3. Invariant under interchange of first pair of indices with second

$$R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}$$

4. Cyclic permutation of last three indices vanish

$$R_{\rho[\sigma\mu\nu]} = R_{\rho\sigma\mu\nu} + R_{\rho\mu\nu\sigma} + R_{\rho\nu\sigma\mu} = 0$$

Since these are tensorial equations, they're valid in any coordinate system beyond the flat one we were considering.

An antisymmetric $n \times n$ matrix has $n(n-1)/2$ independent terms, while a symmetric $n \times n$ matrix has $n(n+1)/2$ independent terms. If we think of the Riemann tensor as the “symmetric” matrix $R_{[\rho\sigma][\mu\nu]}$, then we have

$$\frac{1}{2} \left[\frac{1}{2} n(n-1) \right] \left[\frac{1}{2} n(n-1) + 1 \right] = \frac{1}{8} (n^4 - 2n^3 + 3n^2 - 2n)$$

independent terms. Also notice that $R_{[\rho\sigma\mu\nu]} = 0$. This is another restriction. A totally antisymmetric 4-index tensor has $n(n-1)(n-2)(n-3)/4!$ independent terms, so overall the Riemann tensor has

$$\frac{1}{8} (n^4 - 2n^3 + 3n^2 - 2n) - \frac{n(n-1)(n-2)(n-3)}{4!} = \frac{1}{12} n^2 (n^2 - 1)$$

independent terms. In 4 dimensions, this is 20 independent components.

There's also a differential identity involved. Consider taking the covariant derivative of the Riemann tensor at a locally inertial coordinate.

$$\begin{aligned}\nabla_{\hat{\lambda}} R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}} &= \partial_{\hat{\lambda}} R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}} \\ &= \frac{1}{2} \partial_{\hat{\lambda}} (\partial_{\hat{\mu}} \partial_{\hat{\sigma}} g_{\hat{\rho}\hat{\nu}} - \partial_{\hat{\mu}} \partial_{\hat{\rho}} g_{\hat{\nu}\hat{\sigma}} - \partial_{\hat{\nu}} \partial_{\hat{\sigma}} g_{\hat{\rho}\hat{\mu}} + \partial_{\hat{\nu}} \partial_{\hat{\rho}} g_{\hat{\mu}\hat{\sigma}})\end{aligned}$$

After some algebra, it turns out that

$$\begin{aligned}\nabla_{\hat{\lambda}} R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}} + \nabla_{\hat{\rho}} R_{\hat{\sigma}\hat{\lambda}\hat{\mu}\hat{\nu}} + \nabla_{\hat{\sigma}} R_{\hat{\lambda}\hat{\rho}\hat{\mu}\hat{\nu}} &= 0 \\ \nabla_{[\lambda} R_{\rho\sigma]\mu\nu} &= 0\end{aligned}$$

This is called the **Bianchi identity**.

3.7.1 Tensors, Tensors, Tensors

Start from the Riemann tensor

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}$$

We seek to decompose this into a sum of tensors. The naive way of $R^\rho{}_{\sigma\mu 0}$, $R^\rho{}_{\sigma\mu\nu}$ doesn't work, because it's not coordinate-invariant. The tools we can use (to preserve coordinate-independence) are **contraction** and **decomposition into symmetric/antisymmetric parts**.

First, contract the Riemann tensor to get the **Ricci tensor**.

$$R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$$

For a Riemann tensor formed from the Christoffel connection, all other contractions vanish, so this is the only one. By nature of the Christoffel connection, the Ricci tensor is automatically symmetric. The trace $R = R^\mu{}_\mu = g^{\mu\nu} R_{\mu\nu}$ is called the **Ricci scalar**.

The Ricci tensor, as a symmetric tensor, captures information about the trace of the Riemann tensor. The trace-free part is captured by the **Weyl tensor**, the Riemann tensor with all contractions removed.

$$\begin{aligned}C_{\rho\sigma\mu\nu} &= R_{\rho\sigma\mu\nu} - \frac{2}{n-2} (g_{\rho[\mu} R_{\nu]\sigma} - g_{\sigma[\mu} R_{\nu]\rho}) + \frac{2}{(n-1)(n-2)} g_{\rho[\mu} g_{\nu]\sigma} R \\ C_{\rho\sigma\mu\nu} &= C_{[\rho\sigma][\mu\nu]} \\ C_{\rho\sigma\mu\nu} &= C_{\mu\nu\rho\sigma} \\ C_{\rho[\sigma\mu\nu]} &= 0\end{aligned}$$

All contractions of the Weyl tensor vanish, but the symmetries of the Riemann tensor clearly remain.

Now consider the Bianchi identity contracted twice:

$$\begin{aligned}g^{\nu\sigma} g^{\mu\lambda} (\nabla_\lambda R_{\rho\sigma\mu\nu} + \nabla_\rho R_{\sigma\lambda\mu\nu} + \nabla_\sigma R_{\lambda\rho\mu\nu}) &= 0 \\ \nabla^\mu R_{\rho\mu} - \nabla_\rho R + \nabla^\nu R_{\rho\nu} &= 0 \\ \nabla^\mu R_{\rho\mu} &= \frac{1}{2} \nabla_\rho R\end{aligned}$$

Define the **Einstein tensor**

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$$

This is symmetric due to the symmetry of the Ricci tensor and the metric. As expected from the name, the Einstein tensor will be very useful in general relativity.

Note that the Bianchi identity is now equal to

$$\nabla^\mu G_{\mu\nu} = 0$$

3.7.2 Example: Two-Sphere S^2

Consider the two-sphere S^2 with metric

$$ds^2 = a^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

The nonzero Christoffel symbols are

$$\begin{aligned}\Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta \\ \Gamma_{\theta\phi}^\phi &= \Gamma_{\phi\theta}^\phi = \cot \theta\end{aligned}$$

Compute a promising candidate for the Riemann tensor

$$\begin{aligned}R_{\phi\theta\phi}^\theta &= \partial_\theta \Gamma_{\phi\phi}^\theta - \partial_\phi \Gamma_{\theta\phi}^\theta + \Gamma_{\theta\lambda}^\theta \Gamma_{\phi\phi}^\lambda - \Gamma_{\phi\lambda}^\theta \Gamma_{\theta\phi}^\lambda \\ &= (\sin^2 \theta - \cos^2 \theta) - (-\sin \theta \cos \theta)(\cot \theta) \\ &= \sin^2 \theta\end{aligned}$$

Now lower an index to get

$$R_{\theta\phi\theta\phi} = g_{\theta\lambda} R_{\phi\theta\phi}^\lambda = a^2 \sin^2 \theta$$

Compute the Ricci tensor with $R_{\mu\nu} = g^{\alpha\beta} R_{\alpha\mu\beta\nu}$. The nonzero components are

$$\begin{aligned}R_{\theta\theta} &= g^{\phi\phi} R_{\phi\theta\phi\theta} = 1 \\ R_{\phi\phi} &= g^{\theta\theta} R_{\theta\phi\theta\phi} = \sin^2 \theta\end{aligned}$$

Contract to find the Ricci scalar.

$$R = g^{\mu\nu} R_{\mu\nu} = g^{\theta\theta} R_{\theta\theta} + g^{\phi\phi} R_{\phi\phi} = \frac{2}{a^2}$$

Note that the Ricci scalar (encoding curvature) is constant over S^2 . If there were any “bumps” on the manifold, we would expect it to vary.

3.8 Symmetries and Killing Vectors

3.8.1 Symmetries

It's impossible to describe the entire manifold of spacetime exactly, so we have to approximate. It also turns out that Einstein's equation is nonlinear, making it hard to find exact solutions. So we're especially interested in symmetries of a system. But the notion of “symmetries” is more difficult in curved spacetime.

A manifold M has a symmetry if its geometry is invariant under some transformation that maps M to itself: i.e. the metric is the “same” from one point to another. Symmetries of the metric are called **isometries**.

If $\partial_{\sigma^*} g_{\mu\nu} = 0$, then there's a symmetry along x^{σ^*} .

$$x^{\sigma^*} \rightarrow x^{\sigma^*} + a^{\sigma^*}$$

Let $p^\mu = mU^\mu$. Recall the alternate form of the geodesic equation $p^\lambda \nabla_\lambda p_\mu = 0$. Expanding a bit, we can show that

$$m \frac{dp_\mu}{d\tau} = \frac{1}{2} (\partial_\mu g_{\nu\lambda}) p^\lambda p^\nu$$

Then the symmetry $\partial_{\sigma^*} g_{\mu\nu} = 0$ implies that $dp_{\sigma^*}/d\tau = 0$. I.e. the momentum component p_{σ^*} is a conserved quantity of the motion.

3.8.2 Killing Vectors

Let x^{σ^*} be the coordinate that $g_{\mu\nu}$ is independent of. Consider the vector ∂_{σ^*} .

$$K = \partial_{\sigma^*}$$

$$K^\mu = (\partial_{\sigma^*})^\mu = \delta_{\sigma^*}^\mu$$

We say that K^μ **generates the isometry**. Transformations that keep the geometry invariant are infinitesimal motions in the direction of K^μ . In terms of K , the conserved momentum p_{σ^*} is

$$p_{\sigma^*} = K^\nu p_\nu = K_\nu p^\nu$$

$$\frac{dp_{\sigma^*}}{d\tau} = 0 \leftrightarrow p^\mu \nabla_\mu (K_\nu p^\nu) = 0$$

From the above, we can get

$$\nabla_{(\mu} K_{\nu)} = 0 \leftrightarrow p^\mu \nabla_\mu (K_\nu p^\nu) = 0$$

The left equation is called **Killing's equation**. A vector field that satisfies it is called a **Killing vector field**. If the metric is independent of x^{σ^*} , the vector ∂_{σ^*} will satisfy Killing's equation. Additionally, if K^μ satisfies Killing's equation, it's always possible to find a coordinate system where $K = \partial_{\sigma^*}$.

We can generalize to **Killing tensors**, which contract with l copies of the momentum instead of 1.

$$\nabla_{(\mu} K_{\nu_1 \dots \nu_l)} = 0 \leftrightarrow p^\mu \nabla_\mu (K_{\nu_1 \dots \nu_l} p^{\nu_1} \dots p^{\nu_l}) = 0$$

3.8.3 Finding the Killing Vectors

It can be difficult to solve Killing's equation, but some symmetries are easy to find by inspection. For example, in $\mathbb{R}^3$ with metric $ds^2 = dx^2 + dy^2 + dz^2$, we can find 3 Killing vectors immediately:

$$X^\mu = (1, 0, 0)$$

$$Y^\mu = (0, 1, 0)$$

$$Z^\mu = (0, 0, 1)$$

There are also rotational symmetries:

$$R^\mu = (-y, x, 0)$$

$$S^\mu = (z, 0, -x)$$

$$T^\mu = (0, -z, y)$$

3.9 Maximally Symmetric Spaces

In Euclidean space $\mathbb{R}^n$, we have n translational symmetries and $\frac{1}{2}n(n-1)$ rotational symmetries, for a total of $\frac{1}{2}n(n+1)$ symmetries. Spaces with $\frac{1}{2}n(n+1)$ symmetries are called **maximally symmetric spaces**.

The number of symmetries equals the number of independent Killing vector fields.

If a space is maximally symmetric,

1. Curvature is same everywhere (translation isometry)
2. Curvature is same in every direction (rotational isometry)

In a maximally symmetric space, consider a point p . Construct local coordinates such that $g_{\hat{\mu}\hat{\nu}} = \eta_{\hat{\mu}\hat{\nu}}$. η should remain the same after Lorentz transformations, so g should too. Then the Riemann metric R (which depends on g) should also be Lorentz invariant at p . Because of the symmetries, there's only one option:

$$R_{\hat{\rho}\hat{\sigma}\hat{\mu}\hat{\nu}} \propto g_{\hat{\rho}\hat{\mu}}g_{\hat{\sigma}\hat{\nu}} - g_{\hat{\rho}\hat{\nu}}g_{\hat{\sigma}\hat{\mu}}$$

Contract twice to get $R = n(n-1)$, then

$$R_{\rho\sigma\mu\nu} = \frac{R}{n(n-1)} (g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu})$$

Locally, $g \rightarrow \eta$, so only R matters when classifying the space.

1. $R = 0$: flat spacetime, maximally symmetric (in 2 dimensions)
2. $R > 0$: spheres, etc. de Sitter space.
3. $R < 0$: hyperboloids, etc. anti-de Sitter space.

3.9.1 Example: H^2

Consider the 2D hyperboloid H^2 with metric (note: $g_{xx} = g_{yy} = a^2/y^2$):

$$ds^2 = \frac{a^2}{y^2} (dx^2 + dy^2)$$

1. Use the geodesic equation to get Christoffel symbols

$$\begin{aligned}\Gamma_{\mu\nu}^\sigma &= \frac{1}{2} g^{\sigma\rho} (\partial_\mu g_{\nu\rho} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}) \\ \Gamma_{xy}^x &= \Gamma_{yx}^x = -1/y \\ \Gamma_{xx}^y &= 1/y \\ \Gamma_{yy}^y &= -1/y\end{aligned}$$

2. Find the Riemann tensor

$$\begin{aligned}R_{yy} &= R^x_{yy} = \partial_x \Gamma_{yy}^x - \partial_y \Gamma_{xy}^x + \Gamma_{x\lambda}^x \Gamma_{yy}^\lambda - \Gamma_{y\lambda}^x \Gamma_{xy}^\lambda \\ &= \partial_x \left(-\frac{1}{y}\right) + \left(-\frac{1}{y}\right)^2 - \left(-\frac{1}{y}\right)^2 \\ &= -\frac{1}{y^2}\end{aligned}$$

3. Similar procedures show that

$$R_{xx} = R_{yy} = -\frac{1}{y^2}, \quad R_{xy} = R_{yx} = 0$$

4. Contract twice to get the Ricci scalar (curvature)

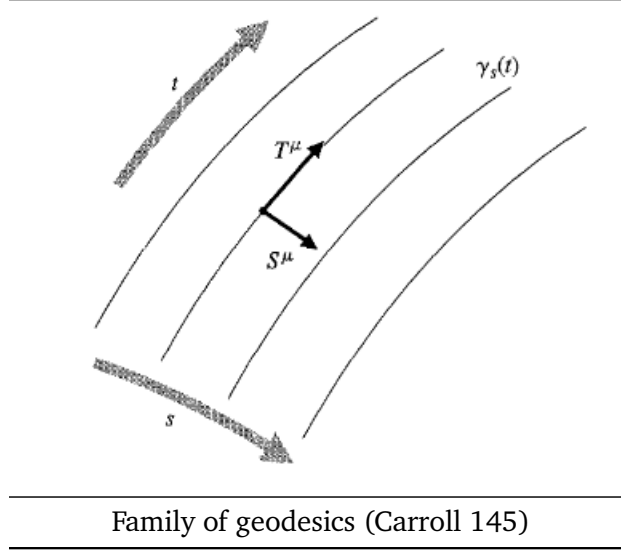
$$R = g^{\mu\nu} R_{\mu\nu} = g^{xx} R_{xx} + g^{yy} R_{yy} = -\frac{2}{a^2}$$

5. Interpret results: this is constant. Since we're in 2 dimensions, our space must be maximally symmetric. Negative curvature as expected.

3.10 Geodesic Deviation

In flat space, geodesics (parallel lines) always stay the same distance apart. In curved spacetime, this isn't true.

Let $\gamma_s(t)$ for $s \in \mathbb{R}$, be a set of geodesics parametrized by t . This collection of geodesics covers a surface with points $x^\mu(s, t) \in M$.



$$\begin{aligned} \text{Tangent vectors } T^\mu &= \frac{\partial x^\mu}{\partial t} \\ \text{Deviation vectors } S^\mu &= \frac{\partial x^\mu}{\partial s} \\ [S, T] &= 0 \end{aligned}$$

Define the relative velocity & acceleration between geodesics:

$$\begin{aligned} \text{Relative velocity } V^\mu &= (\nabla_T S)^\mu = T^\rho \nabla_\rho S^\mu \\ \text{Relative acceleration } A^\mu &= (\nabla_T V)^\mu = T^\rho \nabla_\rho V^\mu \end{aligned}$$

With some manipulation (and using the torsion-free connection), we get the **geodesic deviation equation**:

$$A^\mu = \frac{D^2}{dt^2} S^\mu = R^\mu{}_{\nu\rho\sigma} T^\nu T^\rho S^\sigma$$

As expected, the relative acceleration between geodesics is proportional to the Riemann curvature.

4 Gravitation (pp. 151 – 192)

4.1 The Minimal-Coupling Principle

Previously, the Einstein Equivalence Principle (curved spacetime looks like flat spacetime in small regions) was used to motivate the idea of manifolds, which look like $\mathbb{R}^n$ locally.

Since gravity is high universal, it should be a fundamental feature of the *background* on which matter is placed, not a conventional force.

The **minimal-coupling principle**:

1. Take a law of physics valid in inertial coordinates (flat spacetime).
2. Make its form tensorial, i.e. coordinate-invariant.

$$\begin{aligned}\eta_{\mu\nu} &\rightarrow g_{\mu\nu} \\ \partial_\mu &\rightarrow \nabla_\mu\end{aligned}$$

3. Assert that (2) works for all curved spacetime.

Example: Free-fall particles. In flat spacetime, these follow straight lines. For a path $x^\mu(\lambda)$,

$$\begin{aligned}\frac{d^2 x^\mu}{d\lambda^2} &= 0 \\ \frac{d^2 x^\mu}{d\lambda^2} = \frac{dx^\nu}{d\lambda} \partial_\nu \frac{dx^\mu}{d\lambda} &= 0 \quad \rightarrow \quad \frac{dx^\nu}{d\lambda} \nabla_\nu \frac{dx^\mu}{d\lambda} = \underbrace{\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda}}_{\text{Geodesic equation}} = 0\end{aligned}$$

We can also show that the geodesic equation reduces to Newton's law of gravitation if 1) the object moves slowly, 2) the gravitational field is weak, 3) the field is static. In that case, for $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$,

$$\begin{aligned}h_{00} &= -2\phi \\ g_{00} &= -(1 + 2\phi)\end{aligned}$$

4.2 Einstein Field Equations

The goal is to replace Poisson's equation $\nabla^2 \phi = 4\pi G\rho$ with a tensorial form. Based on the terms, we might guess something like

$$[\nabla^2 g]_{\mu\nu} \propto T_{\mu\nu}$$

The LHS needs to be symmetric, which motivates a use of $R_{\mu\nu}$. This runs into issues with conservation of energy $\nabla^\mu T_{\mu\nu} = 0$. But using the Einstein tensor $G_{\mu\nu}$ happens to work.

$$\begin{aligned}G_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \\ \nabla^\mu G_{\mu\nu} &= 0\end{aligned}$$

This is a guess, but it's backed up by experiment. As physics is.

We guess $G_{\mu\nu} = \kappa T_{\mu\nu}$ for some κ . Contracting, we get

$$\begin{aligned}R &= -\kappa T \\ R_{\mu\nu} &= \kappa \left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu} \right)\end{aligned}$$

To find κ , make the equation reduce to Newtonian gravity in weak, time-dependent field with slow particles. Assume a perfect fluid $T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}$ and work in the rest frame $U^\mu = (U^0, 0, 0, 0)$. Eventually

$$\begin{aligned} T &\approx -\rho \\ R_{00} &\approx \frac{1}{2}\kappa\rho \\ R_{00} = R^i{}_{0i0} &= -\frac{1}{2}\nabla^2 h_{00} \end{aligned}$$

Using $h_{00} = -2\phi$, we recover Poisson's equation if $\kappa = 8\pi G$. Then

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu}$$

Alternatively,

$$\begin{aligned} R_{\mu\nu} &= 8\pi G \left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu} \right) \\ R_{\mu\nu} &= 0 \end{aligned} \quad (\text{vacuum})$$

4.3 Lagrangian Derivation of Einstein's Equation

The action should depend on spacetime fields $\phi^i(x)$ and their derivatives $\nabla_\mu \phi^i(x)$.

$$S = \int \mathcal{L}(\phi^i, \nabla_\mu \phi^i) d^n x$$

$d^n x$ is a tensor density, thus $\mathcal{L}$ must also be a tensor density to make the integrand a tensor. For a scalar $\hat{\mathcal{L}}$, define $\mathcal{L} = \sqrt{-g}\hat{\mathcal{L}}$. The Euler-Lagrange equations are

$$\frac{\partial \hat{\mathcal{L}}}{\partial \phi} - \nabla_\mu \left(\frac{\partial \hat{\mathcal{L}}}{\partial (\nabla_\mu \phi)} \right) = 0$$

We seek to get vary w.r.t. the metric. The metric can have first derivative 0 at any point, using an appropriate coordinate system. To be nonzero, at least the 2nd derivatives of the metric should be used. This screams a Riemann tensor, and the Ricci scalar is the most obvious scalar to use. Hilbert proposed the following action:

$$S_H = \int \sqrt{-g} R d^n x = \int \sqrt{-g} g^{\mu\nu} R_{\mu\nu} d^n x$$

Varying the action shows that (Carroll 4.3)

$$\begin{aligned} \delta S &= \int d^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} + \int d^n x R \delta \sqrt{-g} \\ &= \int d^n x \sqrt{-g} R_{\mu\nu} \delta g^{\mu\nu} - \frac{1}{2} \int d^n x R \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \\ &= \int d^n x \sqrt{-g} \left[R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} \right] \delta g^{\mu\nu} \end{aligned}$$

The Euler-Lagrange equation recovers Einstein's equation in a vacuum:

$$\frac{1}{\sqrt{-g}} \frac{\delta S_H}{\delta g_{\mu\nu}} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0$$

For the non-vacuum equation, the action must account for matter.

$$S = \frac{1}{16\pi G} S_H + S_M$$

An appropriate relation of $T_{\mu\nu}$ and S_M recovers the full Einstein equation:

$$T_{\mu\nu} = -2 \frac{1}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}$$

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

Noether's theorem says that every symmetry of a Lagrangian implies a conserved quantity. For a tensor $S^{\mu\nu}$, this means

$$\partial_\mu S^{\mu\nu} = 0$$

For $S^{\mu\nu}$ depending on ϕ^i and $\partial_\mu \phi^i$, there's an alternate form of the action, but in practice it's not as widely used as the above one in terms of $T_{\mu\nu}$.

$$S^{\mu\nu} = \frac{\delta \mathcal{L}}{\delta(\partial_\mu \phi^i)} \partial^\nu \phi^i - \eta^{\mu\nu} \mathcal{L}$$

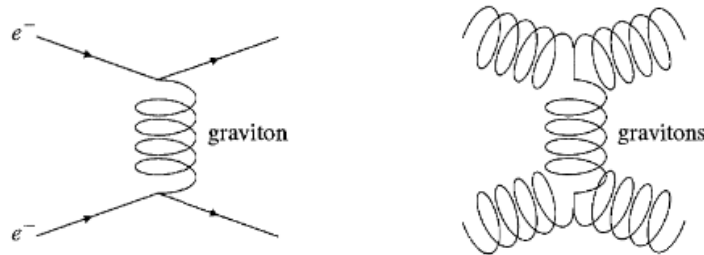
4.4 A Mixed Bag of Cool Stuff

4.4.1 Gravity Couples to Itself

GR is nonlinear: solutions to Einstein's equation can't be superimposed to form another solution.

A reason is that gravity couples to any form of energy-momentum, including gravity itself. You should be getting headaches at this point.

Quantizing Einstein's equations predicts a **graviton**. Gravitons can couple to other gravitons.



Feynman diagram for gravitational attraction of electrons (Carroll 167)

4.4.2 The Cosmological Constant

Einstein's equation depends on the energy-momentum tensor directly, not just the difference in energies. This implies a **vacuum energy**: energy density for empty space.

This vacuum energy should be Lorentz invariant, i.e. proportional to the metric. Predicted:

$$T_{\mu\nu}^{(\text{vac})} = -\rho_{\text{vac}}g_{\mu\nu}$$

Recall the perfect fluid energy-momentum: $T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}$. In this view, vacuum energy is an isotropic perfect fluid with

$$p_{\text{vac}} = -\rho_{\text{vac}}$$

Split $T_{\mu\nu}$ into a matter piece $T_{\mu\nu}^{(\text{M})}$ and a vacuum piece $T_{\mu\nu}^{(\text{vac})} = -\rho_{\text{vac}}g_{\mu\nu}$. Einstein's equation becomes

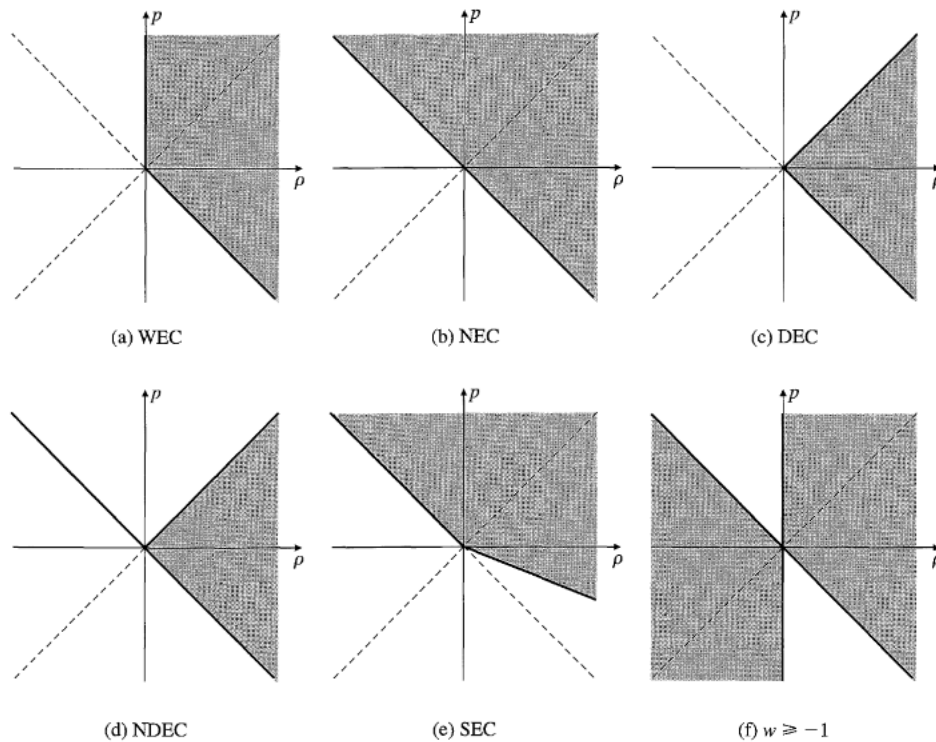
$$\begin{aligned} R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} &= 8\pi G(T_{\mu\nu}^{(\text{M})} - \rho_{\text{vac}}g_{\mu\nu}) \\ R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} &= 8\pi GT_{\mu\nu} \\ \rho_{\text{vac}} &= \frac{\Lambda}{8\pi G} \end{aligned}$$

Λ is the **cosmological constant**. Theoretical results with QFT predict $\rho_{\text{vac}} \sim 10^{112} \text{ erg/cm}^3$. Experimental results show $\rho_{\text{vac}} \sim 10^{-8} \text{ erg/cm}^3$. The difference is 120 orders of magnitude – not good. It remains an unresolved problem.

4.4.3 Energy Conditions

Assuming a perfect fluid $T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}$,

1. **Weak Energy Condition (WEC):** $T_{\mu\nu}t^\mu t^\nu \geq 0$ for all timelike t^μ . Or $\rho \geq 0$ and $\rho + p \geq 0$.
2. **Null Energy Condition (NEC):** $T_{\mu\nu}l^\mu l^\nu \geq 0$ for all lightlike l^μ . Or $\rho + p \geq 0$.
3. **Dominant Energy Condition (DEC):** WEC + condition that $T^{\mu\nu}t_\mu$ is a non-spacelike vector. I.e. $T_{\mu\nu}T^\nu{}_\lambda t^\mu t^\lambda \leq 0$. Or $\rho \geq |p|$.
4. **Null Dominant Energy Condition (NDEC):** DEC for null vectors. Same as above, except negative densities are allowed as long as $p = -\rho$. I.e. allows for negative vacuum energy.
5. **Strong Energy Condition (SEC):** $T_{\mu\nu}t^\mu t^\nu \geq \frac{1}{2}T^\lambda{}_\lambda t^\sigma t_\sigma$. I.e. $\rho + p \geq 0$ and $\rho + 3p \geq 0$.



Geometric interpretation of energy conditions (Carroll 176). $w = p/\rho$, equation of state parameter.

4.4.4 Equivalence Principle

The EP is usually used to justify the following 4 ideas:

1. Laws of physics should be expressed in covariant, coordinate-independent form.
2. Spacetime has a metric, the curvature of which is “gravity.”
3. There aren’t any other fields that resemble gravity.
4. Interaction of matter to fields is minimal. No direct coupling to Riemann tensor or its contractions.

(1) and (2) follow from previous discussion. (3) and (4) may have to be adjusted in the future, but hold up well.

5 The Schwarzschild Solution (pp. 193 – 237)

5.1 The Schwarzschild Metric

We seek the solution for a static, spherically symmetrical gravitational field. This is a black hole solution to Einstein’s equation.

Start in spherical coordinates $\{t, r, \theta, \phi\}$ with $d\Omega^2$ as the unit 2-sphere. Parameters are arbitrary.

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + e^{2\gamma(r)} r^2 d\Omega^2$$

$$d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$$

To get rid of $e^{2\gamma(r)}$, change to radial coordinates where this vanishes. I.e.

$$\bar{r} = e^{\gamma(r)} r$$

$$d\bar{r} = \frac{d}{dr} [e^{\gamma(r)} r] dr = \left(1 + r \frac{d\gamma}{dr}\right) e^{\gamma(r)} dr$$

The metric becomes

$$ds^2 = -e^{2\alpha(r)} dt^2 + \left(1 + r \frac{d\gamma}{dr}\right)^{-2} e^{2\beta(r)-2\gamma(r)} d\bar{r}^2 + \bar{r}^2 d\Omega^2$$

Relabeling gives a simpler form. This isn't exactly equal to the previous metric, because of the relabeling. But it preserves the same geometry.

$$\bar{r} \rightarrow r$$

$$\left(1 + r \frac{d\gamma}{dr}\right)^{-2} e^{2\beta(r)-2\gamma(r)} \rightarrow e^{2\beta(r)}$$

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2 \quad (1)$$

Assume that outside the source, there's no mass, so $R_{\mu\nu} = 0$. We seek to enforce this condition on the metric. Recall the procedure for finding the Ricci tensor:

1. Find the Christoffel symbols according to

$$\Gamma_{\mu\nu}^{\sigma} = \frac{1}{2} g^{\sigma\rho} (\partial_{\mu} g_{\nu\rho} + \partial_{\nu} g_{\rho\mu} - \partial_{\rho} g_{\mu\nu})$$

2. Find the Riemann tensor:

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu} \Gamma_{\nu\sigma}^{\rho} - \partial_{\nu} \Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho} \Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho} \Gamma_{\mu\sigma}^{\lambda}$$

3. Contract the Riemann tensor to get the Ricci tensor:

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}$$

The specific symbols are below (Carroll p. 195). They take some time to find. Symmetries $\Gamma_{\mu\nu}^{\rho} = \Gamma_{\nu\mu}^{\rho}$ are implied.

$\Gamma_{tr}^t = \partial_r \alpha$	$\Gamma_{tt}^r = e^{2(\alpha-\beta)} \partial_r \alpha$	$\Gamma_{rr}^r = \partial_r \beta$
$\Gamma_{r\theta}^{\theta} = \frac{1}{r}$	$\Gamma_{\theta\theta}^r = -r e^{-2\beta}$	$\Gamma_{r\phi}^{\phi} = \frac{1}{r}$
$\Gamma^r \phi \phi = -r e^{-2\beta} \sin^2 \theta$	$\Gamma_{\phi\phi}^{\theta} = -\sin \theta \cos \theta$	$\Gamma_{\theta\phi}^{\phi} = \frac{\cos \theta}{\sin \theta}$

And the Riemann tensor:

$$\begin{aligned}
R^t_{rtr} &= (\partial_r \alpha)(\partial_r \beta) - \partial_r^2 \alpha - (\partial_r \alpha)^2 \\
R^t_{\theta t \theta} &= -r e^{-2\beta} \partial_r \alpha \\
R^t_{\phi t \phi} &= -r e^{-2\beta} \sin^2 \theta \partial_r \alpha \\
R^r_{\theta r \theta} &= r e^{-2\beta} \partial_r \beta \\
R^r_{\phi r \phi} &= r e^{-2\beta} \sin^2 \theta \partial_r \beta \\
R^\theta_{\phi \theta \phi} &= (1 - e^{-2\beta}) \sin^2 \theta
\end{aligned}$$

Now the Ricci tensor:

$$\begin{aligned}
R_{tt} &= e^{2(\alpha-\beta)} \left[\partial_r^2 \alpha + (\partial_r \alpha)^2 - (\partial_r \alpha)(\partial_r \beta) + \frac{2}{r} \partial_r \alpha \right] \\
R_{rr} &= -e^{2(\beta-\alpha)} R_{tt} + \frac{2}{r} (\partial_r \alpha + \partial_r \beta) \\
R_{\theta\theta} &= e^{-2\beta} [r(\partial_r \beta - \partial_r \alpha) - 1] + 1 \\
R_{\phi\phi} &= \sin^2 \theta R_{\theta\theta}
\end{aligned}$$

By requiring that the Ricci tensor vanish, we can simplify to get (for some constant R_S that doesn't depend on r)

$$ds^2 = - \left(1 - \frac{R_S}{r} \right) dt^2 + \left(1 - \frac{R_S}{r} \right)^{-1} dr^2 + r^2 d\Omega^2 \quad (2)$$

In the weak field limit with a point mass, relativity reduces to Newton's gravity and the metric component g_{tt} satisfies

$$g_{tt} = - \left(1 - \frac{2GM}{r} \right)$$

Thus let $R_S = 2GM$ so that the Schwarzschild metric reduces to the weak-field point-mass case when $r \gg 2GM$. The full Schwarzschild metric is

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) dt^2 + \left(1 - \frac{2GM}{r} \right)^{-1} dr^2 + r^2 d\Omega^2$$

- Reduces to Minkowski space as $M \rightarrow 0$
- Reduces to Minkowski space as $r \rightarrow \infty$ (asymptotic flatness)

5.2 Singularities

The Schwarzschild metric “blows up” at $r = 0$ and $r = 2GM$. This isn't worrying on its own. E.g. in polar coordinates $ds^2 = dr^2 + r^2 d\theta^2$, the inverse metric $g^{\theta\theta} = \frac{1}{r^2}$ blows up at $r = 0$, despite being perfectly fine there.

Question of “breakdown of coordinate system” (fine) vs “breakdown of manifold” (uh oh).

Simple check: is the curvature infinite?

- The curvature (a scalar) is coordinate-independent, so if it's infinite, we have a problem.

- Simplest is Ricci curvature $R = g^{\mu\nu} R_{\mu\nu}$, but also $R^{\mu\nu} R_{\mu\nu}$, $R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma}$, etc.

For Schwarzschild solution:

$$R^{\mu\nu\rho\sigma} R_{\mu\nu\rho\sigma} = \frac{48G^2 M^2}{r^6}$$

Thus $r = 0$ is a singularity, and merits investigation. $r = 2GM$ turns out to be fine, but it's interesting as well: “event horizon of a black hole.”

5.3 Geodesics

The geodesic equations are very messy and difficult to solve. Instead, start from the Killing vectors (3 for spherical symmetry, 1 for time translations). Also use constants from the geodesic equation:

$$\text{Killing vectors: } K_\mu \frac{dx^\mu}{d\lambda} = \text{const.}$$

$$\text{From geodesic equation: } \epsilon = -g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}$$

For massive particles, we can set $\lambda = \tau$, then $\epsilon = 1$. For massless particles, $\epsilon = 0$.

Fix the direction of angular momentum in a plane to get $\theta = \pi/2$, then find the remaining killing vectors of energy (from time translations) and angular momentum (from spherical symmetry):

$$\begin{aligned} K^\mu = (\partial_t)^\mu = (1, 0, 0, 0) &\longrightarrow K_\mu = \left(-\left(1 - \frac{2GM}{r}\right), 0, 0, 0 \right) \\ R^\mu = (\partial_\phi)^\mu = (0, 0, 0, 1) &\longrightarrow R_\mu = (0, 0, 0, r^2 \sin^2 \theta) \end{aligned}$$

The conserved quantities (energy, angular momentum) are:

$$\begin{aligned} E &= -K_\mu \frac{dx^\mu}{d\lambda} = \left(1 - \frac{2GM}{r}\right) \frac{dt}{d\lambda} \\ L &= R_\mu \frac{dx^\mu}{d\lambda} = r^2 \frac{d\phi}{d\lambda} \end{aligned}$$

Now expand our equation for ϵ using E and L .

$$\begin{aligned} -\epsilon &= -\left(1 - \frac{2GM}{r}\right) \left(\frac{dt}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r}\right)^{-1} \left(\frac{dr}{d\lambda}\right)^2 + r^2 \left(\frac{d\phi}{d\lambda}\right)^2 \\ -\epsilon \left(1 - \frac{2GM}{r}\right) &= -E^2 + \left(\frac{dr}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r}\right) \frac{L^2}{r^2} \\ 0 &= -E^2 + \left(\frac{dr}{d\lambda}\right)^2 + \left(1 - \frac{2GM}{r}\right) \left(\frac{L^2}{r^2} + \epsilon\right) \end{aligned} \tag{1}$$

Define the following:

$$\begin{aligned} V(r) &= \frac{1}{2}\epsilon - \epsilon \frac{GM}{r} + \frac{L^2}{2r^2} - \underbrace{\frac{GML^2}{r^3}}_{\text{GR term}} \\ \mathcal{E} &= \frac{1}{2}E^2 \end{aligned}$$

Note that as $r \rightarrow \infty$ and $\epsilon \rightarrow 0$, the potential is similar to the Newtonian one. Using these new variables, (1) becomes

$$\frac{1}{2} \left(\frac{dr}{d\lambda} \right)^2 + V(r) = \mathcal{E} \quad (2)$$

Circular orbits occur when $dV/dr = 0$. Thus

$$\epsilon GM r_c^2 - L^2 r_c + 3GM L^2 \gamma = 0$$

where $\gamma = 0$ in Newtonian and $\gamma = 1$ in GR. In Newtonian physics, circular orbits occur at $r_c = L^2/(\epsilon GM)$.

Now assume a massless particle ($\epsilon = 0$) in GR ($\gamma = 1$). Solve the above equation to get $r_c = 3GM$. At this r_c , a photon (e.g.) can orbit in a circle forever. However, after any perturbation, the photon will go to $r \rightarrow 0$ or $r \rightarrow \infty$.

For massive particles ($\epsilon \neq 0$), there are circular orbits at

$$r_c = \frac{L^2 \pm \sqrt{L^4 - 12G^2 M^2 L^2}}{2GM} = \left(\frac{L^2}{GM}, 3GM \right)$$

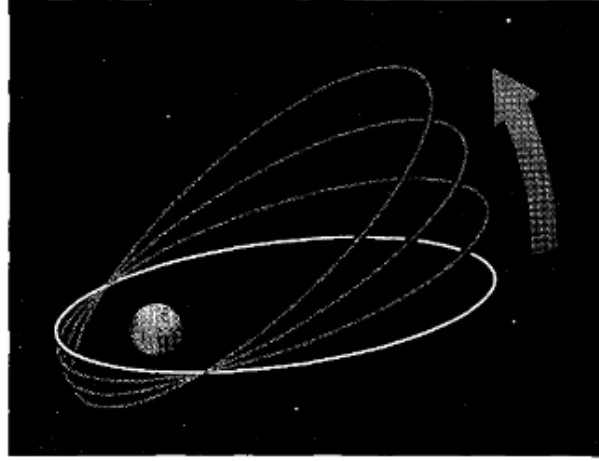
The discriminant vanishes at $L = \sqrt{12}GM$, and corresponding to this, $r_c = 6GM$ is the smallest **stable** circular orbit.

To recap:

- Stable circular orbits at $r_c \geq 6GM$
- Unstable circular orbits at $3GM < r_c < 6GM$

5.4 Experimental Tests

5.4.1 Precession of Mercury's Perihelion



Precession of perihelion (Carroll 213)

Instead of being a closed ellipse, the orbit can precess. To recognize this, describe r in terms of ϕ : $r(\phi)$. For a perfect ellipse, this is perfectly periodic. Not so for a precessing one.

Start from the Schwarzschild geodesic equation and multiply by $(d\phi/d\lambda)^{-2} = (L/r^2)^{-2}$.

$$\begin{aligned} \frac{1}{2} \left(\frac{dr}{d\lambda} \right)^2 + V(r) &= \mathcal{E} \\ \frac{1}{2} \left(\frac{dr}{d\phi} \right)^2 + V(r) \left(\frac{r^4}{L^2} \right) &= \mathcal{E} \left(\frac{r^4}{L^2} \right) \\ \frac{1}{2} \left(\frac{dr}{d\phi} \right)^2 + \left(\frac{1}{2}\epsilon - \epsilon \frac{GM}{r} + \frac{L^2}{2r^2} - \frac{GML^2}{r^3} \right) \frac{r^4}{L^2} &= \mathcal{E} \left(\frac{r^4}{L^2} \right) \end{aligned}$$

Assume a massive particle $\epsilon = 1$:

$$\left(\frac{dr}{d\phi} \right)^2 + \epsilon \frac{r^4}{L^2} - \frac{2\epsilon GM}{L^2} r^3 + r^2 - 2GML = 2\mathcal{E} \frac{r^4}{L^2} \quad (1)$$

Now substitute $x = L^2/(GMr)$ and differentiate with respect to ϕ to simplify the equation (steps not shown):

$$\left(\frac{dx}{d\phi} \right)^2 + \frac{L^2}{G^2 M^2} - 2x + x^2 - \frac{2G^2 M^2}{L^2} x^3 = \frac{2\mathcal{E} L^2}{G^2 M^2} \quad (2)$$

$$\frac{d^2 x}{d\phi^2} - 1 + x = \frac{3G^2 M^2}{L^2} x^2 \quad (3)$$

In Newtonian gravity, the last term would be absent, so we can treat it as a relativistic perturbation. Split x into $x = x_0 + x_1$, where x_0 is the Newtonian contribution and x_1 is the relativistic perturbation.

The result is two differential equations:

$$\begin{aligned}\frac{d^2 x_0}{d\phi^2} - 1 + x_0 &= 0 \quad \longrightarrow \quad x_0 = 1 + e \cos \phi \\ \frac{d^2 x_1}{d\phi^2} + x_1 &= \frac{3G^2 M^2}{L^2} x_0^2 \quad \longrightarrow \quad x_1 = \frac{3G^2 M^2}{L^2} \left[\left(1 + \frac{1}{2}e^2\right) + e\phi \sin \phi - \frac{1}{6}e^2 \cos(2\phi) \right]\end{aligned}$$

In x_1 , there are three terms: a constant, an **accumulation that gathers over time**, and a small oscillation around 0. We're concerned with the second term. Thus only consider the terms in x as follows:

$$\begin{aligned}x &= 1 + e \cos \phi + \frac{3G^2 M^2 e}{L^2} \phi \sin \phi \\ &\approx 1 + e \cos [(1 - \alpha)\phi] + O(\alpha^2), \quad \alpha = \frac{3G^2 M^2}{L^2}\end{aligned}$$

The last approximation can be confirmed by expanding a Taylor series around small α . Overall, each orbit, the perihelion precesses by

$$\Delta\phi = 2\pi\alpha = \frac{6\pi G^2 M^2}{L^2}$$

The precession of Mercury is observed to be 5601''/century.

- 5025''/century due to precession of equinoxes in geocentric frame
- 532''/century due to gravitational pull of other planets (approx with Newtonian gravity)
- 43.0''/century, unexplained

If we calculate the precession of Mercury using GR, we get $\Delta\phi \approx 43.0''/\text{century}$, an amazingly accurate prediction.

5.4.2 Gravitational Redshift Revisited

Recall: redshift of photons was explained using the principle of equivalence. The Schwarzschild solution is a valid solution to Einstein's equations. Thus, in small enough regions of ST, it must predict a redshift.

Consider an observer with 4-velocity U^μ , stationary in Schwarzschild coordinates ($U^i = 0$).

$$U_\mu U^\mu = -1 \quad \longrightarrow \quad U_0 = \left(1 - \frac{2GM}{r}\right)^{-1/2}$$

The frequency of a photon at a null geodesic $x^\mu(\lambda)$ is $\omega = -g_{\mu\nu} U^\mu \frac{dx^\nu}{d\lambda}$. In this case,

$$\omega = \left(1 - \frac{2GM}{r}\right)^{1/2} \frac{dt}{d\lambda} = \left(1 - \frac{2GM}{r}\right)^{-1/2} E$$

As energy is conserved, ω primarily depends on r . Thus,

$$\frac{\omega_2}{\omega_1} = \left(\frac{1 - 2GM/r_1}{1 - 2GM/r_2} \right)^{1/2}$$

If $r \gg 2GM$, we get $\omega_2/\omega_1 \approx 1 - GM/r_1 + GM/r_2 = 1 + \Phi_1 - \Phi_2$, where $\Phi = -GM/r$ is the Newtonian potential. I.e. frequency goes down as Φ increases, or when we climb out of a gravitational field.

This is a redshift, valid in the weak field limit.

5.5 Schwarzschild Black Holes

What happens at $r < 2GM$ (unstable orbit)? To explore the causal structure, we use radial null curves ($\theta = 0, \phi = 0, ds^2 = 0$)

$$0 = - \left(1 - \frac{2GM}{r} \right) dt^2 + \left(1 - \frac{2GM}{r} \right)^{-1} dr^2$$

$$\frac{dt}{dr} = \pm \left(1 - \frac{2GM}{r} \right)^{-1}$$

For large r , slope is ± 1 . But for $r \rightarrow 2GM$, the slope goes to ∞ , making the light cone “close up.”

Thus if a light ray approaches a black hole, an observer sees that it never reaches $r = 2GM$. But this is just from the observer’s POV.

5.5.1 Eddington-Finkelstein Coordinates

Perhaps try another coordinate system that’s better behaved around $r = 2GM$. Use another coordinate time t defined below:

$$r^* = r + 2GM \ln \left(\frac{r}{2GM} - 1 \right)$$

$$t = \pm r^* + \text{const.}$$

r^* is “tortoise coordinates” for $r \geq 2GM$, named so after Zeno’s paradox: as $r^* \rightarrow -\infty, r \rightarrow 2GM$. The Schwarzschild metric is

$$ds^2 = \left(1 - \frac{2GM}{r} \right) (-dt^2 + dr^{*2}) + r^2 d\Omega^2$$

Now define

$$v = t + r^* \quad (v = \text{const: infalling radial geodesic})$$

$$u = t - r^* \quad (u = \text{const: outgoing radial geodesic})$$

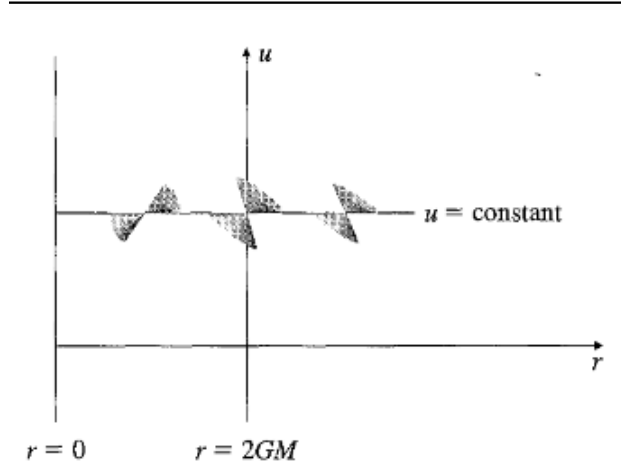
Use v instead of t to get the **Eddington-Finkelstein coordinates** for an ingoing light ray.

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) dv^2 + (dvdr + drdv) + r^2 d\Omega^2$$

The determinant turns out to be $g = -r^4 \sin^2 \theta$, which is nonzero at $r = 2GM$. Thus the metric is invertible at $r = 2GM$, and there's no singularity. $r = 2GM$ was just a coordinate singularity in the Schwarzschild metric.

In Eddington-Finkelstein coordinates, the condition for radial null curves is

$$\frac{dv}{dr} = \begin{cases} 0 & \text{(infalling)} \\ 2 \left(1 - \frac{2GM}{r}\right)^{-1} & \text{(outgoing)} \end{cases}$$



Light cones closing up (Carroll 223)

Light cones seem to “close up” when $r < 2GM$. Thus $r = 2GM$ is the point of no return for an object or light ray, an **event horizon**.

We can't see inside that point from $r > 2GM$, hence the name: **black hole**.

5.6 The Maximally Extended Schwarzschild Solution

5.6.1 Kruskal Coordinates

Previously: we subbed v for t and found that the “singularity” at $r = 2GM$ was not a real singularity.

In (v, r) coordinates, we can pass the event horizon $r = 2GM$ on future-directed curves.

The opposite is true in (u, r) coordinates: we can pass $r = 2GM$ on past-directed curves.

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) du^2 - (dudr + drdu) + r^2 d\Omega^2$$

What if we use (u, v) instead of (t, r) ?

$$ds^2 = -\frac{1}{2} \left(1 - \frac{2GM}{r}\right) (dvdu + dudv) + r^2 d\Omega^2$$

$$\frac{1}{2}(v - u) = r^* = r + 2GM \ln \left(\frac{r}{2GM} - 1\right)$$

But then $r = 2GM$ is at $v = -\infty$ or $u = +\infty$. To pull this back to finite values, scale by an appropriate factor.

$$\begin{aligned} v' = e^{v/4GM} &\longrightarrow \left(\frac{r}{2GM} - 1\right)^{1/2} e^{(r+t)/4GM} \\ u' = -e^{-u/4GM} &\longrightarrow -\left(\frac{r}{2GM} - 1\right)^{1/2} e^{(r-t)/4GM} \end{aligned}$$

Note that v', u' (like v, u) are null coordinates – tracking light. But we’re most comfortable with 1 timelike, 3 spatial coordinates. Thus define

$$\begin{aligned} T = \frac{1}{2} (v' + u') &= \left(\frac{r}{2GM} - 1\right)^{1/2} e^{r/4GM} \sinh\left(\frac{t}{4GM}\right) \\ R = \frac{1}{2} (v' - u') &= \left(\frac{r}{2GM} - 1\right)^{1/2} e^{r/4GM} \cosh\left(\frac{t}{4GM}\right) \end{aligned}$$

The Schwarzschild metric in these **Kruskal coordinates** (T, R, θ, ϕ) is

$$ds^2 = \frac{32G^3M^3}{r} e^{-r/2GM} \left(-dT^2 + dR^2\right) + r^2 d\Omega^2$$

where r is implicitly defined as

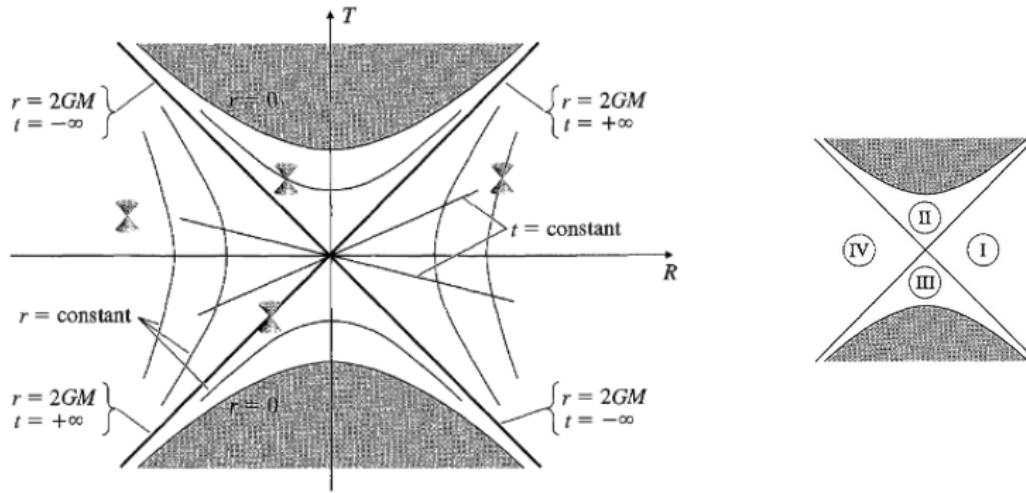
$$T^2 - R^2 = \left(1 - \frac{r}{2GM}\right) e^{r/2GM}$$

5.6.2 Properties of Kruskal Coordinates

Number of convenient features:

- Null curves at $T = \pm R + \text{const.}$, like Minkowski space.
- Event horizon at $r = 2GM$ corresponds to $T = \pm R$.
- Surfaces of $r = \text{const.}$ defined by hyperbolae $T^2 - R^2 = \text{const.}$
- Surfaces of $t = \text{const.}$ defined by $T/R = \tanh(t/4GM)$. Become straight lines of slope ± 1 as $t \rightarrow \pm\infty$, i.e. $T = \pm R$.
- Range $-\infty \leq R \leq \infty$ and $T^2 < R^2 + 1$ to avoid the **real** singularity $r = 0$.

We can represent the Schwarzschild solution on a **Kruskal diagram**:

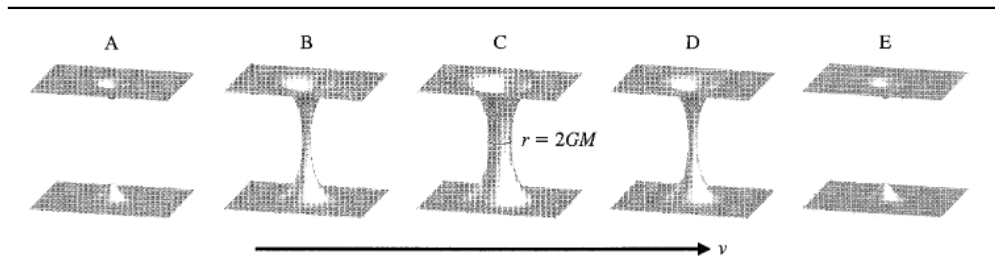


Kruskal Diagram (Carroll 226)

Each point on the diagram represents a 2-sphere.

I represents $r > 2GM$. From I, we reach II by future-directed curves and III by past-directed curves.

- Anything that crosses $I \rightarrow II$ cannot return. II is a **black hole**.
- Things can only cross to I from III, not vice versa. III is a **white hole**.
- IV cannot be reached. Mirror image of our spacetime. Can be thought of as connected to I by a **wormhole** (“Einstein-Rosen bridge”).



Wormhole formation from spacelike slices (Carroll 228)

5.7 Stars and Black Holes

Stars can collapse into black holes. But unlike the Schwarzschild solution, we have a non-vacuum (e.g. nonzero energy-momentum tensor).

1. Use a general spherically symmetric metric.

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\Omega^2$$

2. We model the star as a perfect fluid. Additionally, for static solution, normalize $U^\mu U_\mu = -1$.

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}$$

$$U_\mu = (e^\alpha, 0, 0, 0)$$

3. Use the full form of Einstein's equation (not vacuum):

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi GT_{\mu\nu}$$

$$G_{tt} = \frac{1}{r^2}e^{2(\alpha-\beta)}(2r\partial_r\beta - 1 + e^{2\beta})$$

$$G_{rr} = \frac{1}{r^2}(2r\partial_r\beta - 1 + e^{2\beta})$$

$$G_{\theta\theta} = r^2e^{-2\beta}\left[\partial_r^2\alpha + (\partial_r\alpha)^2 - (\partial_r\alpha)(\partial_r\beta) + \frac{1}{r}(\partial_r\alpha - \partial_r\beta)\right]$$

$$G_{\phi\phi} = \sin^2\theta G_{\theta\theta}$$

5.7.1 Tolman-Oppenheimer-Volkoff Equation

Calculate the energy-momentum tensor with above assumptions:

$$T_{\mu\nu} = \begin{pmatrix} (\rho + p)e^{2\alpha} - pe^{2\alpha} & 0 & 0 & 0 \\ 0 & pe^{2\beta} & 0 & 0 \\ 0 & 0 & r^2p & 0 \\ 0 & 0 & 0 & r^2p\sin^2\theta \end{pmatrix}$$

The Einstein equation reduces to three distinct equations ($G_{\phi\phi}$ is proportional to $G_{\theta\theta}$):

$$\begin{aligned} 1. \quad G_{tt} &= \frac{1}{r^2}e^{2(\alpha-\beta)}(2r\partial_r\beta - 1 + e^{2\beta}) = 8\pi Ge^{2\alpha}\rho \\ 2. \quad G_{rr} &= \frac{1}{r^2}(2r\partial_r\beta - 1 + e^{2\beta}) = 8\pi Ge^{2\beta}p \\ 3. \quad G_{\theta\theta} &= r^2e^{-2\beta}\left[\partial_r^2\alpha + (\partial_r\alpha)^2 - (\partial_r\alpha)(\partial_r\beta) + \frac{1}{r}(\partial_r\alpha - \partial_r\beta)\right] = 8\pi Gr^2p \end{aligned}$$

1. From G_{tt}

Note that it only depends on $\beta(r), \rho(r)$. Make the substitution

$$m(r) = \frac{1}{2G}(r - re^{-r\beta}) \quad , \quad e^{2\beta} = \left[1 - \frac{2Gm(r)}{r}\right]^{1/2}$$

Then the metric becomes very similar to Schwarzschild:

$$ds^2 = -e^{2\alpha(r)}dt^2 + \left[1 - \frac{2Gm(r)}{r}\right]^{-1}dr^2 + r^2d\Omega^2$$

Furthermore, from the pseudo-Schwarzschild metric, this substitution has the nice interpretation that

$$\frac{dm}{dr} = 4\pi r^2\rho \quad \longrightarrow \quad m(r) = 4\pi \int_0^r \rho(r')r'^2 dr'$$

Assuming the star density exists until radius R , and is zero afterwards, we get the “star mass”

$$M = m(R) = 4\pi \int_0^R \rho(r) r^2 dr$$

But note that this isn’t *really* the “star mass” as we would get from a volume integral – that volume integral needs $\sqrt{\gamma} d^3x$, not d^3x . In fact, if we do the calculations with $\sqrt{\gamma} d^3x$, we get a slightly higher “mass” reflecting the gravitational binding energy of a star.

2. From G_{rr}

After substituting $e^{2\beta}$ and simplifying, we find

$$\frac{d\alpha}{dr} = \frac{Gm(r) + 4\pi Gr^3 \rho}{r[r - 2Gm(r)]}$$

Almost there. Use energy-momentum conservation $\nabla_\mu T^{\mu\nu} = 0$ to get the **Tolman-Oppenheimer-Volkoff equation** for hydrostatic equilibrium:

$$\frac{dp}{dr} = -(\rho + p) \frac{d\alpha}{dr} = -\frac{(\rho + p)[Gm(r) + 4\pi Gr^3 p]}{r[r - 2Gm(r)]}$$

5.7.2 The Life of a Star

A good model for a star is to assume constant density ρ_* until the surface R , then 0 density. The values of $m(r)$ are

$$m(r) = \begin{cases} \frac{4}{3}\pi r^3 \rho_* & r < R \\ \frac{4}{3}\pi R^3 \rho_* = M & r > R \end{cases}$$

Assume p is a function of ρ only: $p = p(\rho)$. Usually it’s $p = p(\rho, S)$, where S is the entropy, but assume the entropy is small enough to be neglected. Integrating the TOV equation gives

$$p(r) = \rho_* \left[\frac{R\sqrt{R - 2GM} - \sqrt{R^3 - 2GM r^2}}{\sqrt{R^3 - 2GM r^2} - 3R\sqrt{R - 2GM}} \right]$$

The pressure at the center, $p(0)$, blows up when $M_{\max} = \frac{4}{9G}R$. Stuff any more mass into this radius, and we get no static solution – the star shrinks into a black hole.

Ordinary planets (supported by material pressure) aren’t likely to collapse.

However, massive stars supported by heat of fusion (light nuclei to heavy) can shrink due to gravity when the fuel runs out.

1. This can stop at **Fermi degeneracy pressure**: repulsion between electrons cancels out the gravitational pull. The product is a **white dwarf**.

2. If the mass is high enough (Chandrasekhar limit), it collapses further. Electrons and protons combine to form neutrons and neutrinos through inverse beta decay, and neutrinos fly away, leaving an even smaller **neutron star**.
3. Now it's uncertain – but it's estimated that if the neutron star has mass 3-4 $M_{\odot}$ (Oppenheimer-Volkoff limit), the neutron star will collapse further into a **black hole**.

Black holes can have masses roughly to an order of $M_{\odot}$, these could be endpoints of massive stars.

There are also supermassive black holes, $10^6 - 10^9 M_{\odot}$. The formation of these aren't understood well. Our own galaxy has one: Sagittarius A*, $2 \times 10^6 M_{\odot}$.

Matter falling into a black hole settles in a rotating disk, feeding both energy and angular momentum into the black hole. Thus, black holes spin.

6 More General Black Holes (pp. 238 – 273)

6.1 No Hair!

“A black hole has no hair” – coined by John Archibald Wheeler. If electromagnetism is the only long-range nongravitational field, then we have a **no-hair theorem**:

Stationary, asymptotically flat black hole solutions to general relativity coupled to electromagnetism that are nonsingular outside the event horizon are fully characterized by: mass, electric charge, magnetic charge, and angular momentum.

Contrast this to Earth – the external field depends on the density, the exact mountain ranges, people walking on the surface, etc.

An interesting problem that arises: **information loss paradox**.

1. Take a black hole and stuff a complicated configuration of mass inside it
2. System is now fully described by mass, charge, and spin
3. Information lost?

Classical GR explanation: information is just hidden behind the event horizon.

The real thorniness occurs when quantum field theory comes into play – black holes eventually evaporate, so information is somehow being lost. Does the Hawking radiation encode some information about the black hole? It's unclear.

6.2 A Gentle Overview of Singularities

Let a hypersurface Σ be some section of a manifold where $f(x) = \text{const}$. The gradient $\partial_{\mu}f$ is normal to f .

If $\partial_{\mu}f$ is null, it's tangent to Σ , and we call the hypersurface **null**. Null geodesics $x^{\mu}(\lambda)$ are generators of null hypersurfaces.

Event horizons are surfaces beyond which timelike curves cannot escape to ∞ . Thus, they are null hypersurfaces.

To recognize null hypersurfaces, note that on them, null geodesics are proportional to the hypersurface:

$$\xi^\mu = \frac{dx^\mu}{d\lambda} = h(x)g^{\mu\nu}\partial_\nu f$$

After a choice of $h(x)$ to preserve the affine parametrization,

$$\xi_\mu \xi^\mu = 0 \quad , \quad \xi^\mu \nabla_\mu \xi^\nu = 0$$

Event horizons are usually difficult to find. Some simplifications make it easier.

1. Assume stationary metric: has Killing vector ∂_t that is asymptotically timelike.
2. Adapt metric components to be timelike: $\partial_t g^{\mu\nu} = 0$
3. On a hypersurface $t = \text{const.}$, choose (r, θ, ϕ) that look like Minkowski flat spacetime as $r \rightarrow \infty$.
4. Choose coordinates such that by decreasing r from ∞ , the hypersurface $r = \text{const.}$ can remain timelike until $r = r_H$, after which the surface is null everywhere.
5. $r = r_H$ is the “event horizon.”

Note that $\partial_\mu r$ is a 1-form normal to the hypersurface $r = \text{const.}$ in step 4. The norm is

$$g^{\mu\nu}(\partial_\mu r)(\partial_\nu r) = g^{rr}$$

The norm vanishes at the event horizon, so basically we’re looking $r = r_H$ where $g^{rr}(r_H) = 0$. The Schwarzschild solution satisfies steps 1-5, and clearly $g^r r(2GM) = 1 - 2GM/2GM = 0$.

Event horizons are nearly inevitable in GR, due to 2 results:

1. Singularities are nearly inevitable in GR (singularity theorems of Hawking & Penrose).
2. Singularities are hidden behind event horizons (cosmic censorship conjecture).

6.3 Killing Horizons

6.3.1 Killing Horizons

Previously, for Schwarzschild, we found that the Killing vector $K = \partial_t$ goes from timelike $\rightarrow$ spacelike at the event horizon.

If a Killing vector χ^μ is null along a null hypersurface Σ , then Σ is a **Killing horizon** of χ^μ .

Some tentative classifications:

- Every event horizon Σ in a stationary, asymptotically flat spacetime is a Killing horizon for some Killing vector field χ^μ .
 - If the spacetime is static, χ^μ is the Killing vector field $K^\mu = (\partial_t)^\mu$ representing time translations at ∞ .

- If the spacetime is stationary but not static, it's **axisymmetric** with rotational Killing vector field $R^\mu = (\partial_\phi)^\mu$ and χ^μ is a linear combination for some const. Ω_H :

$$\chi^\mu = K^\mu + \Omega_H R^\mu$$

6.3.2 Surface Gravity

We can associate to each Killing horizon a “surface gravity.” Consider a Killing vector χ^μ with Killing horizon Σ . Because χ^μ is normal to Σ , along the Killing horizon, it obeys the geodesic equation

$$\chi^\mu \nabla_\mu \chi^\nu = -\kappa \chi^\nu$$

The RHS arises because χ^μ may not be affinely parametrized.

Define κ , the **surface gravity**, as

$$\kappa^2 = -\frac{1}{2}(\nabla_\mu \chi^\nu)(\nabla^\mu \chi_\nu)$$

In static spacetime, we interpret κ as the acceleration of a static observer near the event horizon, as measured by a static observer at ∞ . In non-static spacetime, this becomes more nuanced.

A “static observer” is one with 4-velocity U^μ proportional to time-translation Killing vector K^μ : $K^\mu = V(x)U^\mu$. Thus

$$V = \sqrt{-K_\mu K^\mu}$$

$V = 0$ at the Killing horizon, and $V = 1$ at ∞ . Note that a photon has energy $E = -p_\mu K^\mu$, and its frequency as measured by an observer with 4-velocity U^μ is $\omega = -p_\mu U^\mu = E/V$. At two different positions, the redshift is

$$\lambda_2 = \frac{V_2}{V_1} \lambda_1$$

so V is the “redshift factor.” Anyways, from the 4-acceleration a_μ , note that

$$\begin{aligned} a_\mu &= U^\sigma \nabla_\sigma U^\mu = \nabla_\mu \ln(V) \\ a &= \sqrt{a_\mu a^\mu} = \frac{1}{V} \sqrt{\nabla_\mu V \nabla^\mu V} \end{aligned}$$

$a = \infty$ at κ , representing an infinite acceleration needed to keep an object on a static trajectory at the Killing horizon.

We claim that:

$$\kappa = Va = \sqrt{\nabla_\mu V \nabla^\mu V}$$

For the Schwarzschild solution, the time-translation Killing vector and the static (normalized) 4-velocity are

$$K^\mu = (1, 0, 0, 0)$$

$$U^\mu = \left(\left(1 - \frac{2GM}{r}\right)^{-1/2}, 0, 0, 0 \right)$$

The redshift factor is

$$V = \sqrt{1 - \frac{2GM}{r}}$$

The 4-acceleration and its magnitude is

$$a_\mu = \frac{GM}{r^2 \left(1 - \frac{2GM}{r}\right)} \nabla_\mu r \quad , \quad \nabla_\mu r = \delta_\mu^r$$

$$a = \frac{GM}{r^2 \left(1 - \frac{2GM}{r}\right)^{1/2}}$$

From which the surface gravity is $\kappa = Va = GM/r^2$. At the event horizon $r = 2GM$, we have

$$\kappa = \frac{1}{4GM}$$

A larger M creates a weaker surface gravity, due to the inverse square factor of r .

6.4 Mass, Charge, and Spin

These fully characterize a black hole. Their derivation is admittedly going to be very ad hoc, because I didn't fully understand things. Hopefully that will change as I gain familiarity with the subject.

6.4.1 Charge

Let the electric 4-current be J_e^μ . The charge through a spacelike hypersurface Σ is

$$\begin{aligned} Q &= - \int_\Sigma d^3x \sqrt{\gamma} n_\mu J_e^\mu \\ &= - \int_\Sigma d^3x \sqrt{\gamma} n_\mu \nabla_\nu F^{\mu\nu} && \text{(Maxwell's equations)} \\ &= - \int_{\partial\Sigma} d^2x \sqrt{\gamma^{(2)}} n_\mu F^{\mu\nu} && \text{(Stokes' theorem)} \end{aligned}$$

For a magnetic charge, replace $F^{\mu\nu}$ with its dual tensor $*F^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}$. But the electric charge of a black hole is

$$Q = - \int_{\partial\Sigma} d^2x \sqrt{\gamma^{(2)}} n_\mu F^{\mu\nu}$$

6.4.2 Mass (Energy)

For spacetime with a timelike Killing vector, consider the current

$$J_R^\mu = K_\nu R^{\mu\nu} = 8\pi G K_\nu \left(T^{\mu\nu} - \frac{1}{2} T g^{\mu\nu} \right)$$

This is conserved, can be checked by $\nabla_\mu J_R^\mu = 0$ combined with the contracted Bianchi identity $\nabla_\mu R^{\mu\nu} = \frac{1}{2} \nabla^\nu R$. To this conserved current, we might associate a conserved energy:

$$\begin{aligned} E_R &= \frac{1}{4\pi G} \int_\Sigma d^3x \sqrt{\gamma} n_\mu J_R^\mu \\ &= \frac{1}{4\pi G} \int_\Sigma d^3x \sqrt{\gamma} n_\mu \nabla_\nu (\nabla^\mu K^\nu) \\ &= \frac{1}{4\pi G} \int_{\partial\Sigma} d^2x \sqrt{\gamma^{(2)}} n_\mu \sigma_\nu \nabla^\mu K^\nu \end{aligned}$$

Where n_μ, σ_μ are normalized as $n_\mu n^\mu = -1$ and $\sigma_\mu \sigma^\mu = +1$. This is called the **Komar energy** of a black hole.

$$E_R = \frac{1}{4\pi G} \int_{\partial\Sigma} d^2x \sqrt{\gamma^{(2)}} n_\mu \sigma_\nu \nabla^\mu K^\nu$$

Consider a Schwarzschild black hole. Stationary. Then

$$\begin{aligned} n_0 &= - \left(1 - \frac{2GM}{r} \right)^{1/2} \\ \sigma_0 &= \left(1 - \frac{2GM}{r} \right)^{-1/2} \\ n_\mu \sigma_\nu \nabla^\mu K^\nu &= -\nabla^0 K^1 \end{aligned}$$

The time translation Killing vector is $K^\mu = (1, 0, 0, 0)$. Thus

$$\begin{aligned} \nabla^0 K^1 &= g^{00} \nabla_0 K^1 = g^{00} \left(\partial_0 K^1 + \Gamma_{0\lambda}^1 K^\lambda \right) \\ &= g^{00} \left(\partial_0 K^1 + \Gamma_{00}^1 K^0 \right) \\ &= - \left(1 - \frac{2GM}{r} \right)^{-1} \cdot \frac{GM}{r^2} \left(1 - \frac{2GM}{r} \right) = -\frac{GM}{r^2} \end{aligned}$$

Further note that $\gamma_{ij}^{(2)} dx^i dx^j = r^2 (d\theta^2 + \sin^2 \theta d\phi^2)$. Thus $\sqrt{\gamma^{(2)}} = r^2 \sin \theta$. Combining these results, we can calculate the Komar energy of a Schwarzschild black hole:

$$E_R = \frac{1}{4\pi G} \int d\theta d\phi r^2 \sin \theta \left(\frac{GM}{r} \right) = M$$

Mass and energy, yet again.

An alternative formulation of energy: find a Hamiltonian formalism of GR, then set E_{total} to be the generator of time translations. This is called the **ADM energy** after Arnowitt, Deser, and Misner.

For an asymptotically flat spacetime, represent the metric as a perturbation $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. Then

$$E_{ADM} = \frac{1}{16\pi G} \int_{\partial\Sigma} d^2x \sqrt{\gamma^{(2)}} \sigma^j (\partial_j h^j_i - \partial_i h^j_j)$$

6.4.3 Spin

Imagine a rotational Killing vector $R = \partial_\phi$. Define the conserved current as

$$J^\mu_\phi = R_\nu R^{\mu\nu}$$

Then set the conserved angular momentum as the integral of this conserved current over spacelike infinity. We call this the **spin of a black hole**.

$$J = -\frac{1}{8\pi G} \int_{\partial\Sigma} d^2x \sqrt{\gamma^{(2)}} n_\mu \sigma_\nu \nabla^\mu R^\nu$$

6.5 Charged Black Holes (Reissner-Nordström)

Like the Schwarzschild solution, Reissner-Nordström black holes exhibit spherical symmetry. However, there's a nonzero electromagnetic tensor, contributing to a nonzero energy-momentum tensor (i.e. not a vacuum).

$$T_{\mu\nu} = F_{\mu\rho} F_\nu{}^\rho - \frac{1}{4} g_{\mu\nu} F_{\rho\sigma} F^{\rho\sigma}$$

The full metric is:

$$ds^2 = -\Delta dt^2 + \Delta^{-1} dr^2 + r^2 d\Omega^2$$

$$\Delta = 1 - \frac{2GM}{r} + \frac{G(Q^2 + P^2)}{r^2}$$

Q is the total electric charge, while P is the total magnetic charge (NB: if you're against magnetic monopoles, set $P = 0$). The radial fields are

$$E_r = F_{rt} = \frac{Q}{r^2}$$

$$B_r = \frac{F_{\theta\phi}}{r^2 \sin \theta} = \frac{P}{r^2}$$

Find event horizons by seeing where the norm g^{rr} is zero.

$$g^{rr}(r) = \Delta(r) = 1 - \frac{2GM}{r} + \frac{G(Q^2 + P^2)}{r^2} = 0$$

$$r = GM \pm \sqrt{G^2 M^2 - G(Q^2 + P^2)}$$

- **Case 1:** $GM^2 < Q^2 + P^2$

- There are no solutions, thus no event horizons. This is a naked singularity, which goes against the cosmic censorship conjecture.
- But if the electromagnetic field energy is greater than the total energy — it suggests a negative mass, **probably an unphysical solution**.
- **Case 2:** $GM^2 > Q^2 + P^2$
 - There are two solutions, r_+ and r_- .
 - An observer can flip flop between these two event horizons without reaching the real singularity at $r = 0$.
- **Case 3:** $GM^2 = Q^2 + P^2$
 - An **extreme solution** to the Reissner-Nordström black hole.
 - Even a small imbalance causes this to become case 2.

6.6 Rotating Black Holes (Kerr)

The metric:

$$ds^2 = - \left(1 - \frac{2GM}{\rho^2} \right) dt^2 - \frac{2GMa r \sin^2 \theta}{\rho^2} (dt d\phi + d\phi dt) + \frac{\rho^2}{\Delta} dr^2 + \rho^2 d\theta^2 + \frac{\sin^2 \theta}{\rho^2} \left[(r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta \right] d\phi^2$$

The relevant constants are:

$$\begin{aligned} \Delta(r) &= r^2 - 2GM r + a^2 \\ \rho^2(r, \theta) &= r^2 + a^2 \cos^2 \theta \\ a &= J/M \quad , \quad M = \text{Komar energy} \end{aligned}$$

This is pretty complicated, and accordingly, it was discovered much later than Schwarzschild or Reissner-Nordström. To include charge, simply replace $2GM r$ with $2GM r - G(Q^2 + P^2)$.

6.6.1 Singularities

We again try to identify event horizons as where $g^{rr} = 0$. I.e. $\Delta/\rho^2 = 0$.

$$\begin{aligned} \Delta(r) &= r^2 - 2GM r + a^2 = 0 \\ r &= GM \pm \sqrt{G^2 M^2 - a^2} \end{aligned}$$

1. **Case 1:** $GM < a$. This is a naked singularity, probably unphysical.
2. **Case 2:** $GM = a$. Unstable solution that can easily become case 3.
3. **Case 3:** $GM > a$. To analyze this, we use elliptical coordinates (r, θ) , and a is another parameter of the ellipse. The metric coefficients are independent of t, ϕ , so the Killing vectors

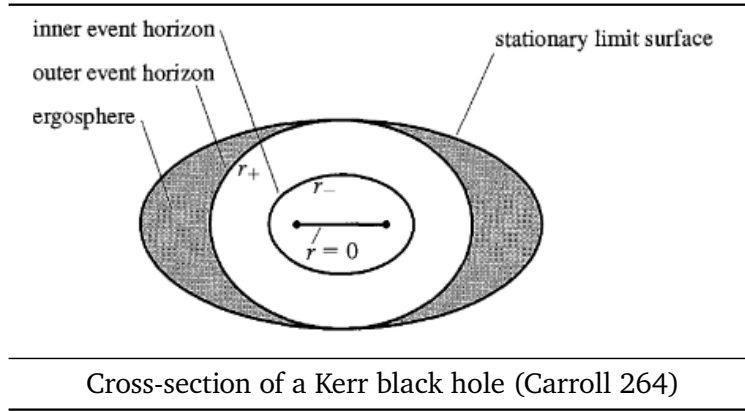
are $K = \partial_t$ and $R = \partial_\phi$. Norm of K^μ :

$$\begin{aligned} K^\mu K_\mu &= \frac{2GMr}{\rho^2} - 1 \\ &= \frac{2GMr - (r^2 + a^2 \cos^2 \theta)}{\rho^2} \\ &= -\frac{1}{\rho^2}(\Delta - a^2 \sin^2 \theta) \end{aligned}$$

This does not vanish at the event horizon $\Delta = 0$, except $\theta = 0, \pi$. Instead, it vanishes at

$$\begin{aligned} \Delta &= a^2 \sin^2 \theta \\ r^2 - 2GMr + a^2 &= a^2 \sin^2 \theta \\ (r - GM)^2 - G^2 M^2 + a^2 \cos^2 \theta &= 0 \end{aligned}$$

The region between this and the outer $\Delta = 0$ is the **ergosphere**.



Further, note that the singularities aren't at $r = 0$, but at $\rho^2 = r^2 + a^2 \cos^2 \theta = 0$. Thus the “singularity” is the disk (in elliptical coordinates)

$$r = 0 \quad , \quad \theta = \frac{\pi}{2}$$

The rotation has “spread out” the singular Schwarzschild singularity into a disk.

6.6.2 Closed Timelike Curves

Consider a region near the disk singularity. Wind around ϕ , with $d\theta = dt = 0$. Keep r a small negative value. Then

$$\begin{aligned} ds^2 &\approx a^2 \left(1 + \frac{2GM}{r}\right) d\phi^2 \\ ds &< 0 \end{aligned}$$

And this is a loop, so we seem to have a closed timelike curve, where you can meet yourself in the past... I'd like to do that, I need a word with my past self.

6.7 Black Hole Thermodynamics

6.7.1 The Penrose Process

AKA how to (theoretically) steal energy from a black hole. Consider a massive particle $p^\mu = m \frac{dx^\mu}{d\tau}$ near a Kerr black hole. Killing vectors are $K = \partial_t$ and $R = \partial_\phi$. Use this to define some conserved quantities (energy, angular momentum):

$$E = -K_\mu p^\mu = m \left(1 - \frac{2GMr}{\rho^2} \right) \frac{dt}{d\tau} + \frac{2mGMar}{\rho^2} \sin^2 \theta \frac{d\phi}{d\tau}$$

$$L = R_\mu p^\mu = -\frac{2mGMar}{\rho^2} \sin^2 \theta \frac{dt}{d\tau} + \frac{m(r^2 + a^2)^2 - m\Delta a^2 \sin^2 \theta}{\rho^2} \sin^2 \theta \frac{d\phi}{d\tau}$$

The Penrose process:

1. Grab rock.
2. Heroically leap into the ergosphere.
3. Throw the rock towards the disk singularity.
4. By conservation of momentum, you drift out of the ergosphere with more energy than you entered.

At the instant you throw the rock, let $p^{(0)\mu}$ be the (you + rock) system, $p^{(1)\mu}$ be you, and $p^{(2)\mu}$ be the rock. By conservation of momentum,

$$p^{(0)\mu} = p^{(1)\mu} + p^{(2)\mu}$$

$$E^{(0)} = E^{(1)} + E^{(2)}$$

$$E^{(2)} < 0 \quad , \quad E^{(1)} > E^{(0)}$$

This is due to the negative sign before E in its definition, which was necessary to ensure a positive energy outside the black hole...

6.7.2 Relation to Classical Thermodynamics

Great, we've extracted energy from a black hole. But this isn't unlimited. It must follow the **area theorem**: the area of the event horizon can't decrease.

- The mass increases (black hole gets a new rock), but to compensate, the angular momentum decreases.
- Thus we can only continue to extract energy from a black hole before it becomes a Schwarzschild black hole (no rotation).

Consider the event horizon r_+ , so $\Delta = 0$. Also set $dt = dr = 0$.

$$\gamma_{ij} dx^i dx^j = (r_+^2 + a^2 \cos^2 \theta) d\theta^2 + \left[\frac{(r_+^2 + a^2)^2 \sin^2 \theta}{r_+^2 + a^2 \cos^2 \theta} \right] d\phi^2$$

$$|\gamma| = (r_+^2 + a^2)^2 \sin^2 \theta$$

The horizon area is

$$A = \int \sqrt{|\gamma|} d\theta d\phi = 4\pi(r_+^2 + a^2)$$

We can use the definition of the **irreducible mass of a black hole**:

$$\begin{aligned} M_{\text{irr}}^2 &= \frac{A}{16\pi G^2} = \frac{1}{4G^2}(r_+^2 + a^2) \\ &= \frac{1}{2} \left(M^2 + \sqrt{M^4 - (Ma/G)^2} \right) \\ &= \frac{1}{2} \left(M^2 + \sqrt{M^4 - (J/G)^2} \right) \\ \delta M_{\text{irr}} &= \frac{a}{4GM_{\text{irr}}\sqrt{G^2M^2 - a^2}} \left(\Omega_H^{-1} \delta M - \delta J \right) \end{aligned}$$

$\delta M_{\text{irr}} > 0$, thus $\delta A > 0$ and by rearranging,

$$\begin{aligned} \delta A &= 8\pi G \frac{a}{\Omega_H \sqrt{G^2M^2 - a^2}} (\delta M - \Omega_H \delta J) \\ \delta M &= \frac{\kappa}{8\pi G} \delta A + \Omega_H \delta J \quad , \quad \kappa = \frac{\sqrt{G^2M^2 - a^2}}{2GM(GM + \sqrt{G^2M^2 - a^2})} \end{aligned}$$

Recall the classical thermodynamics equation:

$$dE = TdS - pdV$$

We might find an equivalence between this and black holes by interpreting $\Omega_H \delta J$ as the work done on the black hole by throwing the rock in it. Then (with $\hbar = c = k = 1$)

$$\begin{aligned} E &\leftrightarrow M \\ S &\leftrightarrow A/4G \\ T &\leftrightarrow \kappa/2\pi \end{aligned}$$

The constants for S and T seem to be chosen arbitrarily. It follows from Hawking's results of applying quantum field theory to gravity. Exciting stuff, but too advanced for my level.

7 Perturbation Theory and Gravitational Radiation (pp. 274 – 320)

7.1 Perturbation of the Metric

We consider a small perturbation of flat spacetime:

$$\begin{aligned} g_{\mu\nu} &= \eta_{\mu\nu} + h_{\mu\nu} \quad , \quad |h_{\mu\nu}| \ll 1 \\ g^{\mu\nu} &= \eta^{\mu\nu} - h^{\mu\nu} \end{aligned}$$

1. Find the Christoffel symbols to 1st order in h .

$$\begin{aligned} \Gamma_{\mu\nu}^\rho &= \frac{1}{2} g^{\rho\lambda} (\partial_\mu g_{\nu\lambda} + \partial_\nu g_{\lambda\mu} - \partial_\lambda g_{\mu\nu}) \\ &= \frac{1}{2} (\eta^{\rho\lambda} - h^{\rho\lambda}) [\partial_\mu (\eta_{\nu\lambda} + h_{\nu\lambda}) + \partial_\nu (\eta_{\lambda\mu} + h_{\lambda\mu}) - \partial_\lambda (\eta_{\mu\nu} + h_{\mu\nu})] \\ &\approx \frac{1}{2} \eta^{\rho\lambda} (\partial_\mu h_{\nu\lambda} + \partial_\nu h_{\lambda\mu} - \partial_\lambda h_{\mu\nu}) \end{aligned}$$

2. Find the Riemannian tensor to 1st order in h .

$$\begin{aligned} R^\rho{}_{\sigma\mu\nu} &= \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma} \\ &\approx \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} \\ R_{\mu\nu\rho\sigma} &= \eta_{\mu\lambda} \partial_\rho \Gamma^\lambda_{\nu\sigma} - \eta_{\mu\lambda} \partial_\sigma \Gamma^\lambda_{\nu\rho} \\ &= \frac{1}{2} [\partial_\rho \partial_\nu h_{\sigma\mu} - \partial_\rho \partial_\mu h_{\nu\sigma} + \partial_\sigma \partial_\mu h_{\nu\rho} - \partial_\sigma \partial_\nu h_{\rho\mu}] \end{aligned}$$

3. Find the Ricci tensor and the Ricci scalar to 1st order in h .

$$\begin{aligned} R_{\mu\nu} &= \frac{1}{2} (\partial_\sigma \partial_\nu h^\sigma{}_\mu + \partial_\sigma \partial_\mu h^\sigma{}_\nu - \partial_\mu \partial_\nu h - \square h_{\mu\nu}) \\ R &= \partial_\mu \partial_\nu h^{\mu\nu} - \square h \\ h &= h^\lambda{}_\lambda, \quad \square = -\partial_t^2 + \partial_x^2 + \partial_y^2 + \partial_z^2 \end{aligned}$$

4. Find the Einstein tensor.

$$\begin{aligned} G_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} R \\ &= \frac{1}{2} (\partial_\sigma \partial_\nu h^\sigma{}_\mu + \partial_\sigma \partial_\mu h^\sigma{}_\nu - \partial_\mu \partial_\nu h - \square h_{\mu\nu}) - \frac{1}{2} (\eta_{\mu\nu} \partial_\rho \partial_\lambda h^{\rho\lambda} - \eta_{\mu\nu} \square h) \end{aligned}$$

And as always, the field equation (even with a small perturbation) is

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

7.1.1 Gauge Invariance

Potential concern: the perturbation $h_{\mu\nu}$ may not be unique. But we have

$$h_{\mu\nu}^{(\epsilon)} = h_{\mu\nu} + 2\epsilon \partial_{(\mu} \xi_{\nu)}$$

This is the change of a metric perturbation h under an infinitesimal diffeomorphism along the vector field $\epsilon \xi^\mu$. Two metric perturbations related by $2\epsilon \partial_{(\mu} \xi_{\nu)}$ denote equivalent spacetime.

7.2 Degrees of Freedom

We dissect the perturbation under spatial rotations. The h_{00} component is a scalar, and h_{0i} components form a 3-vector. The h_{ij} components form a symmetric tensor, which can be further split into a trace and trace-free part.

$$\begin{aligned} h_{00} &= -2\Phi \\ h_{0i} &= w_i \\ h_{ij} &= 2s_{ij} - 2\Psi \delta_{ij} \\ \Psi &= -\frac{1}{6} \delta^{ij} h_{ij}, \quad s_{ij} = \frac{1}{2} \left(h_{ij} - \frac{1}{3} \delta^{kl} h_{kl} \delta_{ij} \right) \end{aligned}$$

The metric becomes

$$ds^2 = -(1 + 2\Phi)dt^2 + w_i(dt dx^i + dx^i dt) + [(1 - 2\Psi)\delta_{ij} + 2s_{ij}] dx^i dx^j$$

By finding the Riemann tensor, Ricci tensor, and Einstein tensor, we get the following field equations:

$$\begin{aligned}\nabla^2 \Psi &= 4\pi G T_{00} - \frac{1}{2} \partial_k \partial_l s^{kl} \\ (\delta_{jk} \nabla^2 - \partial_j \partial_k) w^k &= -16\pi G T_{0j} + 4\partial_0 \partial_j \Psi + 2\partial_0 \partial_k s_j^k \\ (\delta_{ij} \nabla^2 - \partial_i \partial_j) \Phi &= 8\pi G T_{ij} + (\delta_{ij} \nabla^2 - \partial_i \partial_j - 2\delta_{ij} \partial_0^2) \Psi - \delta_{ij} \partial_0 \partial_k w^k + \partial_0 \partial_{(i} w_{j)} + \square s_{ij} \\ &\quad - 2\partial_k \partial_{(i} s_{j)}^k - \delta_{ij} \partial_k \partial_l s^{jl}\end{aligned}$$

7.2.1 Motion of Test Particles

From the 4-momentum of a particle, $p^0 = dx^0/d\lambda = dt/d\lambda = E$. Further, $p^i = E v^i$. Assuming $c = 1$. Then from the geodesic equation, we have

$$\begin{aligned}\frac{dp^\mu}{d\lambda} + \Gamma_{\rho\sigma}^\mu p^\rho p^\sigma &= 0 \\ \frac{dE}{dt} = -\Gamma_{\rho\sigma}^0 \frac{p^\rho p^\sigma}{E} &= -E \left[\partial_0 \Phi + 2(\partial_k \Phi) v^k - \left(\partial_{(j} w_{k)} - \frac{1}{2} \partial_0 h_{jk} \right) v^j v^k \right]\end{aligned}$$

For the spatial components, define the **gravito-electric** and **gravito-magnetic** fields:

$$\begin{aligned}G^i &= -\partial_i \Phi - \partial_0 w_i \\ H^i &= (\nabla \times \vec{w})^i = \epsilon^{ijk} \partial_j w_k\end{aligned}$$

Then from the geodesic equation,

$$\begin{aligned}\frac{dp^i}{dt} &= -E \left[\partial_i \Phi + \partial_0 w_i + 2 \left(\partial_{[i} w_{j]} + \partial_0 h_{ij} \right) v^j + \left(\partial_{(j} h_{k)i} - \frac{1}{2} \partial_i h_{jk} \right) v^j v^k \right] \\ &= E \left[G^i + (\vec{v} \times \vec{H})^i - 2(\partial_0 h_{ij}) v^j - \left(\partial_{(j} h_{k)i} - \frac{1}{2} \partial_i h_{jk} \right) v^j v^k \right]\end{aligned}$$

The motion of the test particle is similar to that of a classical particle under the Lorentz force (first 2 terms). However, there are extra terms that have to do with the perturbation.

7.2.2 Popular Gauges

The Transverse Gauge: Like the Coulomb gauge with $\partial_i A^i = 0$. We set

$$\partial_i s^{ij} = 0 \quad , \quad \partial_i w^i = 0$$

Using

$$\begin{cases} \nabla^2 \xi^j + \frac{1}{3} \partial_j \partial_i \xi^i = -2\partial_i s^{ij} \\ \nabla^2 \xi^0 = \partial_i w^i + \partial_0 \partial_i \xi^i \end{cases}$$

The Einstein field equations become

$$\begin{aligned} G_{00} &= 2\nabla^2\Psi = 8\pi GT_{00} \\ G_{0j} &= -\frac{1}{2}\nabla^2 w_j + 2\partial_0\partial_j\Psi = 8\pi GT_{0j} \\ G_{ij} &= (\delta_{ij}\nabla^2 - \partial_i\partial_j)(\Phi - \Psi) - \partial_0\partial_{(i}w_{j)} + 2\delta_{ij}\partial_0^2\Psi - \square\Psi - \square s_{ij} = 8\pi GT_{ij} \end{aligned}$$

The Synchronous Gauge: Like the temporal electromagnetic gauge $A^0 = 0$. Removes nonspatial components of the perturbation.

$$\Phi = 0 \quad , \quad w^i = 0$$

Using

$$\begin{cases} \partial_0\xi^0 = -\Phi \\ \partial_0\xi^i = -w^i + \partial_i\xi^0 \end{cases}$$

The Lorenz Gauge: Also called the harmonic gauge.

$$\partial_\mu h^\mu{}_\nu - \frac{1}{2}\partial_\nu h = 0$$

7.3 Photons in Newtonian Trajectories

We can model gravitating sources as “dust” (perfect fluid, no pressure). In the frame of dust,

$$T_{\mu\nu} = \rho U_\mu U_\nu = \begin{pmatrix} \rho & & & \\ & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix}$$

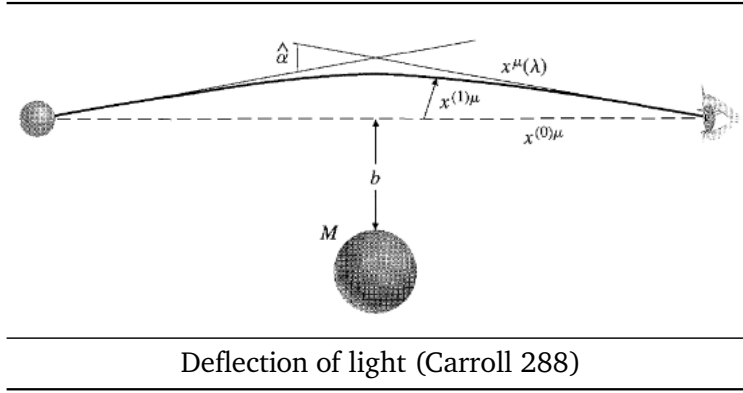
Consider static sources. This can be achieved by Lorentz transforming into the frame of the source (possible since underlying field is Minkowski, and perturbation is small). Use the previous equations for the transverse gauge, but drop time-derivative terms, and plug in $T_{\mu\nu}$ above.

$$\begin{cases} \nabla^2\psi = 8\pi G\rho \\ \nabla^2 w_j = 0 \\ (\delta_{ij}\nabla^2 - \partial_i\partial_j)(\Phi - \Psi) - \nabla^2 s_{ij} = 0 \end{cases}$$

From (2), $w^i = 0$. Trace over δ_{ij} in (3) to get $2\nabla^2(\Phi - \Psi) = 0$, so $\Phi = \Psi$. Finally, $\nabla^2 s_{ij} = 0$, so $s_{ij} = 0$. The metric and perturbation for static sources is

$$\begin{aligned} ds^2 &= -(1 + 2\Phi)dt^2 + (1 - 2\Phi)(dx^2 + dy^2 + dz^2) \\ h_{\mu\nu} &= \begin{pmatrix} -2\Phi & & & \\ & -2\Phi & & \\ & & -2\Phi & \\ & & & -2\Phi \end{pmatrix} \quad , \quad \nabla^2\Phi = 4\pi G\rho \end{aligned}$$

7.3.1 Deflection of Light



Consider the deflection of a photon's geodesic according to above. The overall path can be split into a background path x^0 and a perturbation x^1 .

$$x^\mu(\lambda) = \underbrace{x^{(0)\mu}(\lambda)}_{\text{background}} + \underbrace{x^{(1)\mu}(\lambda)}_{\text{perturbation}}$$

Let $k^\mu = dx^{(0)\mu}/d\lambda$ and $l^\mu = dx^{(1)\mu}/d\lambda$, the derivative of the background wave and the deviation wave. The photon follows a null path, $g^{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0$, so

$$\text{0th order: } \eta_{\mu\nu} k^\mu k^\nu = 0 \longrightarrow (k^0)^2 = \vec{k}^2$$

$$\text{1st order: } 2\eta_{\mu\nu} k^\mu l^\nu + h_{\mu\nu} k^\mu k^\nu = 0 \longrightarrow -kl^0 + \vec{l} \cdot \vec{k} = 2k^2\Phi$$

The geodesic equation is (with $w^i = 0$ and $h_{ij} = -2\Phi\delta_{ij}$)

$$\begin{aligned} \frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} &= 0 \\ \Gamma_{0i}^0 &= \Gamma_{00}^i = \partial_i \Phi \\ \Gamma_{jk}^i &= \delta_{jk} \partial_i \Phi - \delta_{ik} \partial_j \Phi - \delta_{ij} \partial_k \Phi \end{aligned}$$

To 0th order, this is just $dk^\mu/d\lambda = 0$, signifying a straight path. To 1st order, we have

$$\begin{aligned} \frac{dl^\mu}{d\lambda} &= -2k(\vec{k} \cdot \nabla \Phi) \\ \frac{d\vec{l}}{d\lambda} &= -2k^2 \nabla_\perp \Phi = -2k^2 \nabla \Phi + 2(\vec{k} \cdot \nabla \Phi) \vec{k} \end{aligned}$$

The deflection angle is defined by $\hat{\alpha} = -\Delta \vec{l}/k$. Let $s = k\lambda$ be the physical distance traveled. Then

$$\begin{aligned} \hat{\alpha} &= -\frac{1}{k} \int \frac{d\vec{l}}{d\lambda} d\lambda \\ &= 2k \int \nabla_\perp \Phi d\lambda \\ &= 2 \int \nabla_\perp \Phi ds \end{aligned}$$

Or for a point mass with impact parameter $b = |\vec{b}|$ (see diagram),

$$\begin{aligned}\Phi &= -\frac{GM}{r} = -\frac{GM}{(b^2 + x^2)^{1/2}} \\ \nabla_{\perp} \Phi &= \frac{GM}{(b^2 + x^2)^{3/2}} \vec{b} \\ \hat{\alpha} &= 2GMb \int_{-\infty}^{+\infty} \frac{dx}{(b^2 + x^2)^{3/2}} = \frac{4GM}{b}\end{aligned}$$

7.3.2 Gravitational Time-Delay

The total time along a null path is

$$t = \int \frac{dx^0}{d\lambda} d\lambda$$

With the introduction of a Newtonian potential,

$$\begin{aligned}\Delta t &= \int \frac{dx^{(1)0}}{dd\lambda} d\lambda = \int l^0 d\lambda \\ &= -2k \int \Phi d\lambda = -2 \int \Phi ds\end{aligned}$$

That is, for a nonzero potential, photons appear to “slow down” on their trajectory.

7.4 Gravitational Waves

Consider the weak-field transverse gauge again, and keep the time derivatives but set $T_{\mu\nu} = 0$. Then

$$\begin{aligned}\nabla^2 \Psi &= 0 \\ \nabla^2 w_j &= 0 \\ \nabla^2 \Phi &= 0 \quad , \quad \square s_{ij} = 0\end{aligned}$$

Define the transverse traceless gauge ($\Phi = \Psi = w_j = 0$), where s_{ij} is transverse:

$$\begin{aligned}h_{\mu\nu}^{TT} &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & 2s_{ij} & \\ 0 & & & \end{pmatrix} \\ \square h_{\mu\nu}^{TT} &= 0\end{aligned}$$

A useful solution to this wave equation is that of a plane wave, $h_{\mu\nu}^{TT} = C_{\mu\nu} e^{ik_{\sigma} x^{\sigma}}$ for some symmetric $(0, 2)$ tensor $C_{\mu\nu}$. Plugging in, we get

$$\begin{aligned}0 &= \square h_{\mu\nu}^{TT} = \eta^{\rho\sigma} \partial_{\rho} \partial_{\sigma} h_{\mu\nu}^{TT} \\ &= \eta^{\rho\sigma} \partial_{\rho} (i k_{\sigma} h_{\mu\nu}^{TT}) \\ &= \eta^{\rho\sigma} (-k_{\rho} k_{\sigma} h_{\mu\nu}^{TT}) \\ &= -k_{\sigma} k^{\sigma} h_{\mu\nu}^{TT}\end{aligned}$$

I.e. $k_{\sigma} k^{\sigma} = 0$, so the vector is null. That is, gravitational waves propagate at the speed of light.

7.4.1 Physical Effect of Gravitational Waves

Recall the geodesic deviation equation from all the way back (chapter 3):

$$\frac{D^2}{d\tau^2} S^\mu = R^\mu{}_{\nu\rho\sigma} U^\nu U^\rho S^\sigma$$

Find the RHS to 1st order in $h_{\mu\nu}^{TT}$. Since $R^\mu{}_{\nu\rho\sigma}$ is already 1st order in $h_{\mu\nu}^{TT}$, so we only need the 4-velocity to 0th order: $U^\nu = (1, 0, 0, 0)$. Further:

$$\begin{aligned} R_{\mu 00\sigma} &= \frac{1}{2} \left(\partial_0 \partial_0 h_{\mu\sigma}^{TT} + \partial_0 \partial_\mu h_{00}^{TT} - \partial_\sigma \partial_0 h_{\mu 0}^{TT} - \partial_\mu \partial_0 h_{\sigma 0}^{TT} \right) \\ &= \frac{1}{2} \partial_0 \partial_0 h_{\mu\sigma}^{TT} \end{aligned}$$

For slowly moving particles, $\tau = x^0 = t$ to lowest order, and the geodesic deviation is

$$\frac{\partial^2}{\partial t^2} S^\mu = \frac{1}{2} S^\sigma \frac{\partial^2}{\partial t^2} h^{TT\mu}{}_\sigma$$

To analyze this, let $h_+ = C_{11}$ and $h_\times = C_{12}$. Since $C_{\mu\nu}$ is symmetric, we have

$$C_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

1. When $h_\times = 0$, we have (to lowest order)

$$\begin{aligned} \frac{\partial^2}{\partial t^2} S^1 &= \frac{1}{2} S^1 \frac{\partial^2}{\partial t^2} (h_+ e^{ik_\sigma x^\sigma}) \longrightarrow S^1 = \left(1 + \frac{1}{2} h_+ e^{ik_\sigma x^\sigma} \right) S^1(0) \\ \frac{\partial^2}{\partial t^2} S^2 &= -\frac{1}{2} S^2 \frac{\partial^2}{\partial t^2} (h_+ e^{ik_\sigma x^\sigma}) \longrightarrow S^2 = \left(1 - \frac{1}{2} h_+ e^{ik_\sigma x^\sigma} \right) S^2(0) \end{aligned}$$

2. When $h_+ = 0$, we have (to lowest order)

$$\begin{aligned} S^1 &= S^1(0) + \frac{1}{2} h_\times e^{ik_\sigma x^\sigma} S^2(0) \\ S^2 &= S^2(0) + \frac{1}{2} h_\times e^{ik_\sigma x^\sigma} S^1(0) \end{aligned}$$

If $h_\times = 0$, we oscillate in a + pattern. If $h_+ = 0$, we oscillate in a $\times$ pattern. We call h_+ the “plus” polarization and $h_\times$ the “cross” polarization. This has the grace notes of string theory as well.

7.4.2 Quantized Fields and Spin

The “spin” of a field is (roughly) related to how the quantized field transforms under spatial rotations. Generally, we say the spin S is related to θ under which the field is invariant under rotations of $360^\circ/\theta$.

- The electromagnetic field is invariant under $\theta = 360^\circ$, and yields a spin-1 particle (photon) after quantizing.
- The neutrino is invariant under 720° , and is spin- $\frac{1}{2}$.
- From above, the polarization modes of gravitational waves are invariant under 180° , thus we predict a spin-2 particle (“graviton”). This has yet to be observed, and may never be.

7.5 Production of Gravitational Waves

How is gravitational radiation generated by sources?

$$G_{\mu\nu} = 8\pi GT_{\mu\nu}$$

$T_{\mu\nu}$ is nonzero, hence the metric perturbation includes nonzero components. We cannot assume a transverse traceless form of $h_{\mu\nu}$. Instead, we use the full form of $h_{\mu\nu}$, then impose the transverse traceless gauge for sources.

1. Define the trace-reversed perturbation:

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2}h\eta_{\mu\nu} \quad , \quad \bar{h} = \eta^{\mu\nu}\bar{h}_{\mu\nu} = -h$$

2. Under a gauge transformation $h_{\mu\nu}^{(\epsilon)} \rightarrow h_{\mu\nu} + 2\epsilon\partial_{(\mu}\xi_{\nu)}$, we have the “Lorenz gauge”

$$\begin{aligned} \bar{h}_{\mu\nu} &\rightarrow \bar{h}_{\mu\nu} + 2\partial_{(\mu}\xi_{\nu)} - \partial_\lambda\xi^\lambda\eta_{\mu\nu} \\ \partial_\mu\bar{h}^{\mu\nu} &= 0 \quad \text{by choosing} \quad \square\xi_\mu = -\partial_\lambda\bar{h}^\lambda{}_\mu \end{aligned}$$

3. After the procedures outlined before that are tedious and take a long time,

$$\begin{aligned} G_{\mu\nu} &= -\frac{1}{2}\square\bar{h}_{\mu\nu} \\ \square\bar{h}_{\mu\nu} &= -16\pi GT_{\mu\nu} \end{aligned}$$

4. The Green function $G(x^\sigma - y^\sigma)$ for the d’Alembertian is a solution to the wave equation in the presence of a δ -function potential source.

$$\begin{aligned} \square_x G(x^\sigma - y^\sigma) &= \delta^{(4)}(x^\sigma - y^\sigma) \\ \bar{h}_{\mu\nu}(x^\sigma) &= -16\pi G \int G(x^\sigma - y^\sigma) T_{\mu\nu}(y^\sigma) d^4y \end{aligned}$$

5. Green functions can be “retarded” or “advanced” depending on if the wave is traveling backward/forward in time. We use a retarded Green function to see the effect of signals to the past.

$$\begin{aligned} G(x^\sigma - y^\sigma) &= -\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} \delta\left(|\mathbf{x} - \mathbf{y}| - (x^0 - y^0)\right) \theta(x^0 - y^0) \\ \bar{h}_{\mu\nu}(x^\sigma) &= 4G \int \frac{1}{|\mathbf{x} - \mathbf{y}|} T_{\mu\nu}(t - |\mathbf{x} - \mathbf{y}|, \mathbf{y}) d^3y \quad , \quad t = x^0 \end{aligned}$$

We see that the disturbance in a gravitational field at $(t, \mathbf{x})$ is the sum of influences from the past light cone $(t - |\mathbf{x} - \mathbf{y}|, \mathbf{x} - \mathbf{y})$.

(For more details see Carroll 7.5) By Fourier transforming h_{ij} and doing some manipulations, we get

$$\bar{h}_{ij}(t, \mathbf{x}) = \frac{2G}{r} \frac{d^2 I_{ij}}{dt^2}(t_r) \quad , \quad t_r = t - r$$

Where I_{ij} is the **quadrupole moment tensor**:

$$I_{ij}(t) = \int y^i y^j T^{00}(t, \mathbf{y}) d^3 y$$

7.5.1 Gravitational Radiation of a Binary Star

Assume rotation around a common center, a radius of R , and both masses M with speed v . According to the Newtonian approximation, we have a balance of gravity and centrifugal force:

$$\begin{aligned} \frac{GM^2}{(2R)^2} &= \frac{Mv^2}{R} \quad \longrightarrow \quad v = \sqrt{\frac{GM}{4R}} \\ \Omega &= \frac{v}{R} = \sqrt{\frac{GM}{4R^3}} \end{aligned}$$

The paths of stars a and b are given by:

$$\begin{aligned} x_a^1 &= R \cos(\Omega t) \quad , \quad x_a^2 = R \sin(\Omega t) \\ x_b^1 &= -R \cos(\Omega t) \quad , \quad x_b^2 = -R \sin(\Omega t) \end{aligned}$$

From here, we can find the (approximate) energy-momentum tensor and the quadrupole moments:

$$\begin{aligned} T^{00}(t, \mathbf{x}) &= M \delta(x^3) \cdot \left[\delta(x_a^1 - R \cos(\Omega t)) \delta(x_a^2 - R \sin(\Omega t)) + \delta(x_b^1 + R \cos(\Omega t)) \delta(x_b^2 + R \sin(\Omega t)) \right] \\ I_{11} &= 2MR^2 \cos^2(\Omega t) \longrightarrow MR^2(1 + \cos(2\Omega t)) \\ I_{22} &= 2MR^2 \sin^2(\Omega t) \longrightarrow MR^2(1 - \cos(2\Omega t)) \\ I_{12} &= I_{21} = 2MR^2 \cos(\Omega t) \sin(\Omega t) \longrightarrow MR^2 \sin(2\Omega t) \end{aligned}$$

Then the metric perturbation can be found. As always, this is useful for linearizing Einstein's equations, approximating solutions, etc.

$$\bar{h}_{ij}(t, \mathbf{x}) = \frac{8GM}{r} \Omega^2 R^2 \begin{pmatrix} -\cos(2\Omega t_r) & -\sin(2\Omega t_r) & 0 \\ -\sin(2\Omega t_r) & \cos(2\Omega t_r) & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

7.6 Energy Loss Due to Gravitational Radiation

7.6.1 In the Transverse-Traceless Gauge

Expand both the metric and the Ricci tensor to 1st and 2nd order:

$$\begin{aligned} g_{\mu\nu} &= \eta_{\mu\nu} + h_{\mu\nu}^{(1)} + h_{\mu\nu}^{(2)} \\ R_{\mu\nu} &= R_{\mu\nu}^{(0)} + R_{\mu\nu}^{(1)} + R_{\mu\nu}^{(2)} \end{aligned}$$

Consider a flat patch of spacetime (roughly), as well as a vacuum (guaranteeing the transverse-traceless condition). The vacuum equation is $R_{\mu\nu} = 0$. Since the metric is flat, the 0th order $R_{\mu\nu}^{(0)} = 0$ is satisfied. To 1st and 2nd order,

$$\begin{aligned} R_{\mu\nu}^{(1)} [h^{(1)}] &= 0 \\ R_{\mu\nu}^{(2)} [h^{(1)}] + R_{\mu\nu}^{(1)} [h^{(2)}] &= 0 \end{aligned}$$

From the equivalent vacuum condition $G_{\mu\nu} = 0$, we can (to 2nd order in h) define a modified energy-momentum tensor $t_{\mu\nu}$:

$$\begin{aligned} R_{\mu\nu}^{(1)} [h^{(2)}] - \frac{1}{2} \eta^{\rho\sigma} R_{\rho\sigma}^{(1)} [h^{(2)}] \eta_{\mu\nu} &= 8\pi G t_{\mu\nu} \\ t_{\mu\nu} &= -\frac{1}{8\pi G} \left[R_{\mu\nu}^{(2)} [h^{(1)}] - \frac{1}{2} \eta^{\rho\sigma} R_{\rho\sigma}^{(2)} [h^{(1)}] \eta_{\mu\nu} \right] \end{aligned}$$

In the transverse-traceless gauge (for more details, see Carroll 7.6),

$$\begin{aligned} \partial^\mu h_{\mu\nu}^{TT} &= 0 \quad , \quad h^{TT} = 0 \\ t_{\mu\nu} &= \frac{1}{32\pi G} \left\langle (\partial_\mu h_{\rho\sigma}^{TT}) (\partial_\nu h_{TT}^{\rho\sigma}) \right\rangle \end{aligned}$$

7.6.2 Rate of Energy Loss

The total energy of gravitational radiation from a hypersurface Σ is

$$E = \int_{\Sigma} t_{00} d^3x$$

Considering the limit $r \rightarrow \infty$ in spherical coordinates, the total energy radiated to ∞ is

$$\begin{aligned} \Delta E &= \int P dt \\ P &= \int_{S_\infty^2} t_{0\mu} n^\mu r^2 d\Omega \end{aligned}$$

For spherical $\{t, r, \theta, \phi\}$, $n^\mu = (0, 1, 0, 0)$. Now recall the quadrupole formula:

$$\bar{h}_{ij}(t, \mathbf{x}) = \frac{2G}{r} \frac{d^2 I_{ij}}{dt^2}(t_r)$$

This was for a Lorenz gauge, not the transverse-traceless gauge we're interested in. But we can modify the terms to create a transverse-traceless version. Define the **reduced quadrupole moment** J_{ij} :

$$J_{ij} = I_{ij} - \frac{1}{3} \delta_{ij} \delta^{kl} I_{kl}$$

Since h_{ij}^{TT} is traceless, we have

$$\begin{aligned} h_{ij}^{TT} &= \bar{h}_{ij}^{TT} = \frac{2G}{r} \frac{d^2 J_{ij}^{TT}}{dt^2} (t_r) \\ \partial_0 h_{ij}^{TT} &= \frac{2G}{r} \frac{d^3 J_{ij}^{TT}}{dt^3} \\ \partial_r h_{ij}^{TT} &= -\frac{2G}{r} \frac{d^3 J_{ij}^{TT}}{dt^3} - \frac{2G}{r^2} \frac{d^2 J_{ij}^{TT}}{dt^2} \approx -\frac{2G}{r} \frac{d^3 J_{ij}^{TT}}{dt^3} \end{aligned}$$

The revised energy-momentum tensor is

$$t_{0r} = \frac{1}{32\pi G} \left\langle \frac{2G}{r} \frac{d^3 J_{ij}^{TT}}{dt^3} \cdot \left(-\frac{2G}{r} \right) \frac{d^3 J_{ij}^{TT}}{dt^3} \right\rangle = -\frac{G}{8\pi r^2} \left\langle \left(\frac{d^3 J_{ij}^{TT}}{dt^3} \right) \left(\frac{d^3 J_{ij}^{TT}}{dt^3} \right) \right\rangle$$

After some calculations, the radiated power is

$$\begin{aligned} P &= -\frac{G}{8\pi} \int_{S_\infty^2} \left\langle \frac{d^3 J_{ij}}{dt^3} \frac{d^3 J^{ij}}{dt^3} - 2 \frac{d^3 J_i^j}{dt^3} \frac{d^3 J^{ik}}{dt^3} n_j n_k + \frac{1}{2} \frac{d^3 J^{ij}}{dt^3} \frac{d^3 J^{kl}}{dt^3} n_i n_j n_k n_l \right\rangle d\Omega \\ P &= -\frac{G}{5} \left\langle \frac{d^3 J_{ij}}{dt^3} \frac{d^3 J^{ij}}{dt^3} \right\rangle \end{aligned}$$

7.6.3 Binary Star System Revisited

For the binary star system described previously, the reduced quadrupole moment is

$$\begin{aligned} J_{ij} &= \frac{MR^2}{3} \begin{pmatrix} 1 + 3 \cos(2\Omega t) & 3 \sin(2\Omega t) & 0 \\ 3 \sin(2\Omega t) & 1 - 3 \cos(2\Omega t) & 0 \\ 0 & 0 & -2 \end{pmatrix} \\ \frac{d^3 J_{ij}}{dt^3} &= 8MR^2 \Omega^3 \begin{pmatrix} \sin(2\Omega t) & -\cos(2\Omega t) & 0 \\ -\cos(2\Omega t) & -\sin(2\Omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

The total power radiated by this binary star system is

$$\begin{aligned} P &= -\frac{128}{5} GM^2 R^4 \Omega^6, \quad \Omega = \sqrt{\frac{GM}{4R^3}} \\ P &= -\frac{2}{5} \frac{G^4 M^5}{R^5} \end{aligned}$$